

SAPIENZA
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Combinatorial Optimization

Exercises 3 — Dynamic programming

Worked exercises on the chapters of Volume 3 (dynamic programming), each followed by its complete solution. The instances are those of the selected problems, cited by their code.

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Online version and downloadable material:
fabiofurini.github.io/combinatorial-optimization/
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Chapter 16

Principles of dynamic programming

Exercise 16.1 — Optimal substructure and overlapping subproblems

A staircase has $n = 6$ steps. Stepping on step k costs c_k, and from a step we can climb to the next one or skip one step. We start from the ground, step 0, at no cost, and we want to reach step 6 at minimum total cost:

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
c_k	3	1	4	1	5	2

For $k \in \{0, 1, \ldots, n\}$, let $v(k)$ be the minimum cost to reach step k.

Questions

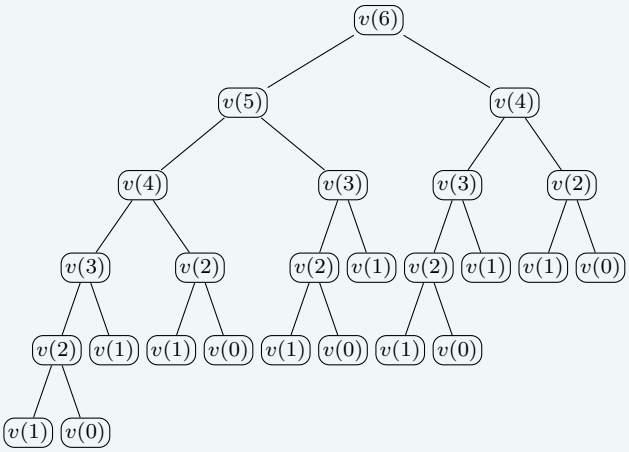
1. Check the optimal substructure: if an optimal way to reach step k arrives from step $k-1$, what can be said about its part from step 0 to step $k-1$? Write the recurrence for $v(k)$, with the base cases.
2. Draw the recursion tree of the top-down evaluation of $v(6)$ down to the base cases. How many times is $v(2)$ evaluated? What property does this show?
3. Fill the lookup table bottom-up, storing also in $s[k]$ the step from which the best way arrives.
4. Reconstruct an optimal solution from the table s.

Solution

1. The part from step 0 to step $k-1$ is a way to reach step $k-1$, and it must be optimal: a cheaper way to reach step $k-1$, followed by the last move, would reach step k at a lower cost. The same holds when the last move comes from step $k-2$. Hence:

$$v(k) = \begin{cases} c_k + \min\{v(k-1),\, v(k-2)\} & \text{if } 2 \le k \le n, \\ c_1 & \text{if } k = 1, \\ 0 & \text{if } k = 0. \end{cases}$$

2. Each call for $k \geq 2$ calls $k-1$ and $k-2$. The tree has 25 calls, and the numbers of evaluations of $v(0), \ldots, v(6)$ are $5, 8, 5, 3, 2, 1, 1$:



The value $v(2)$ is evaluated 5 times: the subproblems **overlap**. With a lookup table each of the 7 values is computed once.

3. The values are computed for increasing k:

k	$\min\{v(k-1), v(k-2)\}$	$v[k]$	$s[k]$	action
0	—	0	—	base case
1	—	3	0	base case: $v[1] \leftarrow c_1 = 3$
2	$\min\{3, 0\}$	$1 + 0 = 1$	0	from step 0
3	$\min\{1, 3\}$	$4 + 1 = 5$	2	from step 2
4	$\min\{5, 1\}$	$1 + 1 = 2$	2	from step 2
5	$\min\{2, 5\}$	$5 + 2 = 7$	4	from step 4
6	$\min\{7, 2\}$	$2 + 2 = 4$	4	from step 4

A tie is broken in favour of step $k-1$.

4. Starting from $k = 6$ and following s backwards: $s[6] = 4$, $s[4] = 2$, $s[2] = 0$. The optimal way is $0 \to 2 \to 4 \to 6$, with cost $1 + 1 + 2 = 4 = v(6)$.

$v(6) = 4$, reached by the steps $0 \to 2 \to 4 \to 6$; the recursion tree evaluates $v(2)$ 5 times, the lookup table once.

Exercise 16.2 — Writing a recurrence: no two consecutive positions

Along a road there are $n = 7$ positions for billboards, numbered in order. A billboard in position k brings a profit p_k, and two billboards cannot be in consecutive positions:

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
p_k	5	8	6	3	7	9	4

We want a set of positions, no two consecutive, of maximum total profit.

Questions

1. Define the subproblems: let $v(k)$ be the optimal value when only the positions $1, \dots, k$ are available. Write a recurrence for $v(k)$, distinguishing whether position k is used, and the base cases.
2. Fill the lookup table bottom-up, storing in $s[k]$ whether position k is used (1) or not (0).
3. Reconstruct an optimal solution.
4. Choose the positions greedily, by non-increasing profit, skipping the positions next to a chosen one. Is the result optimal?

Solution

1. If position k is not used, the best choice among $1, \dots, k$ is the best choice among $1, \dots, k-1$. If it is used, position $k-1$ is forbidden and the rest is an optimal choice among $1, \dots, k-2$ (optimal substructure: a better choice there would improve the whole). Hence:

$$v(k) = \begin{cases} \max\{v(k-1),\ p_k + v(k-2)\} & \text{if } 2 \le k \le n, \\ p_1 & \text{if } k = 1, \\ 0 & \text{if } k = 0. \end{cases}$$

2. The values are computed for increasing k (a tie is broken in favour of not using position k):

k	$v(k-1)$	$p_k + v(k-2)$	$v[k]$	$s[k]$
0	—	—	0	0
1	—	—	5	1
2	5	$8+0=8$	8	1
3	8	$6+5=11$	11	1
4	11	$3+8=11$	11	0
5	11	$7+11=18$	18	1
6	18	$9+11=20$	20	1
7	20	$4+18=22$	22	1

3. Starting from $k = 7$:
   - $s[7] = 1$: position 7 is chosen, go to $k = 5$;
   - $s[5] = 1$: position 5 is chosen, go to $k = 3$;
   - $s[3] = 1$: position 3 is chosen, go to $k = 1$;
   - $s[1] = 1$: position 1 is chosen, go to $k = 0$;

   The optimal solution is $\{1, 3, 5, 7\}$, with profit $5 + 6 + 7 + 4 = 22 = v(7)$.
4. The greedy choice takes the positions $6, 2, 4$ (in this order), of profit $8+3+9 = 20 < 22$: it is not optimal. Taking position 6 first forbids its neighbours, which together are worth more.

$v(7) = 22$ with the positions $\{1, 3, 5, 7\}$; the greedy choice by profit gives only 20.

Exercise 16.3 — Filling a DP table: paths on a grid

A robot moves on a grid of 3 rows and 4 columns, from the cell $(1, 1)$ to the cell $(3, 4)$, one cell down or one cell right at each move. Every cell (i, j) it visits gives a profit a_{ij}:

	$j = 1$	$j = 2$	$j = 3$	$j = 4$
$i = 1$	3	1	4	2
$i = 2$	2	6	1	5
$i = 3$	4	2	3	1

We want a path of maximum total profit.

Questions

1. How many paths are there? Why is the enumeration of the paths not a good idea on large grids?
2. Let $v(i, j)$ be the maximum profit of a path from $(1, 1)$ to (i, j). Write the recurrence and the base cases.
3. Fill the table of $v(i, j)$ row by row, storing in $s(i, j)$ whether the best path arrives from above ($\uparrow$) or from the left ($\leftarrow$); a tie is broken in favour of above.
4. Reconstruct the optimal path.

Solution

1. A path makes 2 moves down and 3 moves right, in any order: $\binom{5}{2} = 10$ paths. On a grid with r rows and q columns there are $\binom{r+q-2}{r-1}$ paths, which grows exponentially, while the table has only rq cells.
2. The last move into (i, j) comes from $(i - 1, j)$ or from $(i, j - 1)$, and the path up to

there must be optimal (optimal substructure):

$$v(i,j) = \begin{cases} a_{ij} + \max\{v(i-1,j),\, v(i,j-1)\} & \text{if } i \geq 2,\ j \geq 2, \\ a_{1j} + v(1,j-1) & \text{if } i = 1,\ j \geq 2, \\ a_{i1} + v(i-1,1) & \text{if } i \geq 2,\ j = 1, \\ a_{11} & \text{if } i = j = 1. \end{cases}$$

3. The tables v and s are:

$v(i,j)$	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	3	4	8	10
$i=2$	5	11	12	17
$i=3$	9	13	16	18

	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	—	$\leftarrow$	$\leftarrow$	$\leftarrow$
$i=2$	$\uparrow$	$\leftarrow$	$\leftarrow$	$\leftarrow$
$i=3$	$\uparrow$	$\uparrow$	$\leftarrow$	$\uparrow$

For example $v(2,2) = a_{22} + \max\{v(1,2), v(2,1)\} = 6 + \max\{4,5\} = 11$.

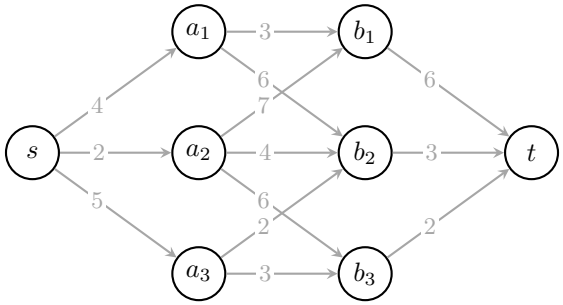
4. From $(3,4)$ we follow the arrows backwards: the path is $(1,1) \to (2,1) \to (2,2) \to (2,3) \to (2,4) \to (3,4)$, with profit $3+2+6+1+5+1 = 18 = v(3,4)$. Its cells are highlighted:

$v(i,j)$	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	3	4	8	10
$i=2$	5	11	12	17
$i=3$	9	13	16	18

The optimal path collects $v(3,4) = 18$. The DP algorithm fills one cell per subproblem, 12 here, while the number of paths grows exponentially with the size of the grid.

Exercise 16.4 — Shortest path in a layered network

We look for a shortest path from s to t in the directed network $G = (N, A)$ below, where every arc $(i,j) \in A$ has a cost c_{ij}:



For every node i, let $v(i)$ be the cost of a shortest path from i to t.

Questions

1. Check the optimal substructure: if a shortest path from s to t passes through node i, what can be said about its part from i to t? Write the recurrence for $v(i)$ and the base case.
2. Which values $v(i)$ are needed by more than one node? What property of DP does this show?
3. Compute the values $v(i)$ backwards, from t to s, storing in $s(i)$ the successor of i on the best path (ties broken by the node listed first).
4. Reconstruct the shortest path and compare its cost with the enumeration of all the paths.

Solution

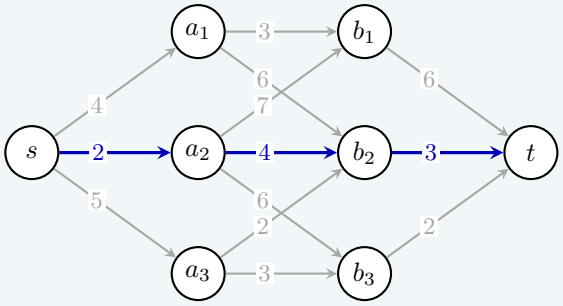
1. The part from i to t must be a shortest path from i to t: a shorter one would give a shorter path from s to t. The first arc out of i goes to some successor j, hence:

$$v(i) = \begin{cases} \min\{c_{ij} + v(j) : (i,j) \in A\} & \text{if } i \neq t, \\ 0 & \text{if } i = t. \end{cases}$$

2. The values $v(b_1)$, $v(b_2)$, $v(b_3)$ are needed by the nodes a_1, a_2, a_3 that point to them: $v(b_1)$ by 2 nodes, $v(b_2)$ by 3 and $v(b_3)$ by 2. The subproblems overlap: with a lookup table each value is computed once.
3. The values, from t backwards:

node i	$c_{ij} + v(j)$ for $(i,j) \in A$	$v(i)$	$s(i)$
t	—	0	—
b_3	$2+0=2$	2	t
b_2	$3+0=3$	3	t
b_1	$6+0=6$	6	t
a_3	$2+3=5$, $3+2=5$	5	b_2
a_2	$7+6=13$, $4+3=7$, $6+2=8$	7	b_2
a_1	$3+6=9$, $6+3=9$	9	b_1
s	$4+9=13$, $2+7=9$, $5+5=10$	9	a_2

4. Following s from s: $s \to a_2 \to b_2 \to t$, of cost $2+4+3=9=v(s)$:



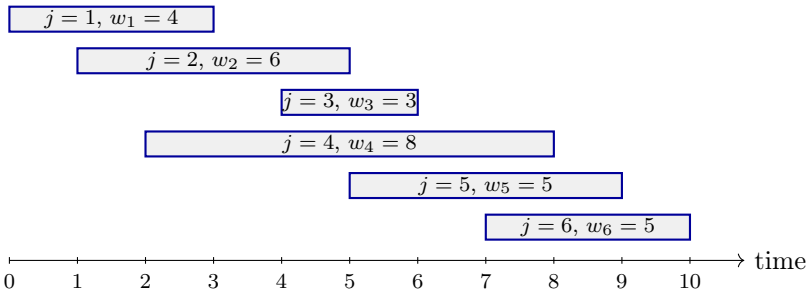
The network has 7 paths from s to t; their enumeration gives the same optimal path, the only one of cost 9.

$v(s) = 9$, with the shortest path $s \to a_2 \to b_2 \to t$; the values of the nodes b_k are computed once and used several times.

Exercise 16.5 — A recurrence for a new problem: compatible requests

A room can host one event at a time. There are $n = 6$ requests: request j occupies the room from time a_j to time b_j and pays w_j; two requests are compatible if one ends before (or when) the other starts. The requests are numbered by non-decreasing end time:

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	$j=6$
a_j	0	1	4	2	5	7
b_j	3	5	6	8	9	10
w_j	4	6	3	8	5	5



We want a set of pairwise compatible requests of maximum total payment.

Questions

1. For every j, let $q(j)$ be the largest index $i < j$ of a request compatible with j ($q(j) = 0$ if there is none). Compute $q(1), \ldots, q(n)$.

2. Let $v(j)$ be the optimal value with the requests $1, \ldots, j$ only. Write a recurrence for $v(j)$, distinguishing whether request j is taken, and the base case.
3. Fill the lookup table, storing in $s[j]$ whether request j is taken; a tie is broken in favour of not taking it.
4. Reconstruct an optimal solution. Does the greedy choice by non-increasing payment find it?

Solution

1. Request j starts at a_j, and the compatible requests with smaller index are those with $b_i \leq a_j$; since the end times are sorted, they are the requests $1, \ldots, q(j)$:

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	$j=6$
$q(j)$	0	0	1	0	2	3

2. If request j is not taken, the best choice is among $1, \ldots, j-1$. If it is taken, the requests $q(j)+1, \ldots, j-1$ overlap with it, and the rest is an optimal choice among $1, \ldots, q(j)$:

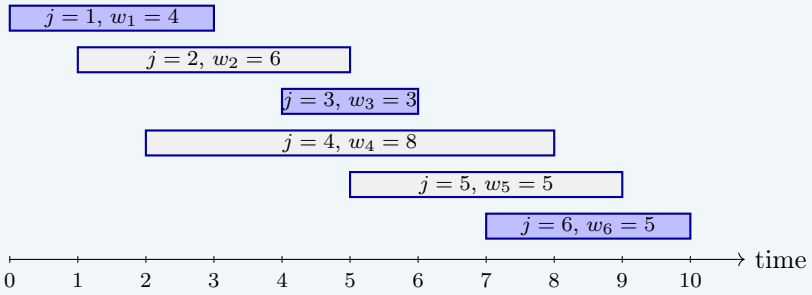
$$v(j) = \begin{cases} \max\{v(j-1),\ w_j + v(q(j))\} & \text{if } 1 \leq j \leq n, \\ 0 & \text{if } j = 0. \end{cases}$$

3. The values are computed for increasing j:

j	$q(j)$	$v(j-1)$	$w_j + v(q(j))$	$v[j]$	$s[j]$
0	—	—	—	0	—
1	0	0	$4+0=4$	4	1
2	0	4	$6+0=6$	6	1
3	1	6	$3+4=7$	7	1
4	0	7	$8+0=8$	8	1
5	2	8	$5+6=11$	11	1
6	3	11	$5+7=12$	12	1

4. Starting from $j=6$:
   - $s[6]=1$: job 6 is taken, go to $q(6)=3$;
   - $s[3]=1$: job 3 is taken, go to $q(3)=1$;
   - $s[1]=1$: job 1 is taken, go to $q(1)=0$;

   The optimal solution is $\{1,3,6\}$, with payment $4+3+5=12=v(6)$:



The greedy choice by payment takes 4, of payment $8 < 12$: request 4, the richest one, overlaps with the requests $1, 3, 6$ of the optimal solution, which together pay 12.

$v(6) = 12$ with the requests $\{1, 3, 6\}$; the greedy choice by payment gives only 8.

Chapter 17

DP for the knapsack problem

Exercise 17.1

Consider the following KP instance, where the knapsack has capacity $c = 6$, and the profits and weights associated with the items are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_i	6	7	8	9
w_i	2	3	4	5

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of k.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
1	0	0	0	$k < w_1$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w_1$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	6	1	$p[1] + v[0][0] = 6 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 6$, $s[1][2] \leftarrow 1$
1	3	6	1	$p[1] + v[0][1] = 6 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 6$, $s[1][3] \leftarrow 1$
1	4	6	1	$p[1] + v[0][2] = 6 + 0 > v[0][4] = 0$: $v[1][4] \leftarrow 6$, $s[1][4] \leftarrow 1$
1	5	6	1	$p[1] + v[0][3] = 6 + 0 > v[0][5] = 0$: $v[1][5] \leftarrow 6$, $s[1][5] \leftarrow 1$
1	6	6	1	$p[1] + v[0][4] = 6 + 0 > v[0][6] = 0$: $v[1][6] \leftarrow 6$, $s[1][6] \leftarrow 1$
2	0	0	0	$k < w_2$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w_2$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	6	0	$k < w_2$: $v[2][2] \leftarrow v[1][2] = 6$, $s[2][2] \leftarrow 0$
2	3	7	1	$p[2] + v[1][0] = 7 + 0 > v[1][3] = 6$: $v[2][3] \leftarrow 7$, $s[2][3] \leftarrow 1$
2	4	7	1	$p[2] + v[1][1] = 7 + 0 > v[1][4] = 6$: $v[2][4] \leftarrow 7$, $s[2][4] \leftarrow 1$
2	5	13	1	$p[2] + v[1][2] = 7 + 6 > v[1][5] = 6$: $v[2][5] \leftarrow 13$, $s[2][5] \leftarrow 1$
2	6	13	1	$p[2] + v[1][3] = 7 + 6 > v[1][6] = 6$: $v[2][6] \leftarrow 13$, $s[2][6] \leftarrow 1$
3	0	0	0	$k < w_3$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w_3$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	6	0	$k < w_3$: $v[3][2] \leftarrow v[2][2] = 6$, $s[3][2] \leftarrow 0$
3	3	7	0	$k < w_3$: $v[3][3] \leftarrow v[2][3] = 7$, $s[3][3] \leftarrow 0$
3	4	8	1	$p[3] + v[2][0] = 8 + 0 > v[2][4] = 7$: $v[3][4] \leftarrow 8$, $s[3][4] \leftarrow 1$
3	5	13	0	$p[3] + v[2][1] = 8 + 0 \leq v[2][5] = 13$: $v[3][5] \leftarrow 13$, $s[3][5] \leftarrow 0$
3	6	14	1	$p[3] + v[2][2] = 8 + 6 > v[2][6] = 13$: $v[3][6] \leftarrow 14$, $s[3][6] \leftarrow 1$
4	0	0	0	$k < w_4$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	0	0	$k < w_4$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	6	0	$k < w_4$: $v[4][2] \leftarrow v[3][2] = 6$, $s[4][2] \leftarrow 0$
4	3	7	0	$k < w_4$: $v[4][3] \leftarrow v[3][3] = 7$, $s[4][3] \leftarrow 0$
4	4	8	0	$k < w_4$: $v[4][4] \leftarrow v[3][4] = 8$, $s[4][4] \leftarrow 0$
4	5	13	0	$p[4] + v[3][0] = 9 + 0 \leq v[3][5] = 13$: $v[4][5] \leftarrow 13$, $s[4][5] \leftarrow 0$
4	6	14	0	$p[4] + v[3][1] = 9 + 0 \leq v[3][6] = 14$: $v[4][6] \leftarrow 14$, $s[4][6] \leftarrow 0$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	6	6	6	6	6
$i=2$	0	0	6	7	7	13	13
$i=3$	0	0	6	7	8	13	14
$i=4$	0	0	6	7	8	13	14

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1
$i=3$	0	0	0	0	1	0	1
$i=4$	0	0	0	0	0	0	0

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1
$i=3$	0	0	0	0	1	0	1
$i=4$	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=4$ and $k=6$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	-	0	0	0	0	$k=6$
1	4	6	0	0	0	0	0	no update
2	3	6	1	0	0	1	0	$x[3] \leftarrow 1$, $k \leftarrow 6-4=2$
3	2	2	0	0	0	1	0	no update
4	1	2	1	1	0	1	0	$x[1] \leftarrow 1$, $k \leftarrow 2-2=0$

Thus, the vector

$$(1,0,1,0)$$

is an optimal solution. The optimal value is

$$v[4][6] = 14.$$

Exercise 17.2

Consider the following KP instance, where the knapsack has capacity $c = 7$, and the profits and weights associated with the items are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_i	4	5	10	11
w_i	2	3	5	6

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of k.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
1	0	0	0	$k < w_1$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w_1$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	4	1	$p[1] + v[0][0] = 4 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 4$, $s[1][2] \leftarrow 1$

1	3	4	1	$p[1]+v[0][1]=4+0>v[0][3]=0$: $v[1][3]\leftarrow 4$, $s[1][3]\leftarrow 1$
1	4	4	1	$p[1]+v[0][2]=4+0>v[0][4]=0$: $v[1][4]\leftarrow 4$, $s[1][4]\leftarrow 1$
1	5	4	1	$p[1]+v[0][3]=4+0>v[0][5]=0$: $v[1][5]\leftarrow 4$, $s[1][5]\leftarrow 1$
1	6	4	1	$p[1]+v[0][4]=4+0>v[0][6]=0$: $v[1][6]\leftarrow 4$, $s[1][6]\leftarrow 1$
1	7	4	1	$p[1]+v[0][5]=4+0>v[0][7]=0$: $v[1][7]\leftarrow 4$, $s[1][7]\leftarrow 1$
2	0	0	0	$k<w_2$: $v[2][0]\leftarrow v[1][0]=0$, $s[2][0]\leftarrow 0$
2	1	0	0	$k<w_2$: $v[2][1]\leftarrow v[1][1]=0$, $s[2][1]\leftarrow 0$
2	2	4	0	$k<w_2$: $v[2][2]\leftarrow v[1][2]=4$, $s[2][2]\leftarrow 0$
2	3	5	1	$p[2]+v[1][0]=5+0>v[1][3]=4$: $v[2][3]\leftarrow 5$, $s[2][3]\leftarrow 1$
2	4	5	1	$p[2]+v[1][1]=5+0>v[1][4]=4$: $v[2][4]\leftarrow 5$, $s[2][4]\leftarrow 1$
2	5	9	1	$p[2]+v[1][2]=5+4>v[1][5]=4$: $v[2][5]\leftarrow 9$, $s[2][5]\leftarrow 1$
2	6	9	1	$p[2]+v[1][3]=5+4>v[1][6]=4$: $v[2][6]\leftarrow 9$, $s[2][6]\leftarrow 1$
2	7	9	1	$p[2]+v[1][4]=5+4>v[1][7]=4$: $v[2][7]\leftarrow 9$, $s[2][7]\leftarrow 1$
3	0	0	0	$k<w_3$: $v[3][0]\leftarrow v[2][0]=0$, $s[3][0]\leftarrow 0$
3	1	0	0	$k<w_3$: $v[3][1]\leftarrow v[2][1]=0$, $s[3][1]\leftarrow 0$
3	2	4	0	$k<w_3$: $v[3][2]\leftarrow v[2][2]=4$, $s[3][2]\leftarrow 0$
3	3	5	0	$k<w_3$: $v[3][3]\leftarrow v[2][3]=5$, $s[3][3]\leftarrow 0$
3	4	5	0	$k<w_3$: $v[3][4]\leftarrow v[2][4]=5$, $s[3][4]\leftarrow 0$
3	5	10	1	$p[3]+v[2][0]=10+0>v[2][5]=9$: $v[3][5]\leftarrow 10$, $s[3][5]\leftarrow 1$
3	6	10	1	$p[3]+v[2][1]=10+0>v[2][6]=9$: $v[3][6]\leftarrow 10$, $s[3][6]\leftarrow 1$
3	7	14	1	$p[3]+v[2][2]=10+4>v[2][7]=9$: $v[3][7]\leftarrow 14$, $s[3][7]\leftarrow 1$
4	0	0	0	$k<w_4$: $v[4][0]\leftarrow v[3][0]=0$, $s[4][0]\leftarrow 0$
4	1	0	0	$k<w_4$: $v[4][1]\leftarrow v[3][1]=0$, $s[4][1]\leftarrow 0$
4	2	4	0	$k<w_4$: $v[4][2]\leftarrow v[3][2]=4$, $s[4][2]\leftarrow 0$
4	3	5	0	$k<w_4$: $v[4][3]\leftarrow v[3][3]=5$, $s[4][3]\leftarrow 0$
4	4	5	0	$k<w_4$: $v[4][4]\leftarrow v[3][4]=5$, $s[4][4]\leftarrow 0$
4	5	10	0	$k<w_4$: $v[4][5]\leftarrow v[3][5]=10$, $s[4][5]\leftarrow 0$
4	6	11	1	$p[4]+v[3][0]=11+0>v[3][6]=10$: $v[4][6]\leftarrow 11$, $s[4][6]\leftarrow 1$
4	7	14	0	$p[4]+v[3][1]=11+0\leq v[3][7]=14$: $v[4][7]\leftarrow 14$, $s[4][7]\leftarrow 0$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	4	4	4	4	4	4
$i=2$	0	0	4	5	5	9	9	9
$i=3$	0	0	4	5	5	10	10	14
$i=4$	0	0	4	5	5	10	11	14

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1	1
$i=3$	0	0	0	0	0	1	1	1
$i=4$	0	0	0	0	0	0	1	0

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1	1
$i=3$	0	0	0	0	0	1	1	1
$i=4$	0	0	0	0	0	0	1	0

We can reconstruct an optimal solution by starting from $i=4$ and $k=7$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	-	0	0	0	0	$k=7$
1	4	7	0	0	0	0	0	no update
2	3	7	1	0	0	1	0	$x[3] \leftarrow 1$, $k \leftarrow 7-5=2$
3	2	2	0	0	0	1	0	no update
4	1	2	1	1	0	1	0	$x[1] \leftarrow 1$, $k \leftarrow 2-2=0$

Thus, the vector

$$(1,0,1,0)$$

is an optimal solution. The optimal value is

$$v[4][7] = 14.$$

Exercise 17.3

Consider the following KP instance, where the knapsack has capacity $c = 8$, and the profits and weights associated with the items are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	3	4	8	8	10
w_i	1	3	4	5	6

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of k.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
1	0	0	0	$k < w_1$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	3	1	$p[1] + v[0][0] = 3 + 0 > v[0][1] = 0$: $v[1][1] \leftarrow 3$, $s[1][1] \leftarrow 1$

1	2	3	1	$p[1]+v[0][1]=3+0>v[0][2]=0$: $v[1][2]\leftarrow 3$, $s[1][2]\leftarrow 1$
1	3	3	1	$p[1]+v[0][2]=3+0>v[0][3]=0$: $v[1][3]\leftarrow 3$, $s[1][3]\leftarrow 1$
1	4	3	1	$p[1]+v[0][3]=3+0>v[0][4]=0$: $v[1][4]\leftarrow 3$, $s[1][4]\leftarrow 1$
1	5	3	1	$p[1]+v[0][4]=3+0>v[0][5]=0$: $v[1][5]\leftarrow 3$, $s[1][5]\leftarrow 1$
1	6	3	1	$p[1]+v[0][5]=3+0>v[0][6]=0$: $v[1][6]\leftarrow 3$, $s[1][6]\leftarrow 1$
1	7	3	1	$p[1]+v[0][6]=3+0>v[0][7]=0$: $v[1][7]\leftarrow 3$, $s[1][7]\leftarrow 1$
1	8	3	1	$p[1]+v[0][7]=3+0>v[0][8]=0$: $v[1][8]\leftarrow 3$, $s[1][8]\leftarrow 1$
2	0	0	0	$k<w_2$: $v[2][0]\leftarrow v[1][0]=0$, $s[2][0]\leftarrow 0$
2	1	3	0	$k<w_2$: $v[2][1]\leftarrow v[1][1]=3$, $s[2][1]\leftarrow 0$
2	2	3	0	$k<w_2$: $v[2][2]\leftarrow v[1][2]=3$, $s[2][2]\leftarrow 0$
2	3	4	1	$p[2]+v[1][0]=4+0>v[1][3]=3$: $v[2][3]\leftarrow 4$, $s[2][3]\leftarrow 1$
2	4	7	1	$p[2]+v[1][1]=4+3>v[1][4]=3$: $v[2][4]\leftarrow 7$, $s[2][4]\leftarrow 1$
2	5	7	1	$p[2]+v[1][2]=4+3>v[1][5]=3$: $v[2][5]\leftarrow 7$, $s[2][5]\leftarrow 1$
2	6	7	1	$p[2]+v[1][3]=4+3>v[1][6]=3$: $v[2][6]\leftarrow 7$, $s[2][6]\leftarrow 1$
2	7	7	1	$p[2]+v[1][4]=4+3>v[1][7]=3$: $v[2][7]\leftarrow 7$, $s[2][7]\leftarrow 1$
2	8	7	1	$p[2]+v[1][5]=4+3>v[1][8]=3$: $v[2][8]\leftarrow 7$, $s[2][8]\leftarrow 1$
3	0	0	0	$k<w_3$: $v[3][0]\leftarrow v[2][0]=0$, $s[3][0]\leftarrow 0$
3	1	3	0	$k<w_3$: $v[3][1]\leftarrow v[2][1]=3$, $s[3][1]\leftarrow 0$
3	2	3	0	$k<w_3$: $v[3][2]\leftarrow v[2][2]=3$, $s[3][2]\leftarrow 0$
3	3	4	0	$k<w_3$: $v[3][3]\leftarrow v[2][3]=4$, $s[3][3]\leftarrow 0$
3	4	8	1	$p[3]+v[2][0]=8+0>v[2][4]=7$: $v[3][4]\leftarrow 8$, $s[3][4]\leftarrow 1$
3	5	11	1	$p[3]+v[2][1]=8+3>v[2][5]=7$: $v[3][5]\leftarrow 11$, $s[3][5]\leftarrow 1$
3	6	11	1	$p[3]+v[2][2]=8+3>v[2][6]=7$: $v[3][6]\leftarrow 11$, $s[3][6]\leftarrow 1$
3	7	12	1	$p[3]+v[2][3]=8+4>v[2][7]=7$: $v[3][7]\leftarrow 12$, $s[3][7]\leftarrow 1$
3	8	15	1	$p[3]+v[2][4]=8+7>v[2][8]=7$: $v[3][8]\leftarrow 15$, $s[3][8]\leftarrow 1$
4	0	0	0	$k<w_4$: $v[4][0]\leftarrow v[3][0]=0$, $s[4][0]\leftarrow 0$
4	1	3	0	$k<w_4$: $v[4][1]\leftarrow v[3][1]=3$, $s[4][1]\leftarrow 0$
4	2	3	0	$k<w_4$: $v[4][2]\leftarrow v[3][2]=3$, $s[4][2]\leftarrow 0$
4	3	4	0	$k<w_4$: $v[4][3]\leftarrow v[3][3]=4$, $s[4][3]\leftarrow 0$
4	4	8	0	$k<w_4$: $v[4][4]\leftarrow v[3][4]=8$, $s[4][4]\leftarrow 0$
4	5	11	0	$p[4]+v[3][0]=8+0\leq v[3][5]=11$: $v[4][5]\leftarrow 11$, $s[4][5]\leftarrow 0$
4	6	11	0	$p[4]+v[3][1]=8+3\leq v[3][6]=11$: $v[4][6]\leftarrow 11$, $s[4][6]\leftarrow 0$
4	7	12	0	$p[4]+v[3][2]=8+3\leq v[3][7]=12$: $v[4][7]\leftarrow 12$, $s[4][7]\leftarrow 0$
4	8	15	0	$p[4]+v[3][3]=8+4\leq v[3][8]=15$: $v[4][8]\leftarrow 15$, $s[4][8]\leftarrow 0$
5	0	0	0	$k<w_5$: $v[5][0]\leftarrow v[4][0]=0$, $s[5][0]\leftarrow 0$
5	1	3	0	$k<w_5$: $v[5][1]\leftarrow v[4][1]=3$, $s[5][1]\leftarrow 0$
5	2	3	0	$k<w_5$: $v[5][2]\leftarrow v[4][2]=3$, $s[5][2]\leftarrow 0$
5	3	4	0	$k<w_5$: $v[5][3]\leftarrow v[4][3]=4$, $s[5][3]\leftarrow 0$
5	4	8	0	$k<w_5$: $v[5][4]\leftarrow v[4][4]=8$, $s[5][4]\leftarrow 0$
5	5	11	0	$k<w_5$: $v[5][5]\leftarrow v[4][5]=11$, $s[5][5]\leftarrow 0$
5	6	11	0	$p[5]+v[4][0]=10+0\leq v[4][6]=11$: $v[5][6]\leftarrow 11$, $s[5][6]\leftarrow 0$
5	7	13	1	$p[5]+v[4][1]=10+3>v[4][7]=12$: $v[5][7]\leftarrow 13$, $s[5][7]\leftarrow 1$
5	8	15	0	$p[5]+v[4][2]=10+3\leq v[4][8]=15$: $v[5][8]\leftarrow 15$, $s[5][8]\leftarrow 0$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	3	3	3	3	3	3	3	3
$i=2$	0	3	3	4	7	7	7	7	7
$i=3$	0	3	3	4	8	11	11	12	15
$i=4$	0	3	3	4	8	11	11	12	15
$i=5$	0	3	3	4	8	11	11	13	15

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1	1	1
$i=3$	0	0	0	0	1	1	1	1	1
$i=4$	0	0	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0	1	0

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	1	1	1	1	1	1
$i=3$	0	0	0	0	1	1	1	1	1
$i=4$	0	0	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0	1	0

We can reconstruct an optimal solution by starting from $i=5$ and $k=8$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	-	0	0	0	0	0	$k=8$
1	5	8	0	0	0	0	0	0	no update
2	4	8	0	0	0	0	0	0	no update
3	3	8	1	0	0	1	0	0	$x[3] \leftarrow 1$, $k \leftarrow 8-4=4$
4	2	4	1	0	1	1	0	0	$x[2] \leftarrow 1$, $k \leftarrow 4-3=1$
5	1	1	1	1	1	1	0	0	$x[1] \leftarrow 1$, $k \leftarrow 1-1=0$

Thus, the vector

$$(1,1,1,0,0)$$

is an optimal solution. The optimal value is

$$v[5][8] = 15.$$

Exercise 17.4

Consider the following KP instance, where the knapsack has capacity $c = 9$, and the profits and weights associated with the items are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	5	6	9	10	13
w_i	2	4	5	6	7

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of k.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
1	0	0	0	$k < w_1$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$

1	1	0	0	$k < w_1$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	5	1	$p[1] + v[0][0] = 5 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 5$, $s[1][2] \leftarrow 1$
1	3	5	1	$p[1] + v[0][1] = 5 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 5$, $s[1][3] \leftarrow 1$
1	4	5	1	$p[1] + v[0][2] = 5 + 0 > v[0][4] = 0$: $v[1][4] \leftarrow 5$, $s[1][4] \leftarrow 1$
1	5	5	1	$p[1] + v[0][3] = 5 + 0 > v[0][5] = 0$: $v[1][5] \leftarrow 5$, $s[1][5] \leftarrow 1$
1	6	5	1	$p[1] + v[0][4] = 5 + 0 > v[0][6] = 0$: $v[1][6] \leftarrow 5$, $s[1][6] \leftarrow 1$
1	7	5	1	$p[1] + v[0][5] = 5 + 0 > v[0][7] = 0$: $v[1][7] \leftarrow 5$, $s[1][7] \leftarrow 1$
1	8	5	1	$p[1] + v[0][6] = 5 + 0 > v[0][8] = 0$: $v[1][8] \leftarrow 5$, $s[1][8] \leftarrow 1$
1	9	5	1	$p[1] + v[0][7] = 5 + 0 > v[0][9] = 0$: $v[1][9] \leftarrow 5$, $s[1][9] \leftarrow 1$
2	0	0	0	$k < w_2$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w_2$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	5	0	$k < w_2$: $v[2][2] \leftarrow v[1][2] = 5$, $s[2][2] \leftarrow 0$
2	3	5	0	$k < w_2$: $v[2][3] \leftarrow v[1][3] = 5$, $s[2][3] \leftarrow 0$
2	4	6	1	$p[2] + v[1][0] = 6 + 0 > v[1][4] = 5$: $v[2][4] \leftarrow 6$, $s[2][4] \leftarrow 1$
2	5	6	1	$p[2] + v[1][1] = 6 + 0 > v[1][5] = 5$: $v[2][5] \leftarrow 6$, $s[2][5] \leftarrow 1$
2	6	11	1	$p[2] + v[1][2] = 6 + 5 > v[1][6] = 5$: $v[2][6] \leftarrow 11$, $s[2][6] \leftarrow 1$
2	7	11	1	$p[2] + v[1][3] = 6 + 5 > v[1][7] = 5$: $v[2][7] \leftarrow 11$, $s[2][7] \leftarrow 1$
2	8	11	1	$p[2] + v[1][4] = 6 + 5 > v[1][8] = 5$: $v[2][8] \leftarrow 11$, $s[2][8] \leftarrow 1$
2	9	11	1	$p[2] + v[1][5] = 6 + 5 > v[1][9] = 5$: $v[2][9] \leftarrow 11$, $s[2][9] \leftarrow 1$
3	0	0	0	$k < w_3$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w_3$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	5	0	$k < w_3$: $v[3][2] \leftarrow v[2][2] = 5$, $s[3][2] \leftarrow 0$
3	3	5	0	$k < w_3$: $v[3][3] \leftarrow v[2][3] = 5$, $s[3][3] \leftarrow 0$
3	4	6	0	$k < w_3$: $v[3][4] \leftarrow v[2][4] = 6$, $s[3][4] \leftarrow 0$
3	5	9	1	$p[3] + v[2][0] = 9 + 0 > v[2][5] = 6$: $v[3][5] \leftarrow 9$, $s[3][5] \leftarrow 1$
3	6	11	0	$p[3] + v[2][1] = 9 + 0 \leq v[2][6] = 11$: $v[3][6] \leftarrow 11$, $s[3][6] \leftarrow 0$
3	7	14	1	$p[3] + v[2][2] = 9 + 5 > v[2][7] = 11$: $v[3][7] \leftarrow 14$, $s[3][7] \leftarrow 1$
3	8	14	1	$p[3] + v[2][3] = 9 + 5 > v[2][8] = 11$: $v[3][8] \leftarrow 14$, $s[3][8] \leftarrow 1$
3	9	15	1	$p[3] + v[2][4] = 9 + 6 > v[2][9] = 11$: $v[3][9] \leftarrow 15$, $s[3][9] \leftarrow 1$
4	0	0	0	$k < w_4$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	0	0	$k < w_4$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	5	0	$k < w_4$: $v[4][2] \leftarrow v[3][2] = 5$, $s[4][2] \leftarrow 0$
4	3	5	0	$k < w_4$: $v[4][3] \leftarrow v[3][3] = 5$, $s[4][3] \leftarrow 0$
4	4	6	0	$k < w_4$: $v[4][4] \leftarrow v[3][4] = 6$, $s[4][4] \leftarrow 0$
4	5	9	0	$k < w_4$: $v[4][5] \leftarrow v[3][5] = 9$, $s[4][5] \leftarrow 0$
4	6	11	0	$p[4] + v[3][0] = 10 + 0 \leq v[3][6] = 11$: $v[4][6] \leftarrow 11$, $s[4][6] \leftarrow 0$
4	7	14	0	$p[4] + v[3][1] = 10 + 0 \leq v[3][7] = 14$: $v[4][7] \leftarrow 14$, $s[4][7] \leftarrow 0$
4	8	15	1	$p[4] + v[3][2] = 10 + 5 > v[3][8] = 14$: $v[4][8] \leftarrow 15$, $s[4][8] \leftarrow 1$
4	9	15	0	$p[4] + v[3][3] = 10 + 5 \leq v[3][9] = 15$: $v[4][9] \leftarrow 15$, $s[4][9] \leftarrow 0$
5	0	0	0	$k < w_5$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	0	0	$k < w_5$: $v[5][1] \leftarrow v[4][1] = 0$, $s[5][1] \leftarrow 0$
5	2	5	0	$k < w_5$: $v[5][2] \leftarrow v[4][2] = 5$, $s[5][2] \leftarrow 0$
5	3	5	0	$k < w_5$: $v[5][3] \leftarrow v[4][3] = 5$, $s[5][3] \leftarrow 0$
5	4	6	0	$k < w_5$: $v[5][4] \leftarrow v[4][4] = 6$, $s[5][4] \leftarrow 0$
5	5	9	0	$k < w_5$: $v[5][5] \leftarrow v[4][5] = 9$, $s[5][5] \leftarrow 0$
5	6	11	0	$k < w_5$: $v[5][6] \leftarrow v[4][6] = 11$, $s[5][6] \leftarrow 0$
5	7	14	0	$p[5] + v[4][0] = 13 + 0 \leq v[4][7] = 14$: $v[5][7] \leftarrow 14$, $s[5][7] \leftarrow 0$
5	8	15	0	$p[5] + v[4][1] = 13 + 0 \leq v[4][8] = 15$: $v[5][8] \leftarrow 15$, $s[5][8] \leftarrow 0$
5	9	18	1	$p[5] + v[4][2] = 13 + 5 > v[4][9] = 15$: $v[5][9] \leftarrow 18$, $s[5][9] \leftarrow 1$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	5	5	5	5	5	5	5	5
$i=2$	0	0	5	5	6	6	11	11	11	11
$i=3$	0	0	5	5	6	9	11	14	14	15
$i=4$	0	0	5	5	6	9	11	14	15	15
$i=5$	0	0	5	5	6	9	11	14	15	18

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	1	1	1	1
$i=3$	0	0	0	0	0	1	0	1	1	1
$i=4$	0	0	0	0	0	0	0	0	1	0
$i=5$	0	0	0	0	0	0	0	0	0	1

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	1	1	1	1
$i=3$	0	0	0	0	0	1	0	1	1	1
$i=4$	0	0	0	0	0	0	0	0	1	0
$i=5$	0	0	0	0	0	0	0	0	0	1

We can reconstruct an optimal solution by starting from $i=5$ and $k=9$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	-	0	0	0	0	0	$k=9$
1	5	9	1	0	0	0	0	1	$x[5] \leftarrow 1$, $k \leftarrow 9-7=2$
2	4	2	0	0	0	0	0	1	no update
3	3	2	0	0	0	0	0	1	no update
4	2	2	0	0	0	0	0	1	no update
5	1	2	1	1	0	0	0	1	$x[1] \leftarrow 1$, $k \leftarrow 2-2=0$

Thus, the vector

$$(1,0,0,0,1)$$

is an optimal solution. The optimal value is

$$v[5][9]=18.$$

Exercise 17.5

Consider the following KP instance, where the knapsack has capacity $c = 10$, and the profits and weights associated with the items are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	2	7	9	12	14
w_i	1	4	5	7	8

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of k.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
0	10	0	0	$v[0][10] \leftarrow 0$, $s[0][10] \leftarrow 0$

1	0	0	0	$k < w_1$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	2	1	$p[1] + v[0][0] = 2 + 0 > v[0][1] = 0$: $v[1][1] \leftarrow 2$, $s[1][1] \leftarrow 1$
1	2	2	1	$p[1] + v[0][1] = 2 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 2$, $s[1][2] \leftarrow 1$
1	3	2	1	$p[1] + v[0][2] = 2 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 2$, $s[1][3] \leftarrow 1$
1	4	2	1	$p[1] + v[0][3] = 2 + 0 > v[0][4] = 0$: $v[1][4] \leftarrow 2$, $s[1][4] \leftarrow 1$
1	5	2	1	$p[1] + v[0][4] = 2 + 0 > v[0][5] = 0$: $v[1][5] \leftarrow 2$, $s[1][5] \leftarrow 1$
1	6	2	1	$p[1] + v[0][5] = 2 + 0 > v[0][6] = 0$: $v[1][6] \leftarrow 2$, $s[1][6] \leftarrow 1$
1	7	2	1	$p[1] + v[0][6] = 2 + 0 > v[0][7] = 0$: $v[1][7] \leftarrow 2$, $s[1][7] \leftarrow 1$
1	8	2	1	$p[1] + v[0][7] = 2 + 0 > v[0][8] = 0$: $v[1][8] \leftarrow 2$, $s[1][8] \leftarrow 1$
1	9	2	1	$p[1] + v[0][8] = 2 + 0 > v[0][9] = 0$: $v[1][9] \leftarrow 2$, $s[1][9] \leftarrow 1$
1	10	2	1	$p[1] + v[0][9] = 2 + 0 > v[0][10] = 0$: $v[1][10] \leftarrow 2$, $s[1][10] \leftarrow 1$
2	0	0	0	$k < w_2$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	2	0	$k < w_2$: $v[2][1] \leftarrow v[1][1] = 2$, $s[2][1] \leftarrow 0$
2	2	2	0	$k < w_2$: $v[2][2] \leftarrow v[1][2] = 2$, $s[2][2] \leftarrow 0$
2	3	2	0	$k < w_2$: $v[2][3] \leftarrow v[1][3] = 2$, $s[2][3] \leftarrow 0$
2	4	7	1	$p[2] + v[1][0] = 7 + 0 > v[1][4] = 2$: $v[2][4] \leftarrow 7$, $s[2][4] \leftarrow 1$
2	5	9	1	$p[2] + v[1][1] = 7 + 2 > v[1][5] = 2$: $v[2][5] \leftarrow 9$, $s[2][5] \leftarrow 1$
2	6	9	1	$p[2] + v[1][2] = 7 + 2 > v[1][6] = 2$: $v[2][6] \leftarrow 9$, $s[2][6] \leftarrow 1$
2	7	9	1	$p[2] + v[1][3] = 7 + 2 > v[1][7] = 2$: $v[2][7] \leftarrow 9$, $s[2][7] \leftarrow 1$
2	8	9	1	$p[2] + v[1][4] = 7 + 2 > v[1][8] = 2$: $v[2][8] \leftarrow 9$, $s[2][8] \leftarrow 1$
2	9	9	1	$p[2] + v[1][5] = 7 + 2 > v[1][9] = 2$: $v[2][9] \leftarrow 9$, $s[2][9] \leftarrow 1$
2	10	9	1	$p[2] + v[1][6] = 7 + 2 > v[1][10] = 2$: $v[2][10] \leftarrow 9$, $s[2][10] \leftarrow 1$
3	0	0	0	$k < w_3$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	2	0	$k < w_3$: $v[3][1] \leftarrow v[2][1] = 2$, $s[3][1] \leftarrow 0$
3	2	2	0	$k < w_3$: $v[3][2] \leftarrow v[2][2] = 2$, $s[3][2] \leftarrow 0$
3	3	2	0	$k < w_3$: $v[3][3] \leftarrow v[2][3] = 2$, $s[3][3] \leftarrow 0$
3	4	7	0	$k < w_3$: $v[3][4] \leftarrow v[2][4] = 7$, $s[3][4] \leftarrow 0$
3	5	9	0	$p[3] + v[2][0] = 9 + 0 \leq v[2][5] = 9$: $v[3][5] \leftarrow 9$, $s[3][5] \leftarrow 0$
3	6	11	1	$p[3] + v[2][1] = 9 + 2 > v[2][6] = 9$: $v[3][6] \leftarrow 11$, $s[3][6] \leftarrow 1$
3	7	11	1	$p[3] + v[2][2] = 9 + 2 > v[2][7] = 9$: $v[3][7] \leftarrow 11$, $s[3][7] \leftarrow 1$
3	8	11	1	$p[3] + v[2][3] = 9 + 2 > v[2][8] = 9$: $v[3][8] \leftarrow 11$, $s[3][8] \leftarrow 1$
3	9	16	1	$p[3] + v[2][4] = 9 + 7 > v[2][9] = 9$: $v[3][9] \leftarrow 16$, $s[3][9] \leftarrow 1$
3	10	18	1	$p[3] + v[2][5] = 9 + 9 > v[2][10] = 9$: $v[3][10] \leftarrow 18$, $s[3][10] \leftarrow 1$
4	0	0	0	$k < w_4$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	2	0	$k < w_4$: $v[4][1] \leftarrow v[3][1] = 2$, $s[4][1] \leftarrow 0$
4	2	2	0	$k < w_4$: $v[4][2] \leftarrow v[3][2] = 2$, $s[4][2] \leftarrow 0$
4	3	2	0	$k < w_4$: $v[4][3] \leftarrow v[3][3] = 2$, $s[4][3] \leftarrow 0$
4	4	7	0	$k < w_4$: $v[4][4] \leftarrow v[3][4] = 7$, $s[4][4] \leftarrow 0$
4	5	9	0	$k < w_4$: $v[4][5] \leftarrow v[3][5] = 9$, $s[4][5] \leftarrow 0$
4	6	11	0	$k < w_4$: $v[4][6] \leftarrow v[3][6] = 11$, $s[4][6] \leftarrow 0$
4	7	12	1	$p[4] + v[3][0] = 12 + 0 > v[3][7] = 11$: $v[4][7] \leftarrow 12$, $s[4][7] \leftarrow 1$
4	8	14	1	$p[4] + v[3][1] = 12 + 2 > v[3][8] = 11$: $v[4][8] \leftarrow 14$, $s[4][8] \leftarrow 1$
4	9	16	0	$p[4] + v[3][2] = 12 + 2 \leq v[3][9] = 16$: $v[4][9] \leftarrow 16$, $s[4][9] \leftarrow 0$
4	10	18	0	$p[4] + v[3][3] = 12 + 2 \leq v[3][10] = 18$: $v[4][10] \leftarrow 18$, $s[4][10] \leftarrow 0$
5	0	0	0	$k < w_5$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	2	0	$k < w_5$: $v[5][1] \leftarrow v[4][1] = 2$, $s[5][1] \leftarrow 0$
5	2	2	0	$k < w_5$: $v[5][2] \leftarrow v[4][2] = 2$, $s[5][2] \leftarrow 0$
5	3	2	0	$k < w_5$: $v[5][3] \leftarrow v[4][3] = 2$, $s[5][3] \leftarrow 0$
5	4	7	0	$k < w_5$: $v[5][4] \leftarrow v[4][4] = 7$, $s[5][4] \leftarrow 0$

5	5	9	0	$k < w_5$: $v[5][5] \leftarrow v[4][5] = 9$, $s[5][5] \leftarrow 0$
5	6	11	0	$k < w_5$: $v[5][6] \leftarrow v[4][6] = 11$, $s[5][6] \leftarrow 0$
5	7	12	0	$k < w_5$: $v[5][7] \leftarrow v[4][7] = 12$, $s[5][7] \leftarrow 0$
5	8	14	0	$p[5] + v[4][0] = 14 + 0 \leq v[4][8] = 14$: $v[5][8] \leftarrow 14$, $s[5][8] \leftarrow 0$
5	9	16	0	$p[5] + v[4][1] = 14 + 2 \leq v[4][9] = 16$: $v[5][9] \leftarrow 16$, $s[5][9] \leftarrow 0$
5	10	18	0	$p[5] + v[4][2] = 14 + 2 \leq v[4][10] = 18$: $v[5][10] \leftarrow 18$, $s[5][10] \leftarrow 0$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	2	2	2	2	2	2	2	2	2	2
$i=2$	0	2	2	2	7	9	9	9	9	9	9
$i=3$	0	2	2	2	7	9	11	11	11	16	18
$i=4$	0	2	2	2	7	9	11	12	14	16	18
$i=5$	0	2	2	2	7	9	11	12	14	16	18

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	1	1	1	1	1
$i=3$	0	0	0	0	0	0	1	1	1	1	1
$i=4$	0	0	0	0	0	0	0	1	1	0	0
$i=5$	0	0	0	0	0	0	0	0	0	0	0

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	1	1	1	1	1
$i=3$	0	0	0	0	0	0	1	1	1	1	1
$i=4$	0	0	0	0	0	0	0	1	1	0	0
$i=5$	0	0	0	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=5$ and $k=10$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	-	0	0	0	0	0	$k=10$
1	5	10	0	0	0	0	0	0	no update
2	4	10	0	0	0	0	0	0	no update
3	3	10	1	0	0	1	0	0	$x[3] \leftarrow 1$, $k \leftarrow 10-5=5$
4	2	5	1	0	1	1	0	0	$x[2] \leftarrow 1$, $k \leftarrow 5-4=1$
5	1	1	1	1	1	1	0	0	$x[1] \leftarrow 1$, $k \leftarrow 1-1=0$

Thus, the vector

$$(1, 1, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5][10] = 18.$$

DP for the unbounded knapsack problem

Exercise 18.1

Consider the following UKP instance, where the knapsack has capacity $c = 7$, each item type is available in infinitely many copies, and the profits and weights associated with the item types are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_i	4	5	10	11
w_i	2	3	5	6

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of k and i, the cells $v[k]$ and $s[k]$, the current value of m, the current value of t, and the operation performed.

k	i	$v[k]$	$s[k]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
1	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
1	1	–	–	0	0	$w[1] = 2 > k = 1$: no update
1	2	–	–	0	0	$w[2] = 3 > k = 1$: no update
1	3	–	–	0	0	$w[3] = 5 > k = 1$: no update
1	4	–	–	0	0	$w[4] = 6 > k = 1$: no update
1	–	0	0	0	0	$v[1] \leftarrow 0$, $s[1] \leftarrow 0$
2	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	4	1	$p[1] + v[0] = 4 + 0 = 4 > m = 0$: $m \leftarrow 4$, $t \leftarrow 1$
2	2	–	–	4	1	$w[2] = 3 > k = 2$: no update
2	3	–	–	4	1	$w[3] = 5 > k = 2$: no update
2	4	–	–	4	1	$w[4] = 6 > k = 2$: no update
2	–	4	1	4	1	$v[2] \leftarrow 4$, $s[2] \leftarrow 1$
3	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	4	1	$p[1] + v[1] = 4 + 0 = 4 > m = 0$: $m \leftarrow 4$, $t \leftarrow 1$
3	2	–	–	5	2	$p[2] + v[0] = 5 + 0 = 5 > m = 4$: $m \leftarrow 5$, $t \leftarrow 2$
3	3	–	–	5	2	$w[3] = 5 > k = 3$: no update
3	4	–	–	5	2	$w[4] = 6 > k = 3$: no update
3	–	5	2	5	2	$v[3] \leftarrow 5$, $s[3] \leftarrow 2$
4	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	8	1	$p[1] + v[2] = 4 + 4 = 8 > m = 0$: $m \leftarrow 8$, $t \leftarrow 1$
4	2	–	–	8	1	$p[2] + v[1] = 5 + 0 = 5 \leq m = 8$: no update
4	3	–	–	8	1	$w[3] = 5 > k = 4$: no update
4	4	–	–	8	1	$w[4] = 6 > k = 4$: no update
4	–	8	1	8	1	$v[4] \leftarrow 8$, $s[4] \leftarrow 1$
5	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	9	1	$p[1] + v[3] = 4 + 5 = 9 > m = 0$: $m \leftarrow 9$, $t \leftarrow 1$
5	2	–	–	9	1	$p[2] + v[2] = 5 + 4 = 9 \leq m = 9$: no update
5	3	–	–	10	3	$p[3] + v[0] = 10 + 0 = 10 > m = 9$: $m \leftarrow 10$, $t \leftarrow 3$
5	4	–	–	10	3	$w[4] = 6 > k = 5$: no update
5	–	10	3	10	3	$v[5] \leftarrow 10$, $s[5] \leftarrow 3$
6	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
6	1	–	–	12	1	$p[1] + v[4] = 4 + 8 = 12 > m = 0$: $m \leftarrow 12$, $t \leftarrow 1$
6	2	–	–	12	1	$p[2] + v[3] = 5 + 5 = 10 \leq m = 12$: no update
6	3	–	–	12	1	$p[3] + v[1] = 10 + 0 = 10 \leq m = 12$: no update
6	4	–	–	12	1	$p[4] + v[0] = 11 + 0 = 11 \leq m = 12$: no update
6	–	12	1	12	1	$v[6] \leftarrow 12$, $s[6] \leftarrow 1$
7	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
7	1	–	–	14	1	$p[1] + v[5] = 4 + 10 = 14 > m = 0$: $m \leftarrow 14$, $t \leftarrow 1$
7	2	–	–	14	1	$p[2] + v[4] = 5 + 8 = 13 \leq m = 14$: no update
7	3	–	–	14	1	$p[3] + v[2] = 10 + 4 = 14 \leq m = 14$: no update
7	4	–	–	14	1	$p[4] + v[1] = 11 + 0 = 11 \leq m = 14$: no update
7	–	14	1	14	1	$v[7] \leftarrow 14$, $s[7] \leftarrow 1$

This is the vector v after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$v[k]$	0	0	4	5	8	10	12	14

This is the vector s after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$s[k]$	0	0	1	2	1	3	1	1

2. We have the following vector s:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$s[k]$	0	0	1	2	1	3	1	1

We can reconstruct an optimal solution by starting from $k=7$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	0	0	0	0	$k=7$
1	7	1	1	0	0	0	$x[1] \leftarrow 1$, $k \leftarrow 7-2=5$
2	5	3	1	0	1	0	$x[3] \leftarrow 1$, $k \leftarrow 5-5=0$

Thus, the vector

$$(1,0,1,0)$$

is an optimal solution. The optimal value is

$$v[7]=14.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. All entries in row $i=0$ are initialized to 0.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
1	0	0	0	$k < w[1]$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w[1]$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	4	1	$p[1] + v[1][0] = 4 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 4$, $s[1][2] \leftarrow 1$
1	3	4	1	$p[1] + v[1][1] = 4 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 4$, $s[1][3] \leftarrow 1$
1	4	8	1	$p[1] + v[1][2] = 4 + 4 > v[0][4] = 0$: $v[1][4] \leftarrow 8$, $s[1][4] \leftarrow 1$
1	5	8	1	$p[1] + v[1][3] = 4 + 4 > v[0][5] = 0$: $v[1][5] \leftarrow 8$, $s[1][5] \leftarrow 1$
1	6	12	1	$p[1] + v[1][4] = 4 + 8 > v[0][6] = 0$: $v[1][6] \leftarrow 12$, $s[1][6] \leftarrow 1$
1	7	12	1	$p[1] + v[1][5] = 4 + 8 > v[0][7] = 0$: $v[1][7] \leftarrow 12$, $s[1][7] \leftarrow 1$
2	0	0	0	$k < w[2]$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w[2]$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	4	0	$k < w[2]$: $v[2][2] \leftarrow v[1][2] = 4$, $s[2][2] \leftarrow 0$
2	3	5	1	$p[2] + v[2][0] = 5 + 0 > v[1][3] = 4$: $v[2][3] \leftarrow 5$, $s[2][3] \leftarrow 1$
2	4	8	0	$p[2] + v[2][1] = 5 + 0 \leq v[1][4] = 8$: $v[2][4] \leftarrow 8$, $s[2][4] \leftarrow 0$
2	5	9	1	$p[2] + v[2][2] = 5 + 4 > v[1][5] = 8$: $v[2][5] \leftarrow 9$, $s[2][5] \leftarrow 1$
2	6	12	0	$p[2] + v[2][3] = 5 + 5 \leq v[1][6] = 12$: $v[2][6] \leftarrow 12$, $s[2][6] \leftarrow 0$
2	7	13	1	$p[2] + v[2][4] = 5 + 8 > v[1][7] = 12$: $v[2][7] \leftarrow 13$, $s[2][7] \leftarrow 1$
3	0	0	0	$k < w[3]$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w[3]$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	4	0	$k < w[3]$: $v[3][2] \leftarrow v[2][2] = 4$, $s[3][2] \leftarrow 0$
3	3	5	0	$k < w[3]$: $v[3][3] \leftarrow v[2][3] = 5$, $s[3][3] \leftarrow 0$
3	4	8	0	$k < w[3]$: $v[3][4] \leftarrow v[2][4] = 8$, $s[3][4] \leftarrow 0$
3	5	10	1	$p[3] + v[3][0] = 10 + 0 > v[2][5] = 9$: $v[3][5] \leftarrow 10$, $s[3][5] \leftarrow 1$
3	6	12	0	$p[3] + v[3][1] = 10 + 0 \leq v[2][6] = 12$: $v[3][6] \leftarrow 12$, $s[3][6] \leftarrow 0$
3	7	14	1	$p[3] + v[3][2] = 10 + 4 > v[2][7] = 13$: $v[3][7] \leftarrow 14$, $s[3][7] \leftarrow 1$
4	0	0	0	$k < w[4]$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	0	0	$k < w[4]$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	4	0	$k < w[4]$: $v[4][2] \leftarrow v[3][2] = 4$, $s[4][2] \leftarrow 0$
4	3	5	0	$k < w[4]$: $v[4][3] \leftarrow v[3][3] = 5$, $s[4][3] \leftarrow 0$
4	4	8	0	$k < w[4]$: $v[4][4] \leftarrow v[3][4] = 8$, $s[4][4] \leftarrow 0$
4	5	10	0	$k < w[4]$: $v[4][5] \leftarrow v[3][5] = 10$, $s[4][5] \leftarrow 0$
4	6	12	0	$p[4] + v[4][0] = 11 + 0 \leq v[3][6] = 12$: $v[4][6] \leftarrow 12$, $s[4][6] \leftarrow 0$
4	7	14	0	$p[4] + v[4][1] = 11 + 0 \leq v[3][7] = 14$: $v[4][7] \leftarrow 14$, $s[4][7] \leftarrow 0$

This is the matrix v after the execution of the second DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	4	4	8	8	12	12
$i=2$	0	0	4	5	8	9	12	13
$i=3$	0	0	4	5	8	10	12	14
$i=4$	0	0	4	5	8	10	12	14

This is the matrix s after the execution of the second DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1
$i=2$	0	0	0	1	0	1	0	1
$i=3$	0	0	0	0	0	1	0	1
$i=4$	0	0	0	0	0	0	0	0

4. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1
$i=2$	0	0	0	1	0	1	0	1
$i=3$	0	0	0	0	0	1	0	1
$i=4$	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=4$ and $k=7$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	-	0	0	0	0	$i=4,\ k=7$
1	4	7	0	0	0	0	0	$i \leftarrow 3$
2	3	7	1	0	0	1	0	$x[3] \leftarrow 1,\ k \leftarrow 7-5=2$
3	3	2	0	0	0	1	0	$i \leftarrow 2$
4	2	2	0	0	0	1	0	$i \leftarrow 1$
5	1	2	1	1	0	1	0	$x[1] \leftarrow 1,\ k \leftarrow 2-2=0$

Thus, the vector

$$(1,0,1,0)$$

is an optimal solution. The optimal value is

$$v[4][7] = 14.$$

Exercise 18.2

Consider the following UKP instance, where the knapsack has capacity $c = 8$, each item type is available in infinitely many copies, and the profits and weights associated with the item types are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	2	5	7	8	10
w_i	1	2	3	4	5

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of k and i, the cells $v[k]$ and $s[k]$, the current value of m, the current value of t, and the operation performed.

k	i	$v[k]$	$s[k]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
1	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
1	1	–	–	2	1	$p[1] + v[0] = 2 + 0 = 2 > m = 0$: $m \leftarrow 2$, $t \leftarrow 1$

1	2	–	–	2	1	$w[2] = 2 > k = 1$: no update
1	3	–	–	2	1	$w[3] = 3 > k = 1$: no update
1	4	–	–	2	1	$w[4] = 4 > k = 1$: no update
1	5	–	–	2	1	$w[5] = 5 > k = 1$: no update
1	–	2	1	2	1	$v[1] \leftarrow 2$, $s[1] \leftarrow 1$
2	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	4	1	$p[1] + v[1] = 2 + 2 = 4 > m = 0$: $m \leftarrow 4$, $t \leftarrow 1$
2	2	–	–	5	2	$p[2] + v[0] = 5 + 0 = 5 > m = 4$: $m \leftarrow 5$, $t \leftarrow 2$
2	3	–	–	5	2	$w[3] = 3 > k = 2$: no update
2	4	–	–	5	2	$w[4] = 4 > k = 2$: no update
2	5	–	–	5	2	$w[5] = 5 > k = 2$: no update
2	–	5	2	5	2	$v[2] \leftarrow 5$, $s[2] \leftarrow 2$
3	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	7	1	$p[1] + v[2] = 2 + 5 = 7 > m = 0$: $m \leftarrow 7$, $t \leftarrow 1$
3	2	–	–	7	1	$p[2] + v[1] = 5 + 2 = 7 \leq m = 7$: no update
3	3	–	–	7	1	$p[3] + v[0] = 7 + 0 = 7 \leq m = 7$: no update
3	4	–	–	7	1	$w[4] = 4 > k = 3$: no update
3	5	–	–	7	1	$w[5] = 5 > k = 3$: no update
3	–	7	1	7	1	$v[3] \leftarrow 7$, $s[3] \leftarrow 1$
4	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	9	1	$p[1] + v[3] = 2 + 7 = 9 > m = 0$: $m \leftarrow 9$, $t \leftarrow 1$
4	2	–	–	10	2	$p[2] + v[2] = 5 + 5 = 10 > m = 9$: $m \leftarrow 10$, $t \leftarrow 2$
4	3	–	–	10	2	$p[3] + v[1] = 7 + 2 = 9 \leq m = 10$: no update
4	4	–	–	10	2	$p[4] + v[0] = 8 + 0 = 8 \leq m = 10$: no update
4	5	–	–	10	2	$w[5] = 5 > k = 4$: no update
4	–	10	2	10	2	$v[4] \leftarrow 10$, $s[4] \leftarrow 2$
5	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	12	1	$p[1] + v[4] = 2 + 10 = 12 > m = 0$: $m \leftarrow 12$, $t \leftarrow 1$
5	2	–	–	12	1	$p[2] + v[3] = 5 + 7 = 12 \leq m = 12$: no update
5	3	–	–	12	1	$p[3] + v[2] = 7 + 5 = 12 \leq m = 12$: no update
5	4	–	–	12	1	$p[4] + v[1] = 8 + 2 = 10 \leq m = 12$: no update
5	5	–	–	12	1	$p[5] + v[0] = 10 + 0 = 10 \leq m = 12$: no update
5	–	12	1	12	1	$v[5] \leftarrow 12$, $s[5] \leftarrow 1$
6	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
6	1	–	–	14	1	$p[1] + v[5] = 2 + 12 = 14 > m = 0$: $m \leftarrow 14$, $t \leftarrow 1$
6	2	–	–	15	2	$p[2] + v[4] = 5 + 10 = 15 > m = 14$: $m \leftarrow 15$, $t \leftarrow 2$
6	3	–	–	15	2	$p[3] + v[3] = 7 + 7 = 14 \leq m = 15$: no update
6	4	–	–	15	2	$p[4] + v[2] = 8 + 5 = 13 \leq m = 15$: no update
6	5	–	–	15	2	$p[5] + v[1] = 10 + 2 = 12 \leq m = 15$: no update
6	–	15	2	15	2	$v[6] \leftarrow 15$, $s[6] \leftarrow 2$
7	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
7	1	–	–	17	1	$p[1] + v[6] = 2 + 15 = 17 > m = 0$: $m \leftarrow 17$, $t \leftarrow 1$
7	2	–	–	17	1	$p[2] + v[5] = 5 + 12 = 17 \leq m = 17$: no update
7	3	–	–	17	1	$p[3] + v[4] = 7 + 10 = 17 \leq m = 17$: no update
7	4	–	–	17	1	$p[4] + v[3] = 8 + 7 = 15 \leq m = 17$: no update
7	5	–	–	17	1	$p[5] + v[2] = 10 + 5 = 15 \leq m = 17$: no update
7	–	17	1	17	1	$v[7] \leftarrow 17$, $s[7] \leftarrow 1$

8	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
8	1	–	–	19	1	$p[1] + v[7] = 2 + 17 = 19 > m = 0$: $m \leftarrow 19$, $t \leftarrow 1$
8	2	–	–	20	2	$p[2] + v[6] = 5 + 15 = 20 > m = 19$: $m \leftarrow 20$, $t \leftarrow 2$
8	3	–	–	20	2	$p[3] + v[5] = 7 + 12 = 19 \leq m = 20$: no update
8	4	–	–	20	2	$p[4] + v[4] = 8 + 10 = 18 \leq m = 20$: no update
8	5	–	–	20	2	$p[5] + v[3] = 10 + 7 = 17 \leq m = 20$: no update
8	–	20	2	20	2	$v[8] \leftarrow 20$, $s[8] \leftarrow 2$

This is the vector v after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$v[k]$	0	2	5	7	10	12	15	17	20

This is the vector s after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$s[k]$	0	1	2	1	2	1	2	1	2

2. We have the following vector s:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$s[k]$	0	1	2	1	2	1	2	1	2

We can reconstruct an optimal solution by starting from $k = 8$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	0	0	0	0	0	$k = 8$
1	8	2	0	1	0	0	0	$x[2] \leftarrow 1$, $k \leftarrow 8 - 2 = 6$
2	6	2	0	2	0	0	0	$x[2] \leftarrow 2$, $k \leftarrow 6 - 2 = 4$
3	4	2	0	3	0	0	0	$x[2] \leftarrow 3$, $k \leftarrow 4 - 2 = 2$
4	2	2	0	4	0	0	0	$x[2] \leftarrow 4$, $k \leftarrow 2 - 2 = 0$

Thus, the vector

$$(0, 4, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[8] = 20.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. All entries in row $i = 0$ are initialized to 0.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
1	0	0	0	$k < w[1]$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	2	1	$p[1] + v[1][0] = 2 + 0 > v[0][1] = 0$: $v[1][1] \leftarrow 2$, $s[1][1] \leftarrow 1$
1	2	4	1	$p[1] + v[1][1] = 2 + 2 > v[0][2] = 0$: $v[1][2] \leftarrow 4$, $s[1][2] \leftarrow 1$
1	3	6	1	$p[1] + v[1][2] = 2 + 4 > v[0][3] = 0$: $v[1][3] \leftarrow 6$, $s[1][3] \leftarrow 1$
1	4	8	1	$p[1] + v[1][3] = 2 + 6 > v[0][4] = 0$: $v[1][4] \leftarrow 8$, $s[1][4] \leftarrow 1$
1	5	10	1	$p[1] + v[1][4] = 2 + 8 > v[0][5] = 0$: $v[1][5] \leftarrow 10$, $s[1][5] \leftarrow 1$
1	6	12	1	$p[1] + v[1][5] = 2 + 10 > v[0][6] = 0$: $v[1][6] \leftarrow 12$, $s[1][6] \leftarrow 1$
1	7	14	1	$p[1] + v[1][6] = 2 + 12 > v[0][7] = 0$: $v[1][7] \leftarrow 14$, $s[1][7] \leftarrow 1$
1	8	16	1	$p[1] + v[1][7] = 2 + 14 > v[0][8] = 0$: $v[1][8] \leftarrow 16$, $s[1][8] \leftarrow 1$
2	0	0	0	$k < w[2]$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	2	0	$k < w[2]$: $v[2][1] \leftarrow v[1][1] = 2$, $s[2][1] \leftarrow 0$
2	2	5	1	$p[2] + v[2][0] = 5 + 0 > v[1][2] = 4$: $v[2][2] \leftarrow 5$, $s[2][2] \leftarrow 1$
2	3	7	1	$p[2] + v[2][1] = 5 + 2 > v[1][3] = 6$: $v[2][3] \leftarrow 7$, $s[2][3] \leftarrow 1$
2	4	10	1	$p[2] + v[2][2] = 5 + 5 > v[1][4] = 8$: $v[2][4] \leftarrow 10$, $s[2][4] \leftarrow 1$
2	5	12	1	$p[2] + v[2][3] = 5 + 7 > v[1][5] = 10$: $v[2][5] \leftarrow 12$, $s[2][5] \leftarrow 1$
2	6	15	1	$p[2] + v[2][4] = 5 + 10 > v[1][6] = 12$: $v[2][6] \leftarrow 15$, $s[2][6] \leftarrow 1$
2	7	17	1	$p[2] + v[2][5] = 5 + 12 > v[1][7] = 14$: $v[2][7] \leftarrow 17$, $s[2][7] \leftarrow 1$
2	8	20	1	$p[2] + v[2][6] = 5 + 15 > v[1][8] = 16$: $v[2][8] \leftarrow 20$, $s[2][8] \leftarrow 1$
3	0	0	0	$k < w[3]$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	2	0	$k < w[3]$: $v[3][1] \leftarrow v[2][1] = 2$, $s[3][1] \leftarrow 0$
3	2	5	0	$k < w[3]$: $v[3][2] \leftarrow v[2][2] = 5$, $s[3][2] \leftarrow 0$
3	3	7	0	$p[3] + v[3][0] = 7 + 0 \leq v[2][3] = 7$: $v[3][3] \leftarrow 7$, $s[3][3] \leftarrow 0$
3	4	10	0	$p[3] + v[3][1] = 7 + 2 \leq v[2][4] = 10$: $v[3][4] \leftarrow 10$, $s[3][4] \leftarrow 0$
3	5	12	0	$p[3] + v[3][2] = 7 + 5 \leq v[2][5] = 12$: $v[3][5] \leftarrow 12$, $s[3][5] \leftarrow 0$
3	6	15	0	$p[3] + v[3][3] = 7 + 7 \leq v[2][6] = 15$: $v[3][6] \leftarrow 15$, $s[3][6] \leftarrow 0$
3	7	17	0	$p[3] + v[3][4] = 7 + 10 \leq v[2][7] = 17$: $v[3][7] \leftarrow 17$, $s[3][7] \leftarrow 0$
3	8	20	0	$p[3] + v[3][5] = 7 + 12 \leq v[2][8] = 20$: $v[3][8] \leftarrow 20$, $s[3][8] \leftarrow 0$
4	0	0	0	$k < w[4]$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	2	0	$k < w[4]$: $v[4][1] \leftarrow v[3][1] = 2$, $s[4][1] \leftarrow 0$
4	2	5	0	$k < w[4]$: $v[4][2] \leftarrow v[3][2] = 5$, $s[4][2] \leftarrow 0$
4	3	7	0	$k < w[4]$: $v[4][3] \leftarrow v[3][3] = 7$, $s[4][3] \leftarrow 0$
4	4	10	0	$p[4] + v[4][0] = 8 + 0 \leq v[3][4] = 10$: $v[4][4] \leftarrow 10$, $s[4][4] \leftarrow 0$
4	5	12	0	$p[4] + v[4][1] = 8 + 2 \leq v[3][5] = 12$: $v[4][5] \leftarrow 12$, $s[4][5] \leftarrow 0$
4	6	15	0	$p[4] + v[4][2] = 8 + 5 \leq v[3][6] = 15$: $v[4][6] \leftarrow 15$, $s[4][6] \leftarrow 0$
4	7	17	0	$p[4] + v[4][3] = 8 + 7 \leq v[3][7] = 17$: $v[4][7] \leftarrow 17$, $s[4][7] \leftarrow 0$
4	8	20	0	$p[4] + v[4][4] = 8 + 10 \leq v[3][8] = 20$: $v[4][8] \leftarrow 20$, $s[4][8] \leftarrow 0$
5	0	0	0	$k < w[5]$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	2	0	$k < w[5]$: $v[5][1] \leftarrow v[4][1] = 2$, $s[5][1] \leftarrow 0$
5	2	5	0	$k < w[5]$: $v[5][2] \leftarrow v[4][2] = 5$, $s[5][2] \leftarrow 0$
5	3	7	0	$k < w[5]$: $v[5][3] \leftarrow v[4][3] = 7$, $s[5][3] \leftarrow 0$

5	4	10	0	$k < w[5]$: $v[5][4] \leftarrow v[4][4] = 10$, $s[5][4] \leftarrow 0$
5	5	12	0	$p[5] + v[5][0] = 10 + 0 \leq v[4][5] = 12$: $v[5][5] \leftarrow 12$, $s[5][5] \leftarrow 0$
5	6	15	0	$p[5] + v[5][1] = 10 + 2 \leq v[4][6] = 15$: $v[5][6] \leftarrow 15$, $s[5][6] \leftarrow 0$
5	7	17	0	$p[5] + v[5][2] = 10 + 5 \leq v[4][7] = 17$: $v[5][7] \leftarrow 17$, $s[5][7] \leftarrow 0$
5	8	20	0	$p[5] + v[5][3] = 10 + 7 \leq v[4][8] = 20$: $v[5][8] \leftarrow 20$, $s[5][8] \leftarrow 0$

This is the matrix v after the execution of the second DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	2	4	6	8	10	12	14	16
$i=2$	0	2	5	7	10	12	15	17	20
$i=3$	0	2	5	7	10	12	15	17	20
$i=4$	0	2	5	7	10	12	15	17	20
$i=5$	0	2	5	7	10	12	15	17	20

This is the matrix s after the execution of the second DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1
$i=2$	0	0	1	1	1	1	1	1	1
$i=3$	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0	0	0

4. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1	1	1
$i=2$	0	0	1	1	1	1	1	1	1
$i=3$	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=5$ and $k=8$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	-	0	0	0	0	0	$i=5, k=8$
1	5	8	0	0	0	0	0	0	$i \leftarrow 4$
2	4	8	0	0	0	0	0	0	$i \leftarrow 3$
3	3	8	0	0	0	0	0	0	$i \leftarrow 2$
4	2	8	1	0	1	0	0	0	$x[2] \leftarrow 1, k \leftarrow 8-2=6$
5	2	6	1	0	2	0	0	0	$x[2] \leftarrow 2, k \leftarrow 6-2=4$
6	2	4	1	0	3	0	0	0	$x[2] \leftarrow 3, k \leftarrow 4-2=2$
7	2	2	1	0	4	0	0	0	$x[2] \leftarrow 4, k \leftarrow 2-2=0$

Thus, the vector

$$(0, 4, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5][8] = 20.$$

Exercise 18.3

Consider the following UKP instance, where the knapsack has capacity $c = 10$, each item type is available in infinitely many copies, and the profits and weights associated with the item types are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_i	6	10	12	13
w_i	3	4	6	7

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of k and i, the cells $v[k]$

and $s[k]$, the current value of m, the current value of t, and the operation performed.

k	i	$v[k]$	$s[k]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
1	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
1	1	–	–	0	0	$w[1] = 3 > k = 1$: no update
1	2	–	–	0	0	$w[2] = 4 > k = 1$: no update
1	3	–	–	0	0	$w[3] = 6 > k = 1$: no update
1	4	–	–	0	0	$w[4] = 7 > k = 1$: no update
1	–	0	0	0	0	$v[1] \leftarrow 0$, $s[1] \leftarrow 0$
2	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	0	0	$w[1] = 3 > k = 2$: no update
2	2	–	–	0	0	$w[2] = 4 > k = 2$: no update
2	3	–	–	0	0	$w[3] = 6 > k = 2$: no update
2	4	–	–	0	0	$w[4] = 7 > k = 2$: no update
2	–	0	0	0	0	$v[2] \leftarrow 0$, $s[2] \leftarrow 0$
3	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	6	1	$p[1] + v[0] = 6 + 0 = 6 > m = 0$: $m \leftarrow 6$, $t \leftarrow 1$
3	2	–	–	6	1	$w[2] = 4 > k = 3$: no update
3	3	–	–	6	1	$w[3] = 6 > k = 3$: no update
3	4	–	–	6	1	$w[4] = 7 > k = 3$: no update
3	–	6	1	6	1	$v[3] \leftarrow 6$, $s[3] \leftarrow 1$
4	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	6	1	$p[1] + v[1] = 6 + 0 = 6 > m = 0$: $m \leftarrow 6$, $t \leftarrow 1$
4	2	–	–	10	2	$p[2] + v[0] = 10 + 0 = 10 > m = 6$: $m \leftarrow 10$, $t \leftarrow 2$
4	3	–	–	10	2	$w[3] = 6 > k = 4$: no update
4	4	–	–	10	2	$w[4] = 7 > k = 4$: no update
4	–	10	2	10	2	$v[4] \leftarrow 10$, $s[4] \leftarrow 2$
5	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	6	1	$p[1] + v[2] = 6 + 0 = 6 > m = 0$: $m \leftarrow 6$, $t \leftarrow 1$
5	2	–	–	10	2	$p[2] + v[1] = 10 + 0 = 10 > m = 6$: $m \leftarrow 10$, $t \leftarrow 2$
5	3	–	–	10	2	$w[3] = 6 > k = 5$: no update
5	4	–	–	10	2	$w[4] = 7 > k = 5$: no update
5	–	10	2	10	2	$v[5] \leftarrow 10$, $s[5] \leftarrow 2$
6	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
6	1	–	–	12	1	$p[1] + v[3] = 6 + 6 = 12 > m = 0$: $m \leftarrow 12$, $t \leftarrow 1$
6	2	–	–	12	1	$p[2] + v[2] = 10 + 0 = 10 \leq m = 12$: no update
6	3	–	–	12	1	$p[3] + v[0] = 12 + 0 = 12 \leq m = 12$: no update
6	4	–	–	12	1	$w[4] = 7 > k = 6$: no update
6	–	12	1	12	1	$v[6] \leftarrow 12$, $s[6] \leftarrow 1$
7	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
7	1	–	–	16	1	$p[1] + v[4] = 6 + 10 = 16 > m = 0$: $m \leftarrow 16$, $t \leftarrow 1$
7	2	–	–	16	1	$p[2] + v[3] = 10 + 6 = 16 \leq m = 16$: no update
7	3	–	–	16	1	$p[3] + v[1] = 12 + 0 = 12 \leq m = 16$: no update
7	4	–	–	16	1	$p[4] + v[0] = 13 + 0 = 13 \leq m = 16$: no update
7	–	16	1	16	1	$v[7] \leftarrow 16$, $s[7] \leftarrow 1$
8	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$

8	1	–	–	16	1	$p[1] + v[5] = 6 + 10 = 16 > m = 0$: $m \leftarrow 16$, $t \leftarrow 1$
8	2	–	–	20	2	$p[2] + v[4] = 10 + 10 = 20 > m = 16$: $m \leftarrow 20$, $t \leftarrow 2$
8	3	–	–	20	2	$p[3] + v[2] = 12 + 0 = 12 \leq m = 20$: no update
8	4	–	–	20	2	$p[4] + v[1] = 13 + 0 = 13 \leq m = 20$: no update
8	–	20	2	20	2	$v[8] \leftarrow 20$, $s[8] \leftarrow 2$
9	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
9	1	–	–	18	1	$p[1] + v[6] = 6 + 12 = 18 > m = 0$: $m \leftarrow 18$, $t \leftarrow 1$
9	2	–	–	20	2	$p[2] + v[5] = 10 + 10 = 20 > m = 18$: $m \leftarrow 20$, $t \leftarrow 2$
9	3	–	–	20	2	$p[3] + v[3] = 12 + 6 = 18 \leq m = 20$: no update
9	4	–	–	20	2	$p[4] + v[2] = 13 + 0 = 13 \leq m = 20$: no update
9	–	20	2	20	2	$v[9] \leftarrow 20$, $s[9] \leftarrow 2$
10	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
10	1	–	–	22	1	$p[1] + v[7] = 6 + 16 = 22 > m = 0$: $m \leftarrow 22$, $t \leftarrow 1$
10	2	–	–	22	1	$p[2] + v[6] = 10 + 12 = 22 \leq m = 22$: no update
10	3	–	–	22	1	$p[3] + v[4] = 12 + 10 = 22 \leq m = 22$: no update
10	4	–	–	22	1	$p[4] + v[3] = 13 + 6 = 19 \leq m = 22$: no update
10	–	22	1	22	1	$v[10] \leftarrow 22$, $s[10] \leftarrow 1$

This is the vector v after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$v[k]$	0	0	0	6	10	10	12	16	20	20	22

This is the vector s after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$s[k]$	0	0	0	1	2	2	1	1	2	2	1

2. We have the following vector s:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$s[k]$	0	0	0	1	2	2	1	1	2	2	1

We can reconstruct an optimal solution by starting from $k = 10$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	0	0	0	0	$k = 10$
1	10	1	1	0	0	0	$x[1] \leftarrow 1$, $k \leftarrow 10 - 3 = 7$
2	7	1	2	0	0	0	$x[1] \leftarrow 2$, $k \leftarrow 7 - 3 = 4$
3	4	2	2	1	0	0	$x[2] \leftarrow 1$, $k \leftarrow 4 - 4 = 0$

Thus, the vector

$$(2, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[10] = 22.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. All entries in row $i = 0$ are initialized to 0.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
0	10	0	0	$v[0][10] \leftarrow 0$, $s[0][10] \leftarrow 0$
1	0	0	0	$k < w[1]$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w[1]$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	0	0	$k < w[1]$: $v[1][2] \leftarrow v[0][2] = 0$, $s[1][2] \leftarrow 0$
1	3	6	1	$p[1] + v[1][0] = 6 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 6$, $s[1][3] \leftarrow 1$
1	4	6	1	$p[1] + v[1][1] = 6 + 0 > v[0][4] = 0$: $v[1][4] \leftarrow 6$, $s[1][4] \leftarrow 1$
1	5	6	1	$p[1] + v[1][2] = 6 + 0 > v[0][5] = 0$: $v[1][5] \leftarrow 6$, $s[1][5] \leftarrow 1$
1	6	12	1	$p[1] + v[1][3] = 6 + 6 > v[0][6] = 0$: $v[1][6] \leftarrow 12$, $s[1][6] \leftarrow 1$
1	7	12	1	$p[1] + v[1][4] = 6 + 6 > v[0][7] = 0$: $v[1][7] \leftarrow 12$, $s[1][7] \leftarrow 1$
1	8	12	1	$p[1] + v[1][5] = 6 + 6 > v[0][8] = 0$: $v[1][8] \leftarrow 12$, $s[1][8] \leftarrow 1$
1	9	18	1	$p[1] + v[1][6] = 6 + 12 > v[0][9] = 0$: $v[1][9] \leftarrow 18$, $s[1][9] \leftarrow 1$
1	10	18	1	$p[1] + v[1][7] = 6 + 12 > v[0][10] = 0$: $v[1][10] \leftarrow 18$, $s[1][10] \leftarrow 1$
2	0	0	0	$k < w[2]$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w[2]$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	0	0	$k < w[2]$: $v[2][2] \leftarrow v[1][2] = 0$, $s[2][2] \leftarrow 0$
2	3	6	0	$k < w[2]$: $v[2][3] \leftarrow v[1][3] = 6$, $s[2][3] \leftarrow 0$
2	4	10	1	$p[2] + v[2][0] = 10 + 0 > v[1][4] = 6$: $v[2][4] \leftarrow 10$, $s[2][4] \leftarrow 1$
2	5	10	1	$p[2] + v[2][1] = 10 + 0 > v[1][5] = 6$: $v[2][5] \leftarrow 10$, $s[2][5] \leftarrow 1$
2	6	12	0	$p[2] + v[2][2] = 10 + 0 \leq v[1][6] = 12$: $v[2][6] \leftarrow 12$, $s[2][6] \leftarrow 0$
2	7	16	1	$p[2] + v[2][3] = 10 + 6 > v[1][7] = 12$: $v[2][7] \leftarrow 16$, $s[2][7] \leftarrow 1$
2	8	20	1	$p[2] + v[2][4] = 10 + 10 > v[1][8] = 12$: $v[2][8] \leftarrow 20$, $s[2][8] \leftarrow 1$
2	9	20	1	$p[2] + v[2][5] = 10 + 10 > v[1][9] = 18$: $v[2][9] \leftarrow 20$, $s[2][9] \leftarrow 1$
2	10	22	1	$p[2] + v[2][6] = 10 + 12 > v[1][10] = 18$: $v[2][10] \leftarrow 22$, $s[2][10] \leftarrow 1$
3	0	0	0	$k < w[3]$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w[3]$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	0	0	$k < w[3]$: $v[3][2] \leftarrow v[2][2] = 0$, $s[3][2] \leftarrow 0$
3	3	6	0	$k < w[3]$: $v[3][3] \leftarrow v[2][3] = 6$, $s[3][3] \leftarrow 0$
3	4	10	0	$k < w[3]$: $v[3][4] \leftarrow v[2][4] = 10$, $s[3][4] \leftarrow 0$
3	5	10	0	$k < w[3]$: $v[3][5] \leftarrow v[2][5] = 10$, $s[3][5] \leftarrow 0$
3	6	12	0	$p[3] + v[3][0] = 12 + 0 \leq v[2][6] = 12$: $v[3][6] \leftarrow 12$, $s[3][6] \leftarrow 0$
3	7	16	0	$p[3] + v[3][1] = 12 + 0 \leq v[2][7] = 16$: $v[3][7] \leftarrow 16$, $s[3][7] \leftarrow 0$
3	8	20	0	$p[3] + v[3][2] = 12 + 0 \leq v[2][8] = 20$: $v[3][8] \leftarrow 20$, $s[3][8] \leftarrow 0$
3	9	20	0	$p[3] + v[3][3] = 12 + 6 \leq v[2][9] = 20$: $v[3][9] \leftarrow 20$, $s[3][9] \leftarrow 0$
3	10	22	0	$p[3] + v[3][4] = 12 + 10 \leq v[2][10] = 22$: $v[3][10] \leftarrow 22$, $s[3][10] \leftarrow 0$
4	0	0	0	$k < w[4]$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	0	0	$k < w[4]$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	0	0	$k < w[4]$: $v[4][2] \leftarrow v[3][2] = 0$, $s[4][2] \leftarrow 0$
4	3	6	0	$k < w[4]$: $v[4][3] \leftarrow v[3][3] = 6$, $s[4][3] \leftarrow 0$
4	4	10	0	$k < w[4]$: $v[4][4] \leftarrow v[3][4] = 10$, $s[4][4] \leftarrow 0$

4	5	10	0	$k < w[4]$: $v[4][5] \leftarrow v[3][5] = 10$, $s[4][5] \leftarrow 0$
4	6	12	0	$k < w[4]$: $v[4][6] \leftarrow v[3][6] = 12$, $s[4][6] \leftarrow 0$
4	7	16	0	$p[4] + v[4][0] = 13 + 0 \leq v[3][7] = 16$: $v[4][7] \leftarrow 16$, $s[4][7] \leftarrow 0$
4	8	20	0	$p[4] + v[4][1] = 13 + 0 \leq v[3][8] = 20$: $v[4][8] \leftarrow 20$, $s[4][8] \leftarrow 0$
4	9	20	0	$p[4] + v[4][2] = 13 + 0 \leq v[3][9] = 20$: $v[4][9] \leftarrow 20$, $s[4][9] \leftarrow 0$
4	10	22	0	$p[4] + v[4][3] = 13 + 6 \leq v[3][10] = 22$: $v[4][10] \leftarrow 22$, $s[4][10] \leftarrow 0$

This is the matrix v after the execution of the second DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	6	6	6	12	12	12	18	18
$i=2$	0	0	0	6	10	10	12	16	20	20	22
$i=3$	0	0	0	6	10	10	12	16	20	20	22
$i=4$	0	0	0	6	10	10	12	16	20	20	22

This is the matrix s after the execution of the second DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	0	1	1	1	1
$i=3$	0	0	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	0	0	0	0

4. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$i=0$	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	1	1	0	1	1	1	1
$i=3$	0	0	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=4$ and $k=10$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	-	0	0	0	0	$i=4,\ k=10$
1	4	10	0	0	0	0	0	$i \leftarrow 3$
2	3	10	0	0	0	0	0	$i \leftarrow 2$
3	2	10	1	0	1	0	0	$x[2] \leftarrow 1,\ k \leftarrow 10-4=6$
4	2	6	0	0	1	0	0	$i \leftarrow 1$
5	1	6	1	1	1	0	0	$x[1] \leftarrow 1,\ k \leftarrow 6-3=3$
6	1	3	1	2	1	0	0	$x[1] \leftarrow 2,\ k \leftarrow 3-3=0$

Thus, the vector

$$(2, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[4][10] = 22.$$

The optimal solution is not unique: $(0, 1, 1, 0)$, with total weight $4 + 6 = 10$ and total profit $10 + 12 = 22$, is also optimal. Both DP algorithms find $(2, 1, 0, 0)$, because they update the stored value only when it strictly improves.

Exercise 18.4

Consider the following UKP instance, where the knapsack has capacity $c = 9$, each item type is available in infinitely many copies, and the profits and weights associated with the item types are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$
p_i	5	6	14	18
w_i	2	3	6	7

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of k and i, the cells $v[k]$ and $s[k]$, the current value of m, the current value of t, and the operation performed.

k	i	$v[k]$	$s[k]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
1	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
1	1	–	–	0	0	$w[1] = 2 > k = 1$: no update
1	2	–	–	0	0	$w[2] = 3 > k = 1$: no update
1	3	–	–	0	0	$w[3] = 6 > k = 1$: no update
1	4	–	–	0	0	$w[4] = 7 > k = 1$: no update
1	–	0	0	0	0	$v[1] \leftarrow 0$, $s[1] \leftarrow 0$
2	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	5	1	$p[1] + v[0] = 5 + 0 = 5 > m = 0$: $m \leftarrow 5$, $t \leftarrow 1$
2	2	–	–	5	1	$w[2] = 3 > k = 2$: no update
2	3	–	–	5	1	$w[3] = 6 > k = 2$: no update
2	4	–	–	5	1	$w[4] = 7 > k = 2$: no update
2	–	5	1	5	1	$v[2] \leftarrow 5$, $s[2] \leftarrow 1$
3	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	5	1	$p[1] + v[1] = 5 + 0 = 5 > m = 0$: $m \leftarrow 5$, $t \leftarrow 1$
3	2	–	–	6	2	$p[2] + v[0] = 6 + 0 = 6 > m = 5$: $m \leftarrow 6$, $t \leftarrow 2$
3	3	–	–	6	2	$w[3] = 6 > k = 3$: no update
3	4	–	–	6	2	$w[4] = 7 > k = 3$: no update
3	–	6	2	6	2	$v[3] \leftarrow 6$, $s[3] \leftarrow 2$
4	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	10	1	$p[1] + v[2] = 5 + 5 = 10 > m = 0$: $m \leftarrow 10$, $t \leftarrow 1$
4	2	–	–	10	1	$p[2] + v[1] = 6 + 0 = 6 \leq m = 10$: no update
4	3	–	–	10	1	$w[3] = 6 > k = 4$: no update
4	4	–	–	10	1	$w[4] = 7 > k = 4$: no update
4	–	10	1	10	1	$v[4] \leftarrow 10$, $s[4] \leftarrow 1$
5	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	11	1	$p[1] + v[3] = 5 + 6 = 11 > m = 0$: $m \leftarrow 11$, $t \leftarrow 1$
5	2	–	–	11	1	$p[2] + v[2] = 6 + 5 = 11 \leq m = 11$: no update
5	3	–	–	11	1	$w[3] = 6 > k = 5$: no update
5	4	–	–	11	1	$w[4] = 7 > k = 5$: no update
5	–	11	1	11	1	$v[5] \leftarrow 11$, $s[5] \leftarrow 1$
6	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
6	1	–	–	15	1	$p[1] + v[4] = 5 + 10 = 15 > m = 0$: $m \leftarrow 15$, $t \leftarrow 1$
6	2	–	–	15	1	$p[2] + v[3] = 6 + 6 = 12 \leq m = 15$: no update
6	3	–	–	15	1	$p[3] + v[0] = 14 + 0 = 14 \leq m = 15$: no update
6	4	–	–	15	1	$w[4] = 7 > k = 6$: no update
6	–	15	1	15	1	$v[6] \leftarrow 15$, $s[6] \leftarrow 1$
7	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
7	1	–	–	16	1	$p[1] + v[5] = 5 + 11 = 16 > m = 0$: $m \leftarrow 16$, $t \leftarrow 1$
7	2	–	–	16	1	$p[2] + v[4] = 6 + 10 = 16 \leq m = 16$: no update
7	3	–	–	16	1	$p[3] + v[1] = 14 + 0 = 14 \leq m = 16$: no update
7	4	–	–	18	4	$p[4] + v[0] = 18 + 0 = 18 > m = 16$: $m \leftarrow 18$, $t \leftarrow 4$
7	–	18	4	18	4	$v[7] \leftarrow 18$, $s[7] \leftarrow 4$
8	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
8	1	–	–	20	1	$p[1] + v[6] = 5 + 15 = 20 > m = 0$: $m \leftarrow 20$, $t \leftarrow 1$

8	2	–	–	20	1	$p[2] + v[5] = 6 + 11 = 17 \leq m = 20$: no update
8	3	–	–	20	1	$p[3] + v[2] = 14 + 5 = 19 \leq m = 20$: no update
8	4	–	–	20	1	$p[4] + v[1] = 18 + 0 = 18 \leq m = 20$: no update
8	–	20	1	20	1	$v[8] \leftarrow 20$, $s[8] \leftarrow 1$
9	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
9	1	–	–	23	1	$p[1] + v[7] = 5 + 18 = 23 > m = 0$: $m \leftarrow 23$, $t \leftarrow 1$
9	2	–	–	23	1	$p[2] + v[6] = 6 + 15 = 21 \leq m = 23$: no update
9	3	–	–	23	1	$p[3] + v[3] = 14 + 6 = 20 \leq m = 23$: no update
9	4	–	–	23	1	$p[4] + v[2] = 18 + 5 = 23 \leq m = 23$: no update
9	–	23	1	23	1	$v[9] \leftarrow 23$, $s[9] \leftarrow 1$

This is the vector v after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$v[k]$	0	0	5	6	10	11	15	18	20	23

This is the vector s after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$s[k]$	0	0	1	2	1	1	1	4	1	1

2. We have the following vector s:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$s[k]$	0	0	1	2	1	1	1	4	1	1

We can reconstruct an optimal solution by starting from $k = 9$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	0	0	0	0	$k = 9$
1	9	1	1	0	0	0	$x[1] \leftarrow 1$, $k \leftarrow 9 - 2 = 7$
2	7	4	1	0	0	1	$x[4] \leftarrow 1$, $k \leftarrow 7 - 7 = 0$

Thus, the vector

$$(1, 0, 0, 1)$$

is an optimal solution. The optimal value is

$$v[9] = 23.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. All entries in row $i = 0$ are initialized to 0.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
1	0	0	0	$k < w[1]$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w[1]$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	5	1	$p[1] + v[1][0] = 5 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 5$, $s[1][2] \leftarrow 1$
1	3	5	1	$p[1] + v[1][1] = 5 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 5$, $s[1][3] \leftarrow 1$
1	4	10	1	$p[1] + v[1][2] = 5 + 5 > v[0][4] = 0$: $v[1][4] \leftarrow 10$, $s[1][4] \leftarrow 1$
1	5	10	1	$p[1] + v[1][3] = 5 + 5 > v[0][5] = 0$: $v[1][5] \leftarrow 10$, $s[1][5] \leftarrow 1$
1	6	15	1	$p[1] + v[1][4] = 5 + 10 > v[0][6] = 0$: $v[1][6] \leftarrow 15$, $s[1][6] \leftarrow 1$
1	7	15	1	$p[1] + v[1][5] = 5 + 10 > v[0][7] = 0$: $v[1][7] \leftarrow 15$, $s[1][7] \leftarrow 1$
1	8	20	1	$p[1] + v[1][6] = 5 + 15 > v[0][8] = 0$: $v[1][8] \leftarrow 20$, $s[1][8] \leftarrow 1$
1	9	20	1	$p[1] + v[1][7] = 5 + 15 > v[0][9] = 0$: $v[1][9] \leftarrow 20$, $s[1][9] \leftarrow 1$
2	0	0	0	$k < w[2]$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w[2]$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	5	0	$k < w[2]$: $v[2][2] \leftarrow v[1][2] = 5$, $s[2][2] \leftarrow 0$
2	3	6	1	$p[2] + v[2][0] = 6 + 0 > v[1][3] = 5$: $v[2][3] \leftarrow 6$, $s[2][3] \leftarrow 1$
2	4	10	0	$p[2] + v[2][1] = 6 + 0 \leq v[1][4] = 10$: $v[2][4] \leftarrow 10$, $s[2][4] \leftarrow 0$
2	5	11	1	$p[2] + v[2][2] = 6 + 5 > v[1][5] = 10$: $v[2][5] \leftarrow 11$, $s[2][5] \leftarrow 1$
2	6	15	0	$p[2] + v[2][3] = 6 + 6 \leq v[1][6] = 15$: $v[2][6] \leftarrow 15$, $s[2][6] \leftarrow 0$
2	7	16	1	$p[2] + v[2][4] = 6 + 10 > v[1][7] = 15$: $v[2][7] \leftarrow 16$, $s[2][7] \leftarrow 1$
2	8	20	0	$p[2] + v[2][5] = 6 + 11 \leq v[1][8] = 20$: $v[2][8] \leftarrow 20$, $s[2][8] \leftarrow 0$
2	9	21	1	$p[2] + v[2][6] = 6 + 15 > v[1][9] = 20$: $v[2][9] \leftarrow 21$, $s[2][9] \leftarrow 1$
3	0	0	0	$k < w[3]$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w[3]$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	5	0	$k < w[3]$: $v[3][2] \leftarrow v[2][2] = 5$, $s[3][2] \leftarrow 0$
3	3	6	0	$k < w[3]$: $v[3][3] \leftarrow v[2][3] = 6$, $s[3][3] \leftarrow 0$
3	4	10	0	$k < w[3]$: $v[3][4] \leftarrow v[2][4] = 10$, $s[3][4] \leftarrow 0$
3	5	11	0	$k < w[3]$: $v[3][5] \leftarrow v[2][5] = 11$, $s[3][5] \leftarrow 0$
3	6	15	0	$p[3] + v[3][0] = 14 + 0 \leq v[2][6] = 15$: $v[3][6] \leftarrow 15$, $s[3][6] \leftarrow 0$
3	7	16	0	$p[3] + v[3][1] = 14 + 0 \leq v[2][7] = 16$: $v[3][7] \leftarrow 16$, $s[3][7] \leftarrow 0$
3	8	20	0	$p[3] + v[3][2] = 14 + 5 \leq v[2][8] = 20$: $v[3][8] \leftarrow 20$, $s[3][8] \leftarrow 0$
3	9	21	0	$p[3] + v[3][3] = 14 + 6 \leq v[2][9] = 21$: $v[3][9] \leftarrow 21$, $s[3][9] \leftarrow 0$
4	0	0	0	$k < w[4]$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	0	0	$k < w[4]$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	5	0	$k < w[4]$: $v[4][2] \leftarrow v[3][2] = 5$, $s[4][2] \leftarrow 0$
4	3	6	0	$k < w[4]$: $v[4][3] \leftarrow v[3][3] = 6$, $s[4][3] \leftarrow 0$
4	4	10	0	$k < w[4]$: $v[4][4] \leftarrow v[3][4] = 10$, $s[4][4] \leftarrow 0$
4	5	11	0	$k < w[4]$: $v[4][5] \leftarrow v[3][5] = 11$, $s[4][5] \leftarrow 0$
4	6	15	0	$k < w[4]$: $v[4][6] \leftarrow v[3][6] = 15$, $s[4][6] \leftarrow 0$
4	7	18	1	$p[4] + v[4][0] = 18 + 0 > v[3][7] = 16$: $v[4][7] \leftarrow 18$, $s[4][7] \leftarrow 1$
4	8	20	0	$p[4] + v[4][1] = 18 + 0 \leq v[3][8] = 20$: $v[4][8] \leftarrow 20$, $s[4][8] \leftarrow 0$

4	9	23	1	$p[4] + v[4][2] = 18 + 5 > v[3][9] = 21$: $v[4][9] \leftarrow 23$, $s[4][9] \leftarrow 1$

This is the matrix v after the execution of the second DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	5	5	10	10	15	15	20	20
$i=2$	0	0	5	6	10	11	15	16	20	21
$i=3$	0	0	5	6	10	11	15	16	20	21
$i=4$	0	0	5	6	10	11	15	18	20	23

This is the matrix s after the execution of the second DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	1	0	1	0	1	0	1
$i=3$	0	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	1	0	1

4. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1
$i=2$	0	0	0	1	0	1	0	1	0	1
$i=3$	0	0	0	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0	1	0	1

We can reconstruct an optimal solution by starting from $i=4$ and $k=9$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	Action
-	-	-	-	0	0	0	0	$i=4$, $k=9$
1	4	9	1	0	0	0	1	$x[4] \leftarrow 1$, $k \leftarrow 9-7=2$
2	4	2	0	0	0	0	1	$i \leftarrow 3$
3	3	2	0	0	0	0	1	$i \leftarrow 2$
4	2	2	0	0	0	0	1	$i \leftarrow 1$
5	1	2	1	1	0	0	1	$x[1] \leftarrow 1$, $k \leftarrow 2-2=0$

Thus, the vector

$$(1,0,0,1)$$

is an optimal solution. The optimal value is

$$v[4][9] = 23.$$

Exercise 18.5

Consider the following UKP instance, where the knapsack has capacity $c = 11$, each item type is available in infinitely many copies, and the profits and weights associated with the item types are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	3	9	10	16	17
w_i	2	5	6	9	10

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of k and i, the cells $v[k]$ and $s[k]$, the current value of m, the current value of t, and the operation performed.

k	i	$v[k]$	$s[k]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
1	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
1	1	–	–	0	0	$w[1] = 2 > k = 1$: no update

1	2	–	–	0	0	$w[2] = 5 > k = 1$: no update
1	3	–	–	0	0	$w[3] = 6 > k = 1$: no update
1	4	–	–	0	0	$w[4] = 9 > k = 1$: no update
1	5	–	–	0	0	$w[5] = 10 > k = 1$: no update
1	–	0	0	0	0	$v[1] \leftarrow 0$, $s[1] \leftarrow 0$
2	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	3	1	$p[1] + v[0] = 3 + 0 = 3 > m = 0$: $m \leftarrow 3$, $t \leftarrow 1$
2	2	–	–	3	1	$w[2] = 5 > k = 2$: no update
2	3	–	–	3	1	$w[3] = 6 > k = 2$: no update
2	4	–	–	3	1	$w[4] = 9 > k = 2$: no update
2	5	–	–	3	1	$w[5] = 10 > k = 2$: no update
2	–	3	1	3	1	$v[2] \leftarrow 3$, $s[2] \leftarrow 1$
3	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	3	1	$p[1] + v[1] = 3 + 0 = 3 > m = 0$: $m \leftarrow 3$, $t \leftarrow 1$
3	2	–	–	3	1	$w[2] = 5 > k = 3$: no update
3	3	–	–	3	1	$w[3] = 6 > k = 3$: no update
3	4	–	–	3	1	$w[4] = 9 > k = 3$: no update
3	5	–	–	3	1	$w[5] = 10 > k = 3$: no update
3	–	3	1	3	1	$v[3] \leftarrow 3$, $s[3] \leftarrow 1$
4	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	6	1	$p[1] + v[2] = 3 + 3 = 6 > m = 0$: $m \leftarrow 6$, $t \leftarrow 1$
4	2	–	–	6	1	$w[2] = 5 > k = 4$: no update
4	3	–	–	6	1	$w[3] = 6 > k = 4$: no update
4	4	–	–	6	1	$w[4] = 9 > k = 4$: no update
4	5	–	–	6	1	$w[5] = 10 > k = 4$: no update
4	–	6	1	6	1	$v[4] \leftarrow 6$, $s[4] \leftarrow 1$
5	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	6	1	$p[1] + v[3] = 3 + 3 = 6 > m = 0$: $m \leftarrow 6$, $t \leftarrow 1$
5	2	–	–	9	2	$p[2] + v[0] = 9 + 0 = 9 > m = 6$: $m \leftarrow 9$, $t \leftarrow 2$
5	3	–	–	9	2	$w[3] = 6 > k = 5$: no update
5	4	–	–	9	2	$w[4] = 9 > k = 5$: no update
5	5	–	–	9	2	$w[5] = 10 > k = 5$: no update
5	–	9	2	9	2	$v[5] \leftarrow 9$, $s[5] \leftarrow 2$
6	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
6	1	–	–	9	1	$p[1] + v[4] = 3 + 6 = 9 > m = 0$: $m \leftarrow 9$, $t \leftarrow 1$
6	2	–	–	9	1	$p[2] + v[1] = 9 + 0 = 9 \leq m = 9$: no update
6	3	–	–	10	3	$p[3] + v[0] = 10 + 0 = 10 > m = 9$: $m \leftarrow 10$, $t \leftarrow 3$
6	4	–	–	10	3	$w[4] = 9 > k = 6$: no update
6	5	–	–	10	3	$w[5] = 10 > k = 6$: no update
6	–	10	3	10	3	$v[6] \leftarrow 10$, $s[6] \leftarrow 3$
7	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
7	1	–	–	12	1	$p[1] + v[5] = 3 + 9 = 12 > m = 0$: $m \leftarrow 12$, $t \leftarrow 1$
7	2	–	–	12	1	$p[2] + v[2] = 9 + 3 = 12 \leq m = 12$: no update
7	3	–	–	12	1	$p[3] + v[1] = 10 + 0 = 10 \leq m = 12$: no update
7	4	–	–	12	1	$w[4] = 9 > k = 7$: no update
7	5	–	–	12	1	$w[5] = 10 > k = 7$: no update
7	–	12	1	12	1	$v[7] \leftarrow 12$, $s[7] \leftarrow 1$
8	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
8	1	–	–	13	1	$p[1] + v[6] = 3 + 10 = 13 > m = 0$: $m \leftarrow 13$, $t \leftarrow 1$

8	2	–	–	13	1	$p[2] + v[3] = 9 + 3 = 12 \leq m = 13$: no update
8	3	–	–	13	1	$p[3] + v[2] = 10 + 3 = 13 \leq m = 13$: no update
8	4	–	–	13	1	$w[4] = 9 > k = 8$: no update
8	5	–	–	13	1	$w[5] = 10 > k = 8$: no update
8	–	13	1	13	1	$v[8] \leftarrow 13$, $s[8] \leftarrow 1$
9	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
9	1	–	–	15	1	$p[1] + v[7] = 3 + 12 = 15 > m = 0$: $m \leftarrow 15$, $t \leftarrow 1$
9	2	–	–	15	1	$p[2] + v[4] = 9 + 6 = 15 \leq m = 15$: no update
9	3	–	–	15	1	$p[3] + v[3] = 10 + 3 = 13 \leq m = 15$: no update
9	4	–	–	16	4	$p[4] + v[0] = 16 + 0 = 16 > m = 15$: $m \leftarrow 16$, $t \leftarrow 4$
9	5	–	–	16	4	$w[5] = 10 > k = 9$: no update
9	–	16	4	16	4	$v[9] \leftarrow 16$, $s[9] \leftarrow 4$
10	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
10	1	–	–	16	1	$p[1] + v[8] = 3 + 13 = 16 > m = 0$: $m \leftarrow 16$, $t \leftarrow 1$
10	2	–	–	18	2	$p[2] + v[5] = 9 + 9 = 18 > m = 16$: $m \leftarrow 18$, $t \leftarrow 2$
10	3	–	–	18	2	$p[3] + v[4] = 10 + 6 = 16 \leq m = 18$: no update
10	4	–	–	18	2	$p[4] + v[1] = 16 + 0 = 16 \leq m = 18$: no update
10	5	–	–	18	2	$p[5] + v[0] = 17 + 0 = 17 \leq m = 18$: no update
10	–	18	2	18	2	$v[10] \leftarrow 18$, $s[10] \leftarrow 2$
11	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
11	1	–	–	19	1	$p[1] + v[9] = 3 + 16 = 19 > m = 0$: $m \leftarrow 19$, $t \leftarrow 1$
11	2	–	–	19	1	$p[2] + v[6] = 9 + 10 = 19 \leq m = 19$: no update
11	3	–	–	19	1	$p[3] + v[5] = 10 + 9 = 19 \leq m = 19$: no update
11	4	–	–	19	1	$p[4] + v[2] = 16 + 3 = 19 \leq m = 19$: no update
11	5	–	–	19	1	$p[5] + v[1] = 17 + 0 = 17 \leq m = 19$: no update
11	–	19	1	19	1	$v[11] \leftarrow 19$, $s[11] \leftarrow 1$

This is the vector v after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$v[k]$	0	0	3	3	6	9	10	12	13	16	18	19

This is the vector s after the execution of the first DP algorithm:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$s[k]$	0	0	1	1	1	2	3	1	1	4	2	1

2. We have the following vector s:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$s[k]$	0	0	1	1	1	2	3	1	1	4	2	1

We can reconstruct an optimal solution by starting from $k = 11$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	0	0	0	0	0	$k = 11$
1	11	1	1	0	0	0	0	$x[1] \leftarrow 1$, $k \leftarrow 11 - 2 = 9$
2	9	4	1	0	0	1	0	$x[4] \leftarrow 1$, $k \leftarrow 9 - 9 = 0$

Thus, the vector

$$(1, 0, 0, 1, 0)$$

is an optimal solution. The optimal value is

$$v[11] = 19.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed. All entries in row $i = 0$ are initialized to 0.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
0	10	0	0	$v[0][10] \leftarrow 0$, $s[0][10] \leftarrow 0$
0	11	0	0	$v[0][11] \leftarrow 0$, $s[0][11] \leftarrow 0$
1	0	0	0	$k < w[1]$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	0	0	$k < w[1]$: $v[1][1] \leftarrow v[0][1] = 0$, $s[1][1] \leftarrow 0$
1	2	3	1	$p[1] + v[1][0] = 3 + 0 > v[0][2] = 0$: $v[1][2] \leftarrow 3$, $s[1][2] \leftarrow 1$
1	3	3	1	$p[1] + v[1][1] = 3 + 0 > v[0][3] = 0$: $v[1][3] \leftarrow 3$, $s[1][3] \leftarrow 1$
1	4	6	1	$p[1] + v[1][2] = 3 + 3 > v[0][4] = 0$: $v[1][4] \leftarrow 6$, $s[1][4] \leftarrow 1$
1	5	6	1	$p[1] + v[1][3] = 3 + 3 > v[0][5] = 0$: $v[1][5] \leftarrow 6$, $s[1][5] \leftarrow 1$
1	6	9	1	$p[1] + v[1][4] = 3 + 6 > v[0][6] = 0$: $v[1][6] \leftarrow 9$, $s[1][6] \leftarrow 1$
1	7	9	1	$p[1] + v[1][5] = 3 + 6 > v[0][7] = 0$: $v[1][7] \leftarrow 9$, $s[1][7] \leftarrow 1$
1	8	12	1	$p[1] + v[1][6] = 3 + 9 > v[0][8] = 0$: $v[1][8] \leftarrow 12$, $s[1][8] \leftarrow 1$
1	9	12	1	$p[1] + v[1][7] = 3 + 9 > v[0][9] = 0$: $v[1][9] \leftarrow 12$, $s[1][9] \leftarrow 1$
1	10	15	1	$p[1] + v[1][8] = 3 + 12 > v[0][10] = 0$: $v[1][10] \leftarrow 15$, $s[1][10] \leftarrow 1$
1	11	15	1	$p[1] + v[1][9] = 3 + 12 > v[0][11] = 0$: $v[1][11] \leftarrow 15$, $s[1][11] \leftarrow 1$
2	0	0	0	$k < w[2]$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	0	0	$k < w[2]$: $v[2][1] \leftarrow v[1][1] = 0$, $s[2][1] \leftarrow 0$
2	2	3	0	$k < w[2]$: $v[2][2] \leftarrow v[1][2] = 3$, $s[2][2] \leftarrow 0$
2	3	3	0	$k < w[2]$: $v[2][3] \leftarrow v[1][3] = 3$, $s[2][3] \leftarrow 0$
2	4	6	0	$k < w[2]$: $v[2][4] \leftarrow v[1][4] = 6$, $s[2][4] \leftarrow 0$
2	5	9	1	$p[2] + v[2][0] = 9 + 0 > v[1][5] = 6$: $v[2][5] \leftarrow 9$, $s[2][5] \leftarrow 1$
2	6	9	0	$p[2] + v[2][1] = 9 + 0 \leq v[1][6] = 9$: $v[2][6] \leftarrow 9$, $s[2][6] \leftarrow 0$
2	7	12	1	$p[2] + v[2][2] = 9 + 3 > v[1][7] = 9$: $v[2][7] \leftarrow 12$, $s[2][7] \leftarrow 1$
2	8	12	0	$p[2] + v[2][3] = 9 + 3 \leq v[1][8] = 12$: $v[2][8] \leftarrow 12$, $s[2][8] \leftarrow 0$
2	9	15	1	$p[2] + v[2][4] = 9 + 6 > v[1][9] = 12$: $v[2][9] \leftarrow 15$, $s[2][9] \leftarrow 1$
2	10	18	1	$p[2] + v[2][5] = 9 + 9 > v[1][10] = 15$: $v[2][10] \leftarrow 18$, $s[2][10] \leftarrow 1$
2	11	18	1	$p[2] + v[2][6] = 9 + 9 > v[1][11] = 15$: $v[2][11] \leftarrow 18$, $s[2][11] \leftarrow 1$
3	0	0	0	$k < w[3]$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	0	0	$k < w[3]$: $v[3][1] \leftarrow v[2][1] = 0$, $s[3][1] \leftarrow 0$
3	2	3	0	$k < w[3]$: $v[3][2] \leftarrow v[2][2] = 3$, $s[3][2] \leftarrow 0$
3	3	3	0	$k < w[3]$: $v[3][3] \leftarrow v[2][3] = 3$, $s[3][3] \leftarrow 0$
3	4	6	0	$k < w[3]$: $v[3][4] \leftarrow v[2][4] = 6$, $s[3][4] \leftarrow 0$
3	5	9	0	$k < w[3]$: $v[3][5] \leftarrow v[2][5] = 9$, $s[3][5] \leftarrow 0$
3	6	10	1	$p[3] + v[3][0] = 10 + 0 > v[2][6] = 9$: $v[3][6] \leftarrow 10$, $s[3][6] \leftarrow 1$
3	7	12	0	$p[3] + v[3][1] = 10 + 0 \leq v[2][7] = 12$: $v[3][7] \leftarrow 12$, $s[3][7] \leftarrow 0$
3	8	13	1	$p[3] + v[3][2] = 10 + 3 > v[2][8] = 12$: $v[3][8] \leftarrow 13$, $s[3][8] \leftarrow 1$
3	9	15	0	$p[3] + v[3][3] = 10 + 3 \leq v[2][9] = 15$: $v[3][9] \leftarrow 15$, $s[3][9] \leftarrow 0$
3	10	18	0	$p[3] + v[3][4] = 10 + 6 \leq v[2][10] = 18$: $v[3][10] \leftarrow 18$, $s[3][10] \leftarrow 0$
3	11	19	1	$p[3] + v[3][5] = 10 + 9 > v[2][11] = 18$: $v[3][11] \leftarrow 19$, $s[3][11] \leftarrow 1$
4	0	0	0	$k < w[4]$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$

4	1	0	0	$k < w[4]$: $v[4][1] \leftarrow v[3][1] = 0$, $s[4][1] \leftarrow 0$
4	2	3	0	$k < w[4]$: $v[4][2] \leftarrow v[3][2] = 3$, $s[4][2] \leftarrow 0$
4	3	3	0	$k < w[4]$: $v[4][3] \leftarrow v[3][3] = 3$, $s[4][3] \leftarrow 0$
4	4	6	0	$k < w[4]$: $v[4][4] \leftarrow v[3][4] = 6$, $s[4][4] \leftarrow 0$
4	5	9	0	$k < w[4]$: $v[4][5] \leftarrow v[3][5] = 9$, $s[4][5] \leftarrow 0$
4	6	10	0	$k < w[4]$: $v[4][6] \leftarrow v[3][6] = 10$, $s[4][6] \leftarrow 0$
4	7	12	0	$k < w[4]$: $v[4][7] \leftarrow v[3][7] = 12$, $s[4][7] \leftarrow 0$
4	8	13	0	$k < w[4]$: $v[4][8] \leftarrow v[3][8] = 13$, $s[4][8] \leftarrow 0$
4	9	16	1	$p[4] + v[4][0] = 16 + 0 > v[3][9] = 15$: $v[4][9] \leftarrow 16$, $s[4][9] \leftarrow 1$
4	10	18	0	$p[4] + v[4][1] = 16 + 0 \leq v[3][10] = 18$: $v[4][10] \leftarrow 18$, $s[4][10] \leftarrow 0$
4	11	19	0	$p[4] + v[4][2] = 16 + 3 \leq v[3][11] = 19$: $v[4][11] \leftarrow 19$, $s[4][11] \leftarrow 0$
5	0	0	0	$k < w[5]$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	0	0	$k < w[5]$: $v[5][1] \leftarrow v[4][1] = 0$, $s[5][1] \leftarrow 0$
5	2	3	0	$k < w[5]$: $v[5][2] \leftarrow v[4][2] = 3$, $s[5][2] \leftarrow 0$
5	3	3	0	$k < w[5]$: $v[5][3] \leftarrow v[4][3] = 3$, $s[5][3] \leftarrow 0$
5	4	6	0	$k < w[5]$: $v[5][4] \leftarrow v[4][4] = 6$, $s[5][4] \leftarrow 0$
5	5	9	0	$k < w[5]$: $v[5][5] \leftarrow v[4][5] = 9$, $s[5][5] \leftarrow 0$
5	6	10	0	$k < w[5]$: $v[5][6] \leftarrow v[4][6] = 10$, $s[5][6] \leftarrow 0$
5	7	12	0	$k < w[5]$: $v[5][7] \leftarrow v[4][7] = 12$, $s[5][7] \leftarrow 0$
5	8	13	0	$k < w[5]$: $v[5][8] \leftarrow v[4][8] = 13$, $s[5][8] \leftarrow 0$
5	9	16	0	$k < w[5]$: $v[5][9] \leftarrow v[4][9] = 16$, $s[5][9] \leftarrow 0$
5	10	18	0	$p[5] + v[5][0] = 17 + 0 \leq v[4][10] = 18$: $v[5][10] \leftarrow 18$, $s[5][10] \leftarrow 0$
5	11	19	0	$p[5] + v[5][1] = 17 + 0 \leq v[4][11] = 19$: $v[5][11] \leftarrow 19$, $s[5][11] \leftarrow 0$

This is the matrix v after the execution of the second DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$i=0$	0	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	3	3	6	6	9	9	12	12	15	15
$i=2$	0	0	3	3	6	9	9	12	12	15	18	18
$i=3$	0	0	3	3	6	9	10	12	13	15	18	19
$i=4$	0	0	3	3	6	9	10	12	13	16	18	19
$i=5$	0	0	3	3	6	9	10	12	13	16	18	19

This is the matrix s after the execution of the second DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$i=0$	0	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	0	1	0	1	0	1	1	1
$i=3$	0	0	0	0	0	0	1	0	1	0	0	1
$i=4$	0	0	0	0	0	0	0	0	0	1	0	0
$i=5$	0	0	0	0	0	0	0	0	0	0	0	0

4. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$	$k=11$
$i=0$	0	0	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	1	1	1	1	1	1	1	1
$i=2$	0	0	0	0	0	1	0	1	0	1	1	1
$i=3$	0	0	0	0	0	0	1	0	1	0	0	1
$i=4$	0	0	0	0	0	0	0	0	0	1	0	0
$i=5$	0	0	0	0	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=5$ and $k=11$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
-	-	-	-	0	0	0	0	0	$i=5, k=11$
1	5	11	0	0	0	0	0	0	$i \leftarrow 4$
2	4	11	0	0	0	0	0	0	$i \leftarrow 3$
3	3	11	1	0	0	1	0	0	$x[3] \leftarrow 1, k \leftarrow 11-6=5$
4	3	5	0	0	0	1	0	0	$i \leftarrow 2$
5	2	5	1	0	1	1	0	0	$x[2] \leftarrow 1, k \leftarrow 5-5=0$

Thus, the vector

$$(0, 1, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5][11] = 19.$$

The optimal solution is not unique, and the two DP algorithms find different ones: the first finds $(1, 0, 0, 1, 0)$, with total weight $2 + 9 = 11$ and total profit $3 + 16 = 19$; the second finds $(0, 1, 1, 0, 0)$, with total weight $5+6 = 11$ and total profit $9+10 = 19$. These are the only two optimal solutions.

Chapter 19

DP for the bounded knapsack problem

Exercise 19.1

Consider the following BKP instance, where the knapsack has capacity $c = 6$, and the profits, weights, and bounds associated with the item types are:

	$i = 1$	$i = 2$
p_i	4	5
w_i	2	3
b_i	2	1

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
1	0	0	0	$\max\{v[0][0]\} = 0$
1	1	0	0	$\max\{v[0][1]\} = 0$
1	2	4	1	$\max\{v[0][2], p[1]+v[0][0]\} = \max\{0, 4\} = 4$
1	3	4	1	$\max\{v[0][3], p[1]+v[0][1]\} = \max\{0, 4\} = 4$
1	4	8	2	$\max\{v[0][4], p[1]+v[0][2], 2p[1]+v[0][0]\} = \max\{0, 4, 8\} = 8$
1	5	8	2	$\max\{v[0][5], p[1]+v[0][3], 2p[1]+v[0][1]\} = \max\{0, 4, 8\} = 8$
1	6	8	2	$\max\{v[0][6], p[1]+v[0][4], 2p[1]+v[0][2]\} = \max\{0, 4, 8\} = 8$
2	0	0	0	$\max\{v[1][0]\} = 0$
2	1	0	0	$\max\{v[1][1]\} = 0$
2	2	4	0	$\max\{v[1][2]\} = 4$
2	3	5	1	$\max\{v[1][3], p[2]+v[1][0]\} = \max\{4, 5\} = 5$
2	4	8	0	$\max\{v[1][4], p[2]+v[1][1]\} = \max\{8, 5\} = 8$
2	5	9	1	$\max\{v[1][5], p[2]+v[1][2]\} = \max\{8, 9\} = 9$
2	6	9	1	$\max\{v[1][6], p[2]+v[1][3]\} = \max\{8, 9\} = 9$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	4	4	8	8	8
$i=2$	0	0	4	5	8	9	9

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	1	1	2	2	2
$i=2$	0	0	0	1	0	1	1

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	0	1	1	2	2	2
$i=2$	0	0	0	1	0	1	1

We can reconstruct an optimal solution by starting from $i=2$ and $k=6$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	Action
-	-	-	-	0	0	$k=6$
1	2	6	1	0	1	$x[2] \leftarrow 1$, $k \leftarrow 6-1\cdot 3=3$
2	1	3	1	1	1	$x[1] \leftarrow 1$, $k \leftarrow 3-1\cdot 2=1$

Thus, the vector

$$(1,1)$$

is an optimal solution. The optimal value is

$$v[2][6]=9.$$

Exercise 19.2

Consider the following BKP instance, where the knapsack has capacity $c = 7$, and the profits, weights, and bounds associated with the item types are:

	$i = 1$	$i = 2$
p_i	3	8
w_i	1	4
b_i	3	1

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
1	0	0	0	$\max\{v[0][0]\} = 0$
1	1	3	1	$\max\{v[0][1], p[1]+v[0][0]\} = \max\{0, 3\} = 3$
1	2	6	2	$\max\{v[0][2], p[1]+v[0][1], 2p[1]+v[0][0]\} = \max\{0, 3, 6\} = 6$

1	3	9	3	$\max\{v[0][3],\ p[1]+v[0][2],\ 2p[1]+v[0][1],\ 3p[1]+v[0][0]\} = \max\{0,3,6,9\} = 9$
1	4	9	3	$\max\{v[0][4],\ p[1]+v[0][3],\ 2p[1]+v[0][2],\ 3p[1]+v[0][1]\} = \max\{0,3,6,9\} = 9$
1	5	9	3	$\max\{v[0][5],\ p[1]+v[0][4],\ 2p[1]+v[0][3],\ 3p[1]+v[0][2]\} = \max\{0,3,6,9\} = 9$
1	6	9	3	$\max\{v[0][6],\ p[1]+v[0][5],\ 2p[1]+v[0][4],\ 3p[1]+v[0][3]\} = \max\{0,3,6,9\} = 9$
1	7	9	3	$\max\{v[0][7],\ p[1]+v[0][6],\ 2p[1]+v[0][5],\ 3p[1]+v[0][4]\} = \max\{0,3,6,9\} = 9$
2	0	0	0	$\max\{v[1][0]\} = 0$
2	1	3	0	$\max\{v[1][1]\} = 3$
2	2	6	0	$\max\{v[1][2]\} = 6$
2	3	9	0	$\max\{v[1][3]\} = 9$
2	4	9	0	$\max\{v[1][4],\ p[2]+v[1][0]\} = \max\{9,8\} = 9$
2	5	11	1	$\max\{v[1][5],\ p[2]+v[1][1]\} = \max\{9,11\} = 11$
2	6	14	1	$\max\{v[1][6],\ p[2]+v[1][2]\} = \max\{9,14\} = 14$
2	7	17	1	$\max\{v[1][7],\ p[2]+v[1][3]\} = \max\{9,17\} = 17$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	3	6	9	9	9	9	9
$i=2$	0	3	6	9	9	11	14	17

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	1	2	3	3	3	3	3
$i=2$	0	0	0	0	0	1	1	1

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$
$i=0$	0	0	0	0	0	0	0	0
$i=1$	0	1	2	3	3	3	3	3
$i=2$	0	0	0	0	0	1	1	1

We can reconstruct an optimal solution by starting from $i=2$ and $k=7$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	Action
-	-	-	-	0	0	$k=7$
1	2	7	1	0	1	$x[2] \leftarrow 1$, $k \leftarrow 7-1\cdot 4=3$
2	1	3	3	3	1	$x[1] \leftarrow 3$, $k \leftarrow 3-3\cdot 1=0$

Thus, the vector

$$(3,1)$$

is an optimal solution. The optimal value is

$$v[2][7]=17.$$

Exercise 19.3

Consider the following BKP instance, where the knapsack has capacity $c = 5$, and the profits, weights, and bounds associated with the item types are:

	$i = 1$	$i = 2$	$i = 3$
p_i	2	3	4
w_i	1	2	3
b_i	1	1	1

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation (x_1, x_2, x_3), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
1	0	0	0	$\max\{v[0][0]\} = 0$
1	1	2	1	$\max\{v[0][1], p[1]+v[0][0]\} = \max\{0, 2\} = 2$
1	2	2	1	$\max\{v[0][2], p[1]+v[0][1]\} = \max\{0, 2\} = 2$
1	3	2	1	$\max\{v[0][3], p[1]+v[0][2]\} = \max\{0, 2\} = 2$
1	4	2	1	$\max\{v[0][4], p[1]+v[0][3]\} = \max\{0, 2\} = 2$

1	5	2	1	$\max\{v[0][5], p[1]+v[0][4]\} = \max\{0, 2\} = 2$
2	0	0	0	$\max\{v[1][0]\} = 0$
2	1	2	0	$\max\{v[1][1]\} = 2$
2	2	3	1	$\max\{v[1][2], p[2]+v[1][0]\} = \max\{2, 3\} = 3$
2	3	5	1	$\max\{v[1][3], p[2]+v[1][1]\} = \max\{2, 5\} = 5$
2	4	5	1	$\max\{v[1][4], p[2]+v[1][2]\} = \max\{2, 5\} = 5$
2	5	5	1	$\max\{v[1][5], p[2]+v[1][3]\} = \max\{2, 5\} = 5$
3	0	0	0	$\max\{v[2][0]\} = 0$
3	1	2	0	$\max\{v[2][1]\} = 2$
3	2	3	0	$\max\{v[2][2]\} = 3$
3	3	5	0	$\max\{v[2][3], p[3]+v[2][0]\} = \max\{5, 4\} = 5$
3	4	6	1	$\max\{v[2][4], p[3]+v[2][1]\} = \max\{5, 6\} = 6$
3	5	7	1	$\max\{v[2][5], p[3]+v[2][2]\} = \max\{5, 7\} = 7$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$i=0$	0	0	0	0	0	0
$i=1$	0	2	2	2	2	2
$i=2$	0	2	3	5	5	5
$i=3$	0	2	3	5	6	7

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$i=0$	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1
$i=2$	0	0	1	1	1	1
$i=3$	0	0	0	0	1	1

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
$i=0$	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1
$i=2$	0	0	1	1	1	1
$i=3$	0	0	0	0	1	1

We can reconstruct an optimal solution by starting from $i=3$ and $k=5$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	Action
-	-	-	-	0	0	0	$k=5$
1	3	5	1	0	0	1	$x[3] \leftarrow 1$, $k \leftarrow 5-1\cdot 3=2$
2	2	2	1	0	1	1	$x[2] \leftarrow 1$, $k \leftarrow 2-1\cdot 2=0$
3	1	0	0	0	1	1	$x[1] \leftarrow 0$, $k \leftarrow 0-0\cdot 1=0$

Thus, the vector

$$(0,1,1)$$

is an optimal solution. The optimal value is

$$v[3][5]=7.$$

Exercise 19.4

Consider the following BKP instance, where the knapsack has capacity $c = 8$, and the profits, weights, and bounds associated with the item types are:

	$i = 1$	$i = 2$
p_i	5	9
w_i	2	5
b_i	3	1

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
1	0	0	0	$\max\{v[0][0]\} = 0$
1	1	0	0	$\max\{v[0][1]\} = 0$

1	2	5	1	$\max\{v[0][2],\ p[1]+v[0][0]\} = \max\{0, 5\} = 5$
1	3	5	1	$\max\{v[0][3],\ p[1]+v[0][1]\} = \max\{0, 5\} = 5$
1	4	10	2	$\max\{v[0][4],\ p[1]+v[0][2],\ 2p[1]+v[0][0]\} = \max\{0, 5, 10\} = 10$
1	5	10	2	$\max\{v[0][5],\ p[1]+v[0][3],\ 2p[1]+v[0][1]\} = \max\{0, 5, 10\} = 10$
1	6	15	3	$\max\{v[0][6],\ p[1]+v[0][4],\ 2p[1]+v[0][2],\ 3p[1]+v[0][0]\} = \max\{0, 5, 10, 15\} = 15$
1	7	15	3	$\max\{v[0][7],\ p[1]+v[0][5],\ 2p[1]+v[0][3],\ 3p[1]+v[0][1]\} = \max\{0, 5, 10, 15\} = 15$
1	8	15	3	$\max\{v[0][8],\ p[1]+v[0][6],\ 2p[1]+v[0][4],\ 3p[1]+v[0][2]\} = \max\{0, 5, 10, 15\} = 15$
2	0	0	0	$\max\{v[1][0]\} = 0$
2	1	0	0	$\max\{v[1][1]\} = 0$
2	2	5	0	$\max\{v[1][2]\} = 5$
2	3	5	0	$\max\{v[1][3]\} = 5$
2	4	10	0	$\max\{v[1][4]\} = 10$
2	5	10	0	$\max\{v[1][5],\ p[2]+v[1][0]\} = \max\{10, 9\} = 10$
2	6	15	0	$\max\{v[1][6],\ p[2]+v[1][1]\} = \max\{15, 9\} = 15$
2	7	15	0	$\max\{v[1][7],\ p[2]+v[1][2]\} = \max\{15, 14\} = 15$
2	8	15	0	$\max\{v[1][8],\ p[2]+v[1][3]\} = \max\{15, 14\} = 15$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	0	5	5	10	10	15	15	15
$i=2$	0	0	5	5	10	10	15	15	15

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	2	2	3	3	3
$i=2$	0	0	0	0	0	0	0	0	0

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$
$i=0$	0	0	0	0	0	0	0	0	0
$i=1$	0	0	1	1	2	2	3	3	3
$i=2$	0	0	0	0	0	0	0	0	0

We can reconstruct an optimal solution by starting from $i=2$ and $k=8$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	Action
-	-	-	-	0	0	$k=8$
1	2	8	0	0	0	$x[2] \leftarrow 0$, $k \leftarrow 8-0\cdot 5=8$
2	1	8	3	3	0	$x[1] \leftarrow 3$, $k \leftarrow 8-3\cdot 2=2$

Thus, the vector

$$(3,0)$$

is an optimal solution. The optimal value is

$$v[2][8]=15.$$

Exercise 19.5

Consider the following BKP instance, where the knapsack has capacity $c = 9$, and the profits, weights, and bounds associated with the item types are:

	$i = 1$	$i = 2$	$i = 3$
p_i	4	5	3
w_i	3	4	2
b_i	2	1	2

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

2. Using the matrix s, reconstruct an optimal solution in integer notation (x_1, x_2, x_3), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of i and k, the cells $v[i][k]$ and $s[i][k]$, and the operation performed.

i	k	$v[i][k]$	$s[i][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	0	0	$v[0][1] \leftarrow 0$, $s[0][1] \leftarrow 0$
0	2	0	0	$v[0][2] \leftarrow 0$, $s[0][2] \leftarrow 0$
0	3	0	0	$v[0][3] \leftarrow 0$, $s[0][3] \leftarrow 0$
0	4	0	0	$v[0][4] \leftarrow 0$, $s[0][4] \leftarrow 0$
0	5	0	0	$v[0][5] \leftarrow 0$, $s[0][5] \leftarrow 0$
0	6	0	0	$v[0][6] \leftarrow 0$, $s[0][6] \leftarrow 0$
0	7	0	0	$v[0][7] \leftarrow 0$, $s[0][7] \leftarrow 0$
0	8	0	0	$v[0][8] \leftarrow 0$, $s[0][8] \leftarrow 0$
0	9	0	0	$v[0][9] \leftarrow 0$, $s[0][9] \leftarrow 0$
1	0	0	0	$\max\{v[0][0]\} = 0$

1	1	0	0	$\max\{v[0][1]\} = 0$
1	2	0	0	$\max\{v[0][2]\} = 0$
1	3	4	1	$\max\{v[0][3], p[1]+v[0][0]\} = \max\{0, 4\} = 4$
1	4	4	1	$\max\{v[0][4], p[1]+v[0][1]\} = \max\{0, 4\} = 4$
1	5	4	1	$\max\{v[0][5], p[1]+v[0][2]\} = \max\{0, 4\} = 4$
1	6	8	2	$\max\{v[0][6], p[1]+v[0][3], 2p[1]+v[0][0]\} = \max\{0, 4, 8\} = 8$
1	7	8	2	$\max\{v[0][7], p[1]+v[0][4], 2p[1]+v[0][1]\} = \max\{0, 4, 8\} = 8$
1	8	8	2	$\max\{v[0][8], p[1]+v[0][5], 2p[1]+v[0][2]\} = \max\{0, 4, 8\} = 8$
1	9	8	2	$\max\{v[0][9], p[1]+v[0][6], 2p[1]+v[0][3]\} = \max\{0, 4, 8\} = 8$
2	0	0	0	$\max\{v[1][0]\} = 0$
2	1	0	0	$\max\{v[1][1]\} = 0$
2	2	0	0	$\max\{v[1][2]\} = 0$
2	3	4	0	$\max\{v[1][3]\} = 4$
2	4	5	1	$\max\{v[1][4], p[2]+v[1][0]\} = \max\{4, 5\} = 5$
2	5	5	1	$\max\{v[1][5], p[2]+v[1][1]\} = \max\{4, 5\} = 5$
2	6	8	0	$\max\{v[1][6], p[2]+v[1][2]\} = \max\{8, 5\} = 8$
2	7	9	1	$\max\{v[1][7], p[2]+v[1][3]\} = \max\{8, 9\} = 9$
2	8	9	1	$\max\{v[1][8], p[2]+v[1][4]\} = \max\{8, 9\} = 9$
2	9	9	1	$\max\{v[1][9], p[2]+v[1][5]\} = \max\{8, 9\} = 9$
3	0	0	0	$\max\{v[2][0]\} = 0$
3	1	0	0	$\max\{v[2][1]\} = 0$
3	2	3	1	$\max\{v[2][2], p[3]+v[2][0]\} = \max\{0, 3\} = 3$
3	3	4	0	$\max\{v[2][3], p[3]+v[2][1]\} = \max\{4, 3\} = 4$
3	4	6	2	$\max\{v[2][4], p[3]+v[2][2], 2p[3]+v[2][0]\} = \max\{5, 3, 6\} = 6$
3	5	7	1	$\max\{v[2][5], p[3]+v[2][3], 2p[3]+v[2][1]\} = \max\{5, 7, 6\} = 7$
3	6	8	0	$\max\{v[2][6], p[3]+v[2][4], 2p[3]+v[2][2]\} = \max\{8, 8, 6\} = 8$
3	7	10	2	$\max\{v[2][7], p[3]+v[2][5], 2p[3]+v[2][3]\} = \max\{9, 8, 10\} = 10$
3	8	11	1	$\max\{v[2][8], p[3]+v[2][6], 2p[3]+v[2][4]\} = \max\{9, 11, 11\} = 11$
3	9	12	1	$\max\{v[2][9], p[3]+v[2][7], 2p[3]+v[2][5]\} = \max\{9, 12, 11\} = 12$

This is the matrix v after the execution of the DP algorithm:

$v[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	4	4	4	8	8	8	8
$i=2$	0	0	0	4	5	5	8	9	9	9
$i=3$	0	0	3	4	6	7	8	10	11	12

This is the matrix s after the execution of the DP algorithm:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	1	1	1	2	2	2	2
$i=2$	0	0	0	0	1	1	0	1	1	1
$i=3$	0	0	1	0	2	1	0	2	1	1

2. We have the following matrix s:

$s[i][k]$	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$i=0$	0	0	0	0	0	0	0	0	0	0
$i=1$	0	0	0	1	1	1	2	2	2	2
$i=2$	0	0	0	0	1	1	0	1	1	1
$i=3$	0	0	1	0	2	1	0	2	1	1

We can reconstruct an optimal solution by starting from $i=3$ and $k=9$ as follows:

Iteration	i	k	$s[i][k]$	$x[1]$	$x[2]$	$x[3]$	Action
-	-	-	-	0	0	0	$k=9$
1	3	9	1	0	0	1	$x[3] \leftarrow 1$, $k \leftarrow 9-1\cdot 2=7$
2	2	7	1	0	1	1	$x[2] \leftarrow 1$, $k \leftarrow 7-1\cdot 4=3$
3	1	3	1	1	1	1	$x[1] \leftarrow 1$, $k \leftarrow 3-1\cdot 3=0$

Thus, the vector

$$(1,1,1)$$

is an optimal solution. The optimal value is

$$v[3][9]=12.$$

Chapter 20

DP for the rod cutting problem

Exercise 20.1

Consider the following RCP instance, where the rod has length $n = 6$, and the profits associated with the possible piece lengths are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
p_i	2	5	7	8	10	13

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of ℓ and i, the cells $v[\ell]$ and $s[\ell]$, the current value of m, and the current value of t. The cell of m is highlighted in red when updated; the cell of t is highlighted in blue when updated; the cells of $v[\ell]$ and $s[\ell]$ are highlighted in blue when assigned.

ℓ	i	$v[\ell]$	$s[\ell]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
1	1	–	–	2	1	$p[1] + v[0] = 2 + 0 = 2 > 0$: $m \leftarrow 2,\ t \leftarrow 1$
	–	2	1	2	1	$v[1] \leftarrow m = 2,\ s[1] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
2	1	–	–	4	1	$p[1] + v[1] = 2 + 2 = 4 > 0$: $m \leftarrow 4,\ t \leftarrow 1$
	2	–	–	5	2	$p[2] + v[0] = 5 + 0 = 5 > 4$: $m \leftarrow 5,\ t \leftarrow 2$
	–	5	2	5	2	$v[2] \leftarrow m = 5,\ s[2] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
3	1	–	–	7	1	$p[1] + v[2] = 2 + 5 = 7 > 0$: $m \leftarrow 7,\ t \leftarrow 1$
	2	–	–	7	1	$p[2] + v[1] = 5 + 2 = 7 = m$: no update
	3	–	–	7	1	$p[3] + v[0] = 7 + 0 = 7 = m$: no update
	–	7	1	7	1	$v[3] \leftarrow m = 7,\ s[3] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
4	1	–	–	9	1	$p[1] + v[3] = 2 + 7 = 9 > 0$: $m \leftarrow 9,\ t \leftarrow 1$
	2	–	–	10	2	$p[2] + v[2] = 5 + 5 = 10 > 9$: $m \leftarrow 10,\ t \leftarrow 2$
	3	–	–	10	2	$p[3] + v[1] = 7 + 2 = 9 < 10$: no update
	4	–	–	10	2	$p[4] + v[0] = 8 + 0 = 8 < 10$: no update
	–	10	2	10	2	$v[4] \leftarrow m = 10,\ s[4] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
5	1	–	–	12	1	$p[1] + v[4] = 2 + 10 = 12 > 0$: $m \leftarrow 12,\ t \leftarrow 1$
	2	–	–	12	1	$p[2] + v[3] = 5 + 7 = 12 = m$: no update
	3	–	–	12	1	$p[3] + v[2] = 7 + 5 = 12 = m$: no update
	4	–	–	12	1	$p[4] + v[1] = 8 + 2 = 10 < 12$: no update
	5	–	–	12	1	$p[5] + v[0] = 10 + 0 = 10 < 12$: no update
	–	12	1	12	1	$v[5] \leftarrow m = 12,\ s[5] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
6	1	–	–	14	1	$p[1] + v[5] = 2 + 12 = 14 > 0$: $m \leftarrow 14,\ t \leftarrow 1$
	2	–	–	15	2	$p[2] + v[4] = 5 + 10 = 15 > 14$: $m \leftarrow 15,\ t \leftarrow 2$
	3	–	–	15	2	$p[3] + v[3] = 7 + 7 = 14 < 15$: no update
	4	–	–	15	2	$p[4] + v[2] = 8 + 5 = 13 < 15$: no update
	5	–	–	15	2	$p[5] + v[1] = 10 + 2 = 12 < 15$: no update
	6	–	–	15	2	$p[6] + v[0] = 13 + 0 = 13 < 15$: no update
	–	15	2	15	2	$v[6] \leftarrow m = 15,\ s[6] \leftarrow t = 2$

This is the vector v after the execution of the DP algorithm:

	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$v[\ell]$	0	2	5	7	10	12	15

This is the vector s after the execution of the DP algorithm:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	1	2	1	2

2. We have the following vector s:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	1	2	1	2

We can reconstruct an optimal solution for a rod of length $n=6$ as follows:

Iteration	i	$s[i]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Update of i
-	-	-	0	0	0	0	0	0	$i=6$
1	6	2	0	1	0	0	0	0	$i=6-2=4$
2	4	2	0	2	0	0	0	0	$i=4-2=2$
3	2	2	0	3	0	0	0	0	$i=2-2=0$

Thus, the vector

$$(0,3,0,0,0,0)$$

is an optimal solution. The optimal value is

$$v[6]=15.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and ℓ, the cells $v[i][\ell]$ and $s[i][\ell]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of ℓ.

i	ℓ	$v[i][\ell]$	$s[i][\ell]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	$-\infty$	0	$v[0][1] \leftarrow -\infty$, $s[0][1] \leftarrow 0$
0	2	$-\infty$	0	$v[0][2] \leftarrow -\infty$, $s[0][2] \leftarrow 0$
0	3	$-\infty$	0	$v[0][3] \leftarrow -\infty$, $s[0][3] \leftarrow 0$
0	4	$-\infty$	0	$v[0][4] \leftarrow -\infty$, $s[0][4] \leftarrow 0$
0	5	$-\infty$	0	$v[0][5] \leftarrow -\infty$, $s[0][5] \leftarrow 0$
0	6	$-\infty$	0	$v[0][6] \leftarrow -\infty$, $s[0][6] \leftarrow 0$
1	0	0	0	$i > \ell$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	2	1	$p[1] + v[1][0] = 2 + 0 > v[0][1] = -\infty$: $v[1][1] \leftarrow 2$, $s[1][1] \leftarrow 1$
1	2	4	1	$p[1] + v[1][1] = 2 + 2 > v[0][2] = -\infty$: $v[1][2] \leftarrow 4$, $s[1][2] \leftarrow 1$
1	3	6	1	$p[1] + v[1][2] = 2 + 4 > v[0][3] = -\infty$: $v[1][3] \leftarrow 6$, $s[1][3] \leftarrow 1$
1	4	8	1	$p[1] + v[1][3] = 2 + 6 > v[0][4] = -\infty$: $v[1][4] \leftarrow 8$, $s[1][4] \leftarrow 1$
1	5	10	1	$p[1] + v[1][4] = 2 + 8 > v[0][5] = -\infty$: $v[1][5] \leftarrow 10$, $s[1][5] \leftarrow 1$
1	6	12	1	$p[1] + v[1][5] = 2 + 10 > v[0][6] = -\infty$: $v[1][6] \leftarrow 12$, $s[1][6] \leftarrow 1$
2	0	0	0	$i > \ell$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	2	0	$i > \ell$: $v[2][1] \leftarrow v[1][1] = 2$, $s[2][1] \leftarrow 0$
2	2	5	1	$p[2] + v[2][0] = 5 + 0 > v[1][2] = 4$: $v[2][2] \leftarrow 5$, $s[2][2] \leftarrow 1$
2	3	7	1	$p[2] + v[2][1] = 5 + 2 > v[1][3] = 6$: $v[2][3] \leftarrow 7$, $s[2][3] \leftarrow 1$
2	4	10	1	$p[2] + v[2][2] = 5 + 5 > v[1][4] = 8$: $v[2][4] \leftarrow 10$, $s[2][4] \leftarrow 1$
2	5	12	1	$p[2] + v[2][3] = 5 + 7 > v[1][5] = 10$: $v[2][5] \leftarrow 12$, $s[2][5] \leftarrow 1$
2	6	15	1	$p[2] + v[2][4] = 5 + 10 > v[1][6] = 12$: $v[2][6] \leftarrow 15$, $s[2][6] \leftarrow 1$
3	0	0	0	$i > \ell$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	2	0	$i > \ell$: $v[3][1] \leftarrow v[2][1] = 2$, $s[3][1] \leftarrow 0$
3	2	5	0	$i > \ell$: $v[3][2] \leftarrow v[2][2] = 5$, $s[3][2] \leftarrow 0$
3	3	7	0	$p[3] + v[3][0] = 7 + 0 \leq v[2][3] = 7$: $v[3][3] \leftarrow v[2][3] = 7$, $s[3][3] \leftarrow 0$
3	4	10	0	$p[3] + v[3][1] = 7 + 2 \leq v[2][4] = 10$: $v[3][4] \leftarrow v[2][4] = 10$, $s[3][4] \leftarrow 0$
3	5	12	0	$p[3] + v[3][2] = 7 + 5 \leq v[2][5] = 12$: $v[3][5] \leftarrow v[2][5] = 12$, $s[3][5] \leftarrow 0$
3	6	15	0	$p[3] + v[3][3] = 7 + 7 \leq v[2][6] = 15$: $v[3][6] \leftarrow v[2][6] = 15$, $s[3][6] \leftarrow 0$
4	0	0	0	$i > \ell$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	2	0	$i > \ell$: $v[4][1] \leftarrow v[3][1] = 2$, $s[4][1] \leftarrow 0$
4	2	5	0	$i > \ell$: $v[4][2] \leftarrow v[3][2] = 5$, $s[4][2] \leftarrow 0$
4	3	7	0	$i > \ell$: $v[4][3] \leftarrow v[3][3] = 7$, $s[4][3] \leftarrow 0$
4	4	10	0	$p[4] + v[4][0] = 8 + 0 \leq v[3][4] = 10$: $v[4][4] \leftarrow v[3][4] = 10$, $s[4][4] \leftarrow 0$
4	5	12	0	$p[4] + v[4][1] = 8 + 2 \leq v[3][5] = 12$: $v[4][5] \leftarrow v[3][5] = 12$, $s[4][5] \leftarrow 0$
4	6	15	0	$p[4] + v[4][2] = 8 + 5 \leq v[3][6] = 15$: $v[4][6] \leftarrow v[3][6] = 15$, $s[4][6] \leftarrow 0$
5	0	0	0	$i > \ell$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	2	0	$i > \ell$: $v[5][1] \leftarrow v[4][1] = 2$, $s[5][1] \leftarrow 0$
5	2	5	0	$i > \ell$: $v[5][2] \leftarrow v[4][2] = 5$, $s[5][2] \leftarrow 0$
5	3	7	0	$i > \ell$: $v[5][3] \leftarrow v[4][3] = 7$, $s[5][3] \leftarrow 0$
5	4	10	0	$i > \ell$: $v[5][4] \leftarrow v[4][4] = 10$, $s[5][4] \leftarrow 0$
5	5	12	0	$p[5] + v[5][0] = 10 + 0 \leq v[4][5] = 12$: $v[5][5] \leftarrow v[4][5] = 12$, $s[5][5] \leftarrow 0$
5	6	15	0	$p[5] + v[5][1] = 10 + 2 \leq v[4][6] = 15$: $v[5][6] \leftarrow v[4][6] = 15$, $s[5][6] \leftarrow 0$
6	0	0	0	$i > \ell$: $v[6][0] \leftarrow v[5][0] = 0$, $s[6][0] \leftarrow 0$
6	1	2	0	$i > \ell$: $v[6][1] \leftarrow v[5][1] = 2$, $s[6][1] \leftarrow 0$
6	2	5	0	$i > \ell$: $v[6][2] \leftarrow v[5][2] = 5$, $s[6][2] \leftarrow 0$
6	3	7	0	$i > \ell$: $v[6][3] \leftarrow v[5][3] = 7$, $s[6][3] \leftarrow 0$
6	4	10	0	$i > \ell$: $v[6][4] \leftarrow v[5][4] = 10$, $s[6][4] \leftarrow 0$

6	5	12	0	$i > \ell$: $v[6][5] \leftarrow v[5][5] = 12$, $s[6][5] \leftarrow 0$
6	6	15	0	$p[6] + v[6][0] = 13 + 0 \leq v[5][6] = 15$: $v[6][6] \leftarrow v[5][6] = 15$, $s[6][6] \leftarrow 0$

These are the matrices v and s obtained during the execution of the DP algorithm:

$v[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$i=1$	0	2	4	6	8	10	12
$i=2$	0	2	5	7	10	12	15
$i=3$	0	2	5	7	10	12	15
$i=4$	0	2	5	7	10	12	15
$i=5$	0	2	5	7	10	12	15
$i=6$	0	2	5	7	10	12	15

$s[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1
$i=2$	0	0	1	1	1	1	1
$i=3$	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0
$i=6$	0	0	0	0	0	0	0

The optimal value of the RCP instance is $v[6][6] = 15$.

4. Using the matrix s, we can reconstruct an optimal solution as follows:

Iteration	i	ℓ	$s[i][\ell]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Action
1	6	6	0	0	0	0	0	0	0	$i \leftarrow 5$
2	5	6	0	0	0	0	0	0	0	$i \leftarrow 4$
3	4	6	0	0	0	0	0	0	0	$i \leftarrow 3$
4	3	6	0	0	0	0	0	0	0	$i \leftarrow 2$
5	2	6	1	0	1	0	0	0	0	$\ell \leftarrow 4$
6	2	4	1	0	2	0	0	0	0	$\ell \leftarrow 2$
7	2	2	1	0	3	0	0	0	0	$\ell \leftarrow 0$

Thus, the vector

$$(0, 3, 0, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[6][6] = 15.$$

Exercise 20.2

Consider the following RCP instance, where the rod has length $n = 5$, and the profits associated with the possible piece lengths are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
p_i	1	4	6	8	9

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of ℓ and i, the cells

$v[\ell]$ and $s[\ell]$, the current value of m, and the current value of t. The cell of m is highlighted in red when updated; the cell of t is highlighted in blue when updated; the cells of $v[\ell]$ and $s[\ell]$ are highlighted in blue when assigned.

ℓ	i	$v[\ell]$	$s[\ell]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
1	1	–	–	1	1	$p[1] + v[0] = 1 + 0 = 1 > 0$: $m \leftarrow 1,\ t \leftarrow 1$
	–	1	1	1	1	$v[1] \leftarrow m = 1,\ s[1] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
2	1	–	–	2	1	$p[1] + v[1] = 1 + 1 = 2 > 0$: $m \leftarrow 2,\ t \leftarrow 1$
	2	–	–	4	2	$p[2] + v[0] = 4 + 0 = 4 > 2$: $m \leftarrow 4,\ t \leftarrow 2$
	–	4	2	4	2	$v[2] \leftarrow m = 4,\ s[2] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
3	1	–	–	5	1	$p[1] + v[2] = 1 + 4 = 5 > 0$: $m \leftarrow 5,\ t \leftarrow 1$
	2	–	–	5	1	$p[2] + v[1] = 4 + 1 = 5 = m$: no update
	3	–	–	6	3	$p[3] + v[0] = 6 + 0 = 6 > 5$: $m \leftarrow 6,\ t \leftarrow 3$
	–	6	3	6	3	$v[3] \leftarrow m = 6,\ s[3] \leftarrow t = 3$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
4	1	–	–	7	1	$p[1] + v[3] = 1 + 6 = 7 > 0$: $m \leftarrow 7,\ t \leftarrow 1$
	2	–	–	8	2	$p[2] + v[2] = 4 + 4 = 8 > 7$: $m \leftarrow 8,\ t \leftarrow 2$
	3	–	–	8	2	$p[3] + v[1] = 6 + 1 = 7 < 8$: no update
	4	–	–	8	2	$p[4] + v[0] = 8 + 0 = 8 = m$: no update
	–	8	2	8	2	$v[4] \leftarrow m = 8,\ s[4] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
5	1	–	–	9	1	$p[1] + v[4] = 1 + 8 = 9 > 0$: $m \leftarrow 9,\ t \leftarrow 1$
	2	–	–	10	2	$p[2] + v[3] = 4 + 6 = 10 > 9$: $m \leftarrow 10,\ t \leftarrow 2$
	3	–	–	10	2	$p[3] + v[2] = 6 + 4 = 10 = m$: no update
	4	–	–	10	2	$p[4] + v[1] = 8 + 1 = 9 < 10$: no update
	5	–	–	10	2	$p[5] + v[0] = 9 + 0 = 9 < 10$: no update
	–	10	2	10	2	$v[5] \leftarrow m = 10,\ s[5] \leftarrow t = 2$

This is the vector v after the execution of the DP algorithm:

	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$v[\ell]$	0	1	4	6	8	10

This is the vector s after the execution of the DP algorithm:

	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$s[\ell]$	1	2	3	2	2

2. We have the following vector s:

	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$s[\ell]$	1	2	3	2	2

We can reconstruct an optimal solution for a rod of length $n = 5$ as follows:

Iteration	i	$s[i]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Update of i
-	-	-	0	0	0	0	0	$i = 5$
1	5	2	0	1	0	0	0	$i = 5 - 2 = 3$
2	3	3	0	1	1	0	0	$i = 3 - 3 = 0$

Thus, the vector

$$(0, 1, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5] = 10.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and ℓ, the cells $v[i][\ell]$ and $s[i][\ell]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of ℓ.

i	ℓ	$v[i][\ell]$	$s[i][\ell]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	$-\infty$	0	$v[0][1] \leftarrow -\infty$, $s[0][1] \leftarrow 0$
0	2	$-\infty$	0	$v[0][2] \leftarrow -\infty$, $s[0][2] \leftarrow 0$
0	3	$-\infty$	0	$v[0][3] \leftarrow -\infty$, $s[0][3] \leftarrow 0$
0	4	$-\infty$	0	$v[0][4] \leftarrow -\infty$, $s[0][4] \leftarrow 0$
0	5	$-\infty$	0	$v[0][5] \leftarrow -\infty$, $s[0][5] \leftarrow 0$
1	0	0	0	$i > \ell$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	1	1	$p[1] + v[1][0] = 1 + 0 > v[0][1] = -\infty$: $v[1][1] \leftarrow 1$, $s[1][1] \leftarrow 1$
1	2	2	1	$p[1] + v[1][1] = 1 + 1 > v[0][2] = -\infty$: $v[1][2] \leftarrow 2$, $s[1][2] \leftarrow 1$
1	3	3	1	$p[1] + v[1][2] = 1 + 2 > v[0][3] = -\infty$: $v[1][3] \leftarrow 3$, $s[1][3] \leftarrow 1$
1	4	4	1	$p[1] + v[1][3] = 1 + 3 > v[0][4] = -\infty$: $v[1][4] \leftarrow 4$, $s[1][4] \leftarrow 1$
1	5	5	1	$p[1] + v[1][4] = 1 + 4 > v[0][5] = -\infty$: $v[1][5] \leftarrow 5$, $s[1][5] \leftarrow 1$
2	0	0	0	$i > \ell$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	1	0	$i > \ell$: $v[2][1] \leftarrow v[1][1] = 1$, $s[2][1] \leftarrow 0$
2	2	4	1	$p[2] + v[2][0] = 4 + 0 > v[1][2] = 2$: $v[2][2] \leftarrow 4$, $s[2][2] \leftarrow 1$
2	3	5	1	$p[2] + v[2][1] = 4 + 1 > v[1][3] = 3$: $v[2][3] \leftarrow 5$, $s[2][3] \leftarrow 1$
2	4	8	1	$p[2] + v[2][2] = 4 + 4 > v[1][4] = 4$: $v[2][4] \leftarrow 8$, $s[2][4] \leftarrow 1$
2	5	9	1	$p[2] + v[2][3] = 4 + 5 > v[1][5] = 5$: $v[2][5] \leftarrow 9$, $s[2][5] \leftarrow 1$
3	0	0	0	$i > \ell$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	1	0	$i > \ell$: $v[3][1] \leftarrow v[2][1] = 1$, $s[3][1] \leftarrow 0$
3	2	4	0	$i > \ell$: $v[3][2] \leftarrow v[2][2] = 4$, $s[3][2] \leftarrow 0$
3	3	6	1	$p[3] + v[3][0] = 6 + 0 > v[2][3] = 5$: $v[3][3] \leftarrow 6$, $s[3][3] \leftarrow 1$
3	4	8	0	$p[3] + v[3][1] = 6 + 1 \leq v[2][4] = 8$: $v[3][4] \leftarrow v[2][4] = 8$, $s[3][4] \leftarrow 0$
3	5	10	1	$p[3] + v[3][2] = 6 + 4 > v[2][5] = 9$: $v[3][5] \leftarrow 10$, $s[3][5] \leftarrow 1$
4	0	0	0	$i > \ell$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	1	0	$i > \ell$: $v[4][1] \leftarrow v[3][1] = 1$, $s[4][1] \leftarrow 0$
4	2	4	0	$i > \ell$: $v[4][2] \leftarrow v[3][2] = 4$, $s[4][2] \leftarrow 0$
4	3	6	0	$i > \ell$: $v[4][3] \leftarrow v[3][3] = 6$, $s[4][3] \leftarrow 0$
4	4	8	0	$p[4] + v[4][0] = 8 + 0 \leq v[3][4] = 8$: $v[4][4] \leftarrow v[3][4] = 8$, $s[4][4] \leftarrow 0$
4	5	10	0	$p[4] + v[4][1] = 8 + 1 \leq v[3][5] = 10$: $v[4][5] \leftarrow v[3][5] = 10$, $s[4][5] \leftarrow 0$
5	0	0	0	$i > \ell$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	1	0	$i > \ell$: $v[5][1] \leftarrow v[4][1] = 1$, $s[5][1] \leftarrow 0$
5	2	4	0	$i > \ell$: $v[5][2] \leftarrow v[4][2] = 4$, $s[5][2] \leftarrow 0$
5	3	6	0	$i > \ell$: $v[5][3] \leftarrow v[4][3] = 6$, $s[5][3] \leftarrow 0$
5	4	8	0	$i > \ell$: $v[5][4] \leftarrow v[4][4] = 8$, $s[5][4] \leftarrow 0$
5	5	10	0	$p[5] + v[5][0] = 9 + 0 \leq v[4][5] = 10$: $v[5][5] \leftarrow v[4][5] = 10$, $s[5][5] \leftarrow 0$

These are the matrices v and s obtained during the execution of the DP algorithm:

$v[i][\ell]$	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$i = 0$	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$i = 1$	0	1	2	3	4	5
$i = 2$	0	1	4	5	8	9
$i = 3$	0	1	4	6	8	10
$i = 4$	0	1	4	6	8	10
$i = 5$	0	1	4	6	8	10

$s[i][\ell]$	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$i = 0$	0	0	0	0	0	0
$i = 1$	0	1	1	1	1	1
$i = 2$	0	0	1	1	1	1
$i = 3$	0	0	0	1	0	1
$i = 4$	0	0	0	0	0	0
$i = 5$	0	0	0	0	0	0

The optimal value of the RCP instance is $v[5][5] = 10$.

4. Using the matrix s, we can reconstruct an optimal solution as follows:

Iteration	i	ℓ	$s[i][\ell]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
1	5	5	0	0	0	0	0	0	$i \leftarrow 4$
2	4	5	0	0	0	0	0	0	$i \leftarrow 3$
3	3	5	1	0	0	1	0	0	$\ell \leftarrow 2$
4	3	2	0	0	0	1	0	0	$i \leftarrow 2$
5	2	2	1	0	1	1	0	0	$\ell \leftarrow 0$

Thus, the vector

$$(0, 1, 1, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5][5] = 10.$$

Exercise 20.3

Consider the following RCP instance, where the rod has length $n = 5$, and the profits associated with the possible piece lengths are:

	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
p_i	3	5	8	9	10

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of ℓ and i, the cells $v[\ell]$ and $s[\ell]$, the current value of m, and the current value of t. The cell of m is highlighted in red when updated; the cell of t is highlighted in blue when updated; the cells of $v[\ell]$ and $s[\ell]$ are highlighted in blue when assigned.

ℓ	i	$v[\ell]$	$s[\ell]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$

1	1	–	–	3	1	$p[1] + v[0] = 3 + 0 = 3 > 0$: $m \leftarrow 3$, $t \leftarrow 1$
	–	3	1	3	1	$v[1] \leftarrow m = 3$, $s[1] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
2	1	–	–	6	1	$p[1] + v[1] = 3 + 3 = 6 > 0$: $m \leftarrow 6$, $t \leftarrow 1$
	2	–	–	6	1	$p[2] + v[0] = 5 + 0 = 5 < 6$: no update
	–	6	1	6	1	$v[2] \leftarrow m = 6$, $s[2] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
3	1	–	–	9	1	$p[1] + v[2] = 3 + 6 = 9 > 0$: $m \leftarrow 9$, $t \leftarrow 1$
	2	–	–	9	1	$p[2] + v[1] = 5 + 3 = 8 < 9$: no update
	3	–	–	9	1	$p[3] + v[0] = 8 + 0 = 8 < 9$: no update
	–	9	1	9	1	$v[3] \leftarrow m = 9$, $s[3] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
4	1	–	–	12	1	$p[1] + v[3] = 3 + 9 = 12 > 0$: $m \leftarrow 12$, $t \leftarrow 1$
	2	–	–	12	1	$p[2] + v[2] = 5 + 6 = 11 < 12$: no update
	3	–	–	12	1	$p[3] + v[1] = 8 + 3 = 11 < 12$: no update
	4	–	–	12	1	$p[4] + v[0] = 9 + 0 = 9 < 12$: no update
	–	12	1	12	1	$v[4] \leftarrow m = 12$, $s[4] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0$, $t \leftarrow 0$
5	1	–	–	15	1	$p[1] + v[4] = 3 + 12 = 15 > 0$: $m \leftarrow 15$, $t \leftarrow 1$
	2	–	–	15	1	$p[2] + v[3] = 5 + 9 = 14 < 15$: no update
	3	–	–	15	1	$p[3] + v[2] = 8 + 6 = 14 < 15$: no update
	4	–	–	15	1	$p[4] + v[1] = 9 + 3 = 12 < 15$: no update
	5	–	–	15	1	$p[5] + v[0] = 10 + 0 = 10 < 15$: no update
	–	15	1	15	1	$v[5] \leftarrow m = 15$, $s[5] \leftarrow t = 1$

This is the vector v after the execution of the DP algorithm:

	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$
$v[\ell]$	0	3	6	9	12	15

This is the vector s after the execution of the DP algorithm:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$
$s[\ell]$	1	1	1	1	1

2. We have the following vector s:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$
$s[\ell]$	1	1	1	1	1

We can reconstruct an optimal solution for a rod of length $n=5$ as follows:

Iteration	i	$s[i]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Update of i
-	-	-	0	0	0	0	0	$i=5$
1	5	1	1	0	0	0	0	$i=5-1=4$
2	4	1	2	0	0	0	0	$i=4-1=3$
3	3	1	3	0	0	0	0	$i=3-1=2$
4	2	1	4	0	0	0	0	$i=2-1=1$
5	1	1	5	0	0	0	0	$i=1-1=0$

Thus, the vector

$$(5,0,0,0,0)$$

is an optimal solution. The optimal value is

$$v[5]=15.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and ℓ, the cells $v[i][\ell]$ and $s[i][\ell]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of ℓ.

i	ℓ	$v[i][\ell]$	$s[i][\ell]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	$-\infty$	0	$v[0][1] \leftarrow -\infty$, $s[0][1] \leftarrow 0$
0	2	$-\infty$	0	$v[0][2] \leftarrow -\infty$, $s[0][2] \leftarrow 0$
0	3	$-\infty$	0	$v[0][3] \leftarrow -\infty$, $s[0][3] \leftarrow 0$
0	4	$-\infty$	0	$v[0][4] \leftarrow -\infty$, $s[0][4] \leftarrow 0$
0	5	$-\infty$	0	$v[0][5] \leftarrow -\infty$, $s[0][5] \leftarrow 0$
1	0	0	0	$i > \ell$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	3	1	$p[1] + v[1][0] = 3 + 0 > v[0][1] = -\infty$: $v[1][1] \leftarrow 3$, $s[1][1] \leftarrow 1$
1	2	6	1	$p[1] + v[1][1] = 3 + 3 > v[0][2] = -\infty$: $v[1][2] \leftarrow 6$, $s[1][2] \leftarrow 1$
1	3	9	1	$p[1] + v[1][2] = 3 + 6 > v[0][3] = -\infty$: $v[1][3] \leftarrow 9$, $s[1][3] \leftarrow 1$
1	4	12	1	$p[1] + v[1][3] = 3 + 9 > v[0][4] = -\infty$: $v[1][4] \leftarrow 12$, $s[1][4] \leftarrow 1$
1	5	15	1	$p[1] + v[1][4] = 3 + 12 > v[0][5] = -\infty$: $v[1][5] \leftarrow 15$, $s[1][5] \leftarrow 1$
2	0	0	0	$i > \ell$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	3	0	$i > \ell$: $v[2][1] \leftarrow v[1][1] = 3$, $s[2][1] \leftarrow 0$
2	2	6	0	$p[2] + v[2][0] = 5 + 0 \leq v[1][2] = 6$: $v[2][2] \leftarrow v[1][2] = 6$, $s[2][2] \leftarrow 0$
2	3	9	0	$p[2] + v[2][1] = 5 + 3 \leq v[1][3] = 9$: $v[2][3] \leftarrow v[1][3] = 9$, $s[2][3] \leftarrow 0$
2	4	12	0	$p[2] + v[2][2] = 5 + 6 \leq v[1][4] = 12$: $v[2][4] \leftarrow v[1][4] = 12$, $s[2][4] \leftarrow 0$
2	5	15	0	$p[2] + v[2][3] = 5 + 9 \leq v[1][5] = 15$: $v[2][5] \leftarrow v[1][5] = 15$, $s[2][5] \leftarrow 0$
3	0	0	0	$i > \ell$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	3	0	$i > \ell$: $v[3][1] \leftarrow v[2][1] = 3$, $s[3][1] \leftarrow 0$
3	2	6	0	$i > \ell$: $v[3][2] \leftarrow v[2][2] = 6$, $s[3][2] \leftarrow 0$
3	3	9	0	$p[3] + v[3][0] = 8 + 0 \leq v[2][3] = 9$: $v[3][3] \leftarrow v[2][3] = 9$, $s[3][3] \leftarrow 0$
3	4	12	0	$p[3] + v[3][1] = 8 + 3 \leq v[2][4] = 12$: $v[3][4] \leftarrow v[2][4] = 12$, $s[3][4] \leftarrow 0$
3	5	15	0	$p[3] + v[3][2] = 8 + 6 \leq v[2][5] = 15$: $v[3][5] \leftarrow v[2][5] = 15$, $s[3][5] \leftarrow 0$
4	0	0	0	$i > \ell$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	3	0	$i > \ell$: $v[4][1] \leftarrow v[3][1] = 3$, $s[4][1] \leftarrow 0$
4	2	6	0	$i > \ell$: $v[4][2] \leftarrow v[3][2] = 6$, $s[4][2] \leftarrow 0$
4	3	9	0	$i > \ell$: $v[4][3] \leftarrow v[3][3] = 9$, $s[4][3] \leftarrow 0$
4	4	12	0	$p[4] + v[4][0] = 9 + 0 \leq v[3][4] = 12$: $v[4][4] \leftarrow v[3][4] = 12$, $s[4][4] \leftarrow 0$
4	5	15	0	$p[4] + v[4][1] = 9 + 3 \leq v[3][5] = 15$: $v[4][5] \leftarrow v[3][5] = 15$, $s[4][5] \leftarrow 0$
5	0	0	0	$i > \ell$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	3	0	$i > \ell$: $v[5][1] \leftarrow v[4][1] = 3$, $s[5][1] \leftarrow 0$
5	2	6	0	$i > \ell$: $v[5][2] \leftarrow v[4][2] = 6$, $s[5][2] \leftarrow 0$
5	3	9	0	$i > \ell$: $v[5][3] \leftarrow v[4][3] = 9$, $s[5][3] \leftarrow 0$
5	4	12	0	$i > \ell$: $v[5][4] \leftarrow v[4][4] = 12$, $s[5][4] \leftarrow 0$
5	5	15	0	$p[5] + v[5][0] = 10 + 0 \leq v[4][5] = 15$: $v[5][5] \leftarrow v[4][5] = 15$, $s[5][5] \leftarrow 0$

These are the matrices v and s obtained during the execution of the DP algorithm:

$v[i][\ell]$	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$i = 0$	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$i = 1$	0	3	6	9	12	15
$i = 2$	0	3	6	9	12	15
$i = 3$	0	3	6	9	12	15
$i = 4$	0	3	6	9	12	15
$i = 5$	0	3	6	9	12	15

$s[i][\ell]$	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$i = 0$	0	0	0	0	0	0
$i = 1$	0	1	1	1	1	1
$i = 2$	0	0	0	0	0	0
$i = 3$	0	0	0	0	0	0
$i = 4$	0	0	0	0	0	0
$i = 5$	0	0	0	0	0	0

The optimal value of the RCP instance is $v[5][5] = 15$.

4. Using the matrix s, we can reconstruct an optimal solution as follows:

Iteration	i	ℓ	$s[i][\ell]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	Action
1	5	5	0	0	0	0	0	0	$i \leftarrow 4$
2	4	5	0	0	0	0	0	0	$i \leftarrow 3$
3	3	5	0	0	0	0	0	0	$i \leftarrow 2$
4	2	5	0	0	0	0	0	0	$i \leftarrow 1$
5	1	5	1	1	0	0	0	0	$\ell \leftarrow 4$
6	1	4	1	2	0	0	0	0	$\ell \leftarrow 3$
7	1	3	1	3	0	0	0	0	$\ell \leftarrow 2$
8	1	2	1	4	0	0	0	0	$\ell \leftarrow 1$
9	1	1	1	5	0	0	0	0	$\ell \leftarrow 0$

Thus, the vector

$$(5, 0, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[5][5] = 15.$$

Exercise 20.4

Consider the following RCP instance, where the rod has length $n = 6$, and the profits associated with the possible piece lengths are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
p_i	2	6	7	11	13	17

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of ℓ and i, the cells

$v[\ell]$ and $s[\ell]$, the current value of m, and the current value of t. The cell of m is highlighted in red when updated; the cell of t is highlighted in blue when updated; the cells of $v[\ell]$ and $s[\ell]$ are highlighted in blue when assigned.

ℓ	i	$v[\ell]$	$s[\ell]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
1	1	–	–	2	1	$p[1] + v[0] = 2 + 0 = 2 > 0$: $m \leftarrow 2,\ t \leftarrow 1$
	–	2	1	2	1	$v[1] \leftarrow m = 2,\ s[1] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
2	1	–	–	4	1	$p[1] + v[1] = 2 + 2 = 4 > 0$: $m \leftarrow 4,\ t \leftarrow 1$
	2	–	–	6	2	$p[2] + v[0] = 6 + 0 = 6 > 4$: $m \leftarrow 6,\ t \leftarrow 2$
	–	6	2	6	2	$v[2] \leftarrow m = 6,\ s[2] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
3	1	–	–	8	1	$p[1] + v[2] = 2 + 6 = 8 > 0$: $m \leftarrow 8,\ t \leftarrow 1$
	2	–	–	8	1	$p[2] + v[1] = 6 + 2 = 8 = m$: no update
	3	–	–	8	1	$p[3] + v[0] = 7 + 0 = 7 < 8$: no update
	–	8	1	8	1	$v[3] \leftarrow m = 8,\ s[3] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
4	1	–	–	10	1	$p[1] + v[3] = 2 + 8 = 10 > 0$: $m \leftarrow 10,\ t \leftarrow 1$
	2	–	–	12	2	$p[2] + v[2] = 6 + 6 = 12 > 10$: $m \leftarrow 12,\ t \leftarrow 2$
	3	–	–	12	2	$p[3] + v[1] = 7 + 2 = 9 < 12$: no update
	4	–	–	12	2	$p[4] + v[0] = 11 + 0 = 11 < 12$: no update
	–	12	2	12	2	$v[4] \leftarrow m = 12,\ s[4] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
5	1	–	–	14	1	$p[1] + v[4] = 2 + 12 = 14 > 0$: $m \leftarrow 14,\ t \leftarrow 1$
	2	–	–	14	1	$p[2] + v[3] = 6 + 8 = 14 = m$: no update
	3	–	–	14	1	$p[3] + v[2] = 7 + 6 = 13 < 14$: no update
	4	–	–	14	1	$p[4] + v[1] = 11 + 2 = 13 < 14$: no update
	5	–	–	14	1	$p[5] + v[0] = 13 + 0 = 13 < 14$: no update
	–	14	1	14	1	$v[5] \leftarrow m = 14,\ s[5] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
6	1	–	–	16	1	$p[1] + v[5] = 2 + 14 = 16 > 0$: $m \leftarrow 16,\ t \leftarrow 1$
	2	–	–	18	2	$p[2] + v[4] = 6 + 12 = 18 > 16$: $m \leftarrow 18,\ t \leftarrow 2$
	3	–	–	18	2	$p[3] + v[3] = 7 + 8 = 15 < 18$: no update
	4	–	–	18	2	$p[4] + v[2] = 11 + 6 = 17 < 18$: no update
	5	–	–	18	2	$p[5] + v[1] = 13 + 2 = 15 < 18$: no update
	6	–	–	18	2	$p[6] + v[0] = 17 + 0 = 17 < 18$: no update
	–	18	2	18	2	$v[6] \leftarrow m = 18,\ s[6] \leftarrow t = 2$

This is the vector v after the execution of the DP algorithm:

	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$v[\ell]$	0	2	6	8	12	14	18

This is the vector s after the execution of the DP algorithm:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	1	2	1	2

2. We have the following vector s:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	1	2	1	2

We can reconstruct an optimal solution for a rod of length $n=6$ as follows:

Iteration	i	$s[i]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Update of i
-	-	-	0	0	0	0	0	0	$i=6$
1	6	2	0	1	0	0	0	0	$i=6-2=4$
2	4	2	0	2	0	0	0	0	$i=4-2=2$
3	2	2	0	3	0	0	0	0	$i=2-2=0$

Thus, the vector

$$(0,3,0,0,0,0)$$

is an optimal solution. The optimal value is

$$v[6]=18.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and ℓ, the cells $v[i][\ell]$ and $s[i][\ell]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of ℓ.

i	ℓ	$v[i][\ell]$	$s[i][\ell]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	$-\infty$	0	$v[0][1] \leftarrow -\infty$, $s[0][1] \leftarrow 0$
0	2	$-\infty$	0	$v[0][2] \leftarrow -\infty$, $s[0][2] \leftarrow 0$
0	3	$-\infty$	0	$v[0][3] \leftarrow -\infty$, $s[0][3] \leftarrow 0$
0	4	$-\infty$	0	$v[0][4] \leftarrow -\infty$, $s[0][4] \leftarrow 0$
0	5	$-\infty$	0	$v[0][5] \leftarrow -\infty$, $s[0][5] \leftarrow 0$
0	6	$-\infty$	0	$v[0][6] \leftarrow -\infty$, $s[0][6] \leftarrow 0$
1	0	0	0	$i > \ell$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	2	1	$p[1] + v[1][0] = 2 + 0 > v[0][1] = -\infty$: $v[1][1] \leftarrow 2$, $s[1][1] \leftarrow 1$
1	2	4	1	$p[1] + v[1][1] = 2 + 2 > v[0][2] = -\infty$: $v[1][2] \leftarrow 4$, $s[1][2] \leftarrow 1$
1	3	6	1	$p[1] + v[1][2] = 2 + 4 > v[0][3] = -\infty$: $v[1][3] \leftarrow 6$, $s[1][3] \leftarrow 1$
1	4	8	1	$p[1] + v[1][3] = 2 + 6 > v[0][4] = -\infty$: $v[1][4] \leftarrow 8$, $s[1][4] \leftarrow 1$
1	5	10	1	$p[1] + v[1][4] = 2 + 8 > v[0][5] = -\infty$: $v[1][5] \leftarrow 10$, $s[1][5] \leftarrow 1$
1	6	12	1	$p[1] + v[1][5] = 2 + 10 > v[0][6] = -\infty$: $v[1][6] \leftarrow 12$, $s[1][6] \leftarrow 1$
2	0	0	0	$i > \ell$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	2	0	$i > \ell$: $v[2][1] \leftarrow v[1][1] = 2$, $s[2][1] \leftarrow 0$
2	2	6	1	$p[2] + v[2][0] = 6 + 0 > v[1][2] = 4$: $v[2][2] \leftarrow 6$, $s[2][2] \leftarrow 1$
2	3	8	1	$p[2] + v[2][1] = 6 + 2 > v[1][3] = 6$: $v[2][3] \leftarrow 8$, $s[2][3] \leftarrow 1$
2	4	12	1	$p[2] + v[2][2] = 6 + 6 > v[1][4] = 8$: $v[2][4] \leftarrow 12$, $s[2][4] \leftarrow 1$
2	5	14	1	$p[2] + v[2][3] = 6 + 8 > v[1][5] = 10$: $v[2][5] \leftarrow 14$, $s[2][5] \leftarrow 1$
2	6	18	1	$p[2] + v[2][4] = 6 + 12 > v[1][6] = 12$: $v[2][6] \leftarrow 18$, $s[2][6] \leftarrow 1$
3	0	0	0	$i > \ell$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	2	0	$i > \ell$: $v[3][1] \leftarrow v[2][1] = 2$, $s[3][1] \leftarrow 0$
3	2	6	0	$i > \ell$: $v[3][2] \leftarrow v[2][2] = 6$, $s[3][2] \leftarrow 0$
3	3	8	0	$p[3] + v[3][0] = 7 + 0 \leq v[2][3] = 8$: $v[3][3] \leftarrow v[2][3] = 8$, $s[3][3] \leftarrow 0$
3	4	12	0	$p[3] + v[3][1] = 7 + 2 \leq v[2][4] = 12$: $v[3][4] \leftarrow v[2][4] = 12$, $s[3][4] \leftarrow 0$
3	5	14	0	$p[3] + v[3][2] = 7 + 6 \leq v[2][5] = 14$: $v[3][5] \leftarrow v[2][5] = 14$, $s[3][5] \leftarrow 0$
3	6	18	0	$p[3] + v[3][3] = 7 + 8 \leq v[2][6] = 18$: $v[3][6] \leftarrow v[2][6] = 18$, $s[3][6] \leftarrow 0$
4	0	0	0	$i > \ell$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	2	0	$i > \ell$: $v[4][1] \leftarrow v[3][1] = 2$, $s[4][1] \leftarrow 0$
4	2	6	0	$i > \ell$: $v[4][2] \leftarrow v[3][2] = 6$, $s[4][2] \leftarrow 0$
4	3	8	0	$i > \ell$: $v[4][3] \leftarrow v[3][3] = 8$, $s[4][3] \leftarrow 0$
4	4	12	0	$p[4] + v[4][0] = 11 + 0 \leq v[3][4] = 12$: $v[4][4] \leftarrow v[3][4] = 12$, $s[4][4] \leftarrow 0$
4	5	14	0	$p[4] + v[4][1] = 11 + 2 \leq v[3][5] = 14$: $v[4][5] \leftarrow v[3][5] = 14$, $s[4][5] \leftarrow 0$
4	6	18	0	$p[4] + v[4][2] = 11 + 6 \leq v[3][6] = 18$: $v[4][6] \leftarrow v[3][6] = 18$, $s[4][6] \leftarrow 0$
5	0	0	0	$i > \ell$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	2	0	$i > \ell$: $v[5][1] \leftarrow v[4][1] = 2$, $s[5][1] \leftarrow 0$
5	2	6	0	$i > \ell$: $v[5][2] \leftarrow v[4][2] = 6$, $s[5][2] \leftarrow 0$
5	3	8	0	$i > \ell$: $v[5][3] \leftarrow v[4][3] = 8$, $s[5][3] \leftarrow 0$
5	4	12	0	$i > \ell$: $v[5][4] \leftarrow v[4][4] = 12$, $s[5][4] \leftarrow 0$
5	5	14	0	$p[5] + v[5][0] = 13 + 0 \leq v[4][5] = 14$: $v[5][5] \leftarrow v[4][5] = 14$, $s[5][5] \leftarrow 0$
5	6	18	0	$p[5] + v[5][1] = 13 + 2 \leq v[4][6] = 18$: $v[5][6] \leftarrow v[4][6] = 18$, $s[5][6] \leftarrow 0$
6	0	0	0	$i > \ell$: $v[6][0] \leftarrow v[5][0] = 0$, $s[6][0] \leftarrow 0$
6	1	2	0	$i > \ell$: $v[6][1] \leftarrow v[5][1] = 2$, $s[6][1] \leftarrow 0$
6	2	6	0	$i > \ell$: $v[6][2] \leftarrow v[5][2] = 6$, $s[6][2] \leftarrow 0$
6	3	8	0	$i > \ell$: $v[6][3] \leftarrow v[5][3] = 8$, $s[6][3] \leftarrow 0$
6	4	12	0	$i > \ell$: $v[6][4] \leftarrow v[5][4] = 12$, $s[6][4] \leftarrow 0$

6	5	14	0	$i > \ell$: $v[6][5] \leftarrow v[5][5] = 14$, $s[6][5] \leftarrow 0$
6	6	18	0	$p[6]+v[6][0] = 17+0 \leq v[5][6] = 18$: $v[6][6] \leftarrow v[5][6] = 18$, $s[6][6] \leftarrow 0$

These are the matrices v and s obtained during the execution of the DP algorithm:

$v[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$i=1$	0	2	4	6	8	10	12
$i=2$	0	2	6	8	12	14	18
$i=3$	0	2	6	8	12	14	18
$i=4$	0	2	6	8	12	14	18
$i=5$	0	2	6	8	12	14	18
$i=6$	0	2	6	8	12	14	18

$s[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1
$i=2$	0	0	1	1	1	1	1
$i=3$	0	0	0	0	0	0	0
$i=4$	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0
$i=6$	0	0	0	0	0	0	0

The optimal value of the RCP instance is $v[6][6] = 18$.

4. Using the matrix s, we can reconstruct an optimal solution as follows:

Iteration	i	ℓ	$s[i][\ell]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Action
1	6	6	0	0	0	0	0	0	0	$i \leftarrow 5$
2	5	6	0	0	0	0	0	0	0	$i \leftarrow 4$
3	4	6	0	0	0	0	0	0	0	$i \leftarrow 3$
4	3	6	0	0	0	0	0	0	0	$i \leftarrow 2$
5	2	6	1	0	1	0	0	0	0	$\ell \leftarrow 4$
6	2	4	1	0	2	0	0	0	0	$\ell \leftarrow 2$
7	2	2	1	0	3	0	0	0	0	$\ell \leftarrow 0$

Thus, the vector

$$(0, 3, 0, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[6][6] = 18.$$

Exercise 20.5

Consider the following RCP instance, where the rod has length $n = 6$, and the profits associated with the possible piece lengths are:

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$
p_i	1	5	8	9	12	14

Questions

1. Show the execution of the first bottom-up DP algorithm by reporting, for each iteration, the values of the vector v and the values of the vector s.

2. Using the vector s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

3. Show the execution of the second bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and s.

4. Using the matrix s, reconstruct an optimal solution in integer notation $(x_1, x_2, \ldots, x_n)$, and report the optimal value.

Solution

1. Step-by-step execution of the first bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of ℓ and i, the cells

$v[\ell]$ and $s[\ell]$, the current value of m, and the current value of t. The cell of m is highlighted in red when updated; the cell of t is highlighted in blue when updated; the cells of $v[\ell]$ and $s[\ell]$ are highlighted in blue when assigned.

ℓ	i	$v[\ell]$	$s[\ell]$	m	t	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
1	1	–	–	1	1	$p[1] + v[0] = 1 + 0 = 1 > 0$: $m \leftarrow 1,\ t \leftarrow 1$
	–	1	1	1	1	$v[1] \leftarrow m = 1,\ s[1] \leftarrow t = 1$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
2	1	–	–	2	1	$p[1] + v[1] = 1 + 1 = 2 > 0$: $m \leftarrow 2,\ t \leftarrow 1$
	2	–	–	5	2	$p[2] + v[0] = 5 + 0 = 5 > 2$: $m \leftarrow 5,\ t \leftarrow 2$
	–	5	2	5	2	$v[2] \leftarrow m = 5,\ s[2] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
3	1	–	–	6	1	$p[1] + v[2] = 1 + 5 = 6 > 0$: $m \leftarrow 6,\ t \leftarrow 1$
	2	–	–	6	1	$p[2] + v[1] = 5 + 1 = 6 = m$: no update
	3	–	–	8	3	$p[3] + v[0] = 8 + 0 = 8 > 6$: $m \leftarrow 8,\ t \leftarrow 3$
	–	8	3	8	3	$v[3] \leftarrow m = 8,\ s[3] \leftarrow t = 3$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
4	1	–	–	9	1	$p[1] + v[3] = 1 + 8 = 9 > 0$: $m \leftarrow 9,\ t \leftarrow 1$
	2	–	–	10	2	$p[2] + v[2] = 5 + 5 = 10 > 9$: $m \leftarrow 10,\ t \leftarrow 2$
	3	–	–	10	2	$p[3] + v[1] = 8 + 1 = 9 < 10$: no update
	4	–	–	10	2	$p[4] + v[0] = 9 + 0 = 9 < 10$: no update
	–	10	2	10	2	$v[4] \leftarrow m = 10,\ s[4] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
5	1	–	–	11	1	$p[1] + v[4] = 1 + 10 = 11 > 0$: $m \leftarrow 11,\ t \leftarrow 1$
	2	–	–	13	2	$p[2] + v[3] = 5 + 8 = 13 > 11$: $m \leftarrow 13,\ t \leftarrow 2$
	3	–	–	13	2	$p[3] + v[2] = 8 + 5 = 13 = m$: no update
	4	–	–	13	2	$p[4] + v[1] = 9 + 1 = 10 < 13$: no update
	5	–	–	13	2	$p[5] + v[0] = 12 + 0 = 12 < 13$: no update
	–	13	2	13	2	$v[5] \leftarrow m = 13,\ s[5] \leftarrow t = 2$
	–	–	–	0	0	$m \leftarrow 0,\ t \leftarrow 0$
6	1	–	–	14	1	$p[1] + v[5] = 1 + 13 = 14 > 0$: $m \leftarrow 14,\ t \leftarrow 1$
	2	–	–	15	2	$p[2] + v[4] = 5 + 10 = 15 > 14$: $m \leftarrow 15,\ t \leftarrow 2$
	3	–	–	16	3	$p[3] + v[3] = 8 + 8 = 16 > 15$: $m \leftarrow 16,\ t \leftarrow 3$
	4	–	–	16	3	$p[4] + v[2] = 9 + 5 = 14 < 16$: no update
	5	–	–	16	3	$p[5] + v[1] = 12 + 1 = 13 < 16$: no update
	6	–	–	16	3	$p[6] + v[0] = 14 + 0 = 14 < 16$: no update
	–	16	3	16	3	$v[6] \leftarrow m = 16,\ s[6] \leftarrow t = 3$

This is the vector v after the execution of the DP algorithm:

	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$v[\ell]$	0	1	5	8	10	13	16

This is the vector s after the execution of the DP algorithm:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	3	2	2	3

2. We have the following vector s:

	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$s[\ell]$	1	2	3	2	2	3

We can reconstruct an optimal solution for a rod of length $n=6$ as follows:

Iteration	i	$s[i]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Update of i
-	-	-	0	0	0	0	0	0	$i=6$
1	6	3	0	0	1	0	0	0	$i=6-3=3$
2	3	3	0	0	2	0	0	0	$i=3-3=0$

Thus, the vector

$$(0,0,2,0,0,0)$$

is an optimal solution. The optimal value is

$$v[6]=16.$$

3. Step-by-step execution of the second bottom-up DP algorithm

For each step of the DP algorithm, the table shows the values of i and ℓ, the cells $v[i][\ell]$ and $s[i][\ell]$, and the operation performed. The cells are computed for increasing values of i and, for each fixed value of i, for increasing values of ℓ.

i	ℓ	$v[i][\ell]$	$s[i][\ell]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $s[0][0] \leftarrow 0$
0	1	$-\infty$	0	$v[0][1] \leftarrow -\infty$, $s[0][1] \leftarrow 0$
0	2	$-\infty$	0	$v[0][2] \leftarrow -\infty$, $s[0][2] \leftarrow 0$
0	3	$-\infty$	0	$v[0][3] \leftarrow -\infty$, $s[0][3] \leftarrow 0$
0	4	$-\infty$	0	$v[0][4] \leftarrow -\infty$, $s[0][4] \leftarrow 0$
0	5	$-\infty$	0	$v[0][5] \leftarrow -\infty$, $s[0][5] \leftarrow 0$
0	6	$-\infty$	0	$v[0][6] \leftarrow -\infty$, $s[0][6] \leftarrow 0$
1	0	0	0	$i > \ell$: $v[1][0] \leftarrow v[0][0] = 0$, $s[1][0] \leftarrow 0$
1	1	1	1	$p[1] + v[1][0] = 1 + 0 > v[0][1] = -\infty$: $v[1][1] \leftarrow 1$, $s[1][1] \leftarrow 1$
1	2	2	1	$p[1] + v[1][1] = 1 + 1 > v[0][2] = -\infty$: $v[1][2] \leftarrow 2$, $s[1][2] \leftarrow 1$
1	3	3	1	$p[1] + v[1][2] = 1 + 2 > v[0][3] = -\infty$: $v[1][3] \leftarrow 3$, $s[1][3] \leftarrow 1$
1	4	4	1	$p[1] + v[1][3] = 1 + 3 > v[0][4] = -\infty$: $v[1][4] \leftarrow 4$, $s[1][4] \leftarrow 1$
1	5	5	1	$p[1] + v[1][4] = 1 + 4 > v[0][5] = -\infty$: $v[1][5] \leftarrow 5$, $s[1][5] \leftarrow 1$
1	6	6	1	$p[1] + v[1][5] = 1 + 5 > v[0][6] = -\infty$: $v[1][6] \leftarrow 6$, $s[1][6] \leftarrow 1$
2	0	0	0	$i > \ell$: $v[2][0] \leftarrow v[1][0] = 0$, $s[2][0] \leftarrow 0$
2	1	1	0	$i > \ell$: $v[2][1] \leftarrow v[1][1] = 1$, $s[2][1] \leftarrow 0$
2	2	5	1	$p[2] + v[2][0] = 5 + 0 > v[1][2] = 2$: $v[2][2] \leftarrow 5$, $s[2][2] \leftarrow 1$
2	3	6	1	$p[2] + v[2][1] = 5 + 1 > v[1][3] = 3$: $v[2][3] \leftarrow 6$, $s[2][3] \leftarrow 1$
2	4	10	1	$p[2] + v[2][2] = 5 + 5 > v[1][4] = 4$: $v[2][4] \leftarrow 10$, $s[2][4] \leftarrow 1$
2	5	11	1	$p[2] + v[2][3] = 5 + 6 > v[1][5] = 5$: $v[2][5] \leftarrow 11$, $s[2][5] \leftarrow 1$
2	6	15	1	$p[2] + v[2][4] = 5 + 10 > v[1][6] = 6$: $v[2][6] \leftarrow 15$, $s[2][6] \leftarrow 1$
3	0	0	0	$i > \ell$: $v[3][0] \leftarrow v[2][0] = 0$, $s[3][0] \leftarrow 0$
3	1	1	0	$i > \ell$: $v[3][1] \leftarrow v[2][1] = 1$, $s[3][1] \leftarrow 0$
3	2	5	0	$i > \ell$: $v[3][2] \leftarrow v[2][2] = 5$, $s[3][2] \leftarrow 0$
3	3	8	1	$p[3] + v[3][0] = 8 + 0 > v[2][3] = 6$: $v[3][3] \leftarrow 8$, $s[3][3] \leftarrow 1$
3	4	10	0	$p[3] + v[3][1] = 8 + 1 \leq v[2][4] = 10$: $v[3][4] \leftarrow v[2][4] = 10$, $s[3][4] \leftarrow 0$
3	5	13	1	$p[3] + v[3][2] = 8 + 5 > v[2][5] = 11$: $v[3][5] \leftarrow 13$, $s[3][5] \leftarrow 1$
3	6	16	1	$p[3] + v[3][3] = 8 + 8 > v[2][6] = 15$: $v[3][6] \leftarrow 16$, $s[3][6] \leftarrow 1$
4	0	0	0	$i > \ell$: $v[4][0] \leftarrow v[3][0] = 0$, $s[4][0] \leftarrow 0$
4	1	1	0	$i > \ell$: $v[4][1] \leftarrow v[3][1] = 1$, $s[4][1] \leftarrow 0$
4	2	5	0	$i > \ell$: $v[4][2] \leftarrow v[3][2] = 5$, $s[4][2] \leftarrow 0$
4	3	8	0	$i > \ell$: $v[4][3] \leftarrow v[3][3] = 8$, $s[4][3] \leftarrow 0$
4	4	10	0	$p[4] + v[4][0] = 9 + 0 \leq v[3][4] = 10$: $v[4][4] \leftarrow v[3][4] = 10$, $s[4][4] \leftarrow 0$
4	5	13	0	$p[4] + v[4][1] = 9 + 1 \leq v[3][5] = 13$: $v[4][5] \leftarrow v[3][5] = 13$, $s[4][5] \leftarrow 0$
4	6	16	0	$p[4] + v[4][2] = 9 + 5 \leq v[3][6] = 16$: $v[4][6] \leftarrow v[3][6] = 16$, $s[4][6] \leftarrow 0$
5	0	0	0	$i > \ell$: $v[5][0] \leftarrow v[4][0] = 0$, $s[5][0] \leftarrow 0$
5	1	1	0	$i > \ell$: $v[5][1] \leftarrow v[4][1] = 1$, $s[5][1] \leftarrow 0$
5	2	5	0	$i > \ell$: $v[5][2] \leftarrow v[4][2] = 5$, $s[5][2] \leftarrow 0$
5	3	8	0	$i > \ell$: $v[5][3] \leftarrow v[4][3] = 8$, $s[5][3] \leftarrow 0$
5	4	10	0	$i > \ell$: $v[5][4] \leftarrow v[4][4] = 10$, $s[5][4] \leftarrow 0$
5	5	13	0	$p[5] + v[5][0] = 12 + 0 \leq v[4][5] = 13$: $v[5][5] \leftarrow v[4][5] = 13$, $s[5][5] \leftarrow 0$
5	6	16	0	$p[5] + v[5][1] = 12 + 1 \leq v[4][6] = 16$: $v[5][6] \leftarrow v[4][6] = 16$, $s[5][6] \leftarrow 0$
6	0	0	0	$i > \ell$: $v[6][0] \leftarrow v[5][0] = 0$, $s[6][0] \leftarrow 0$
6	1	1	0	$i > \ell$: $v[6][1] \leftarrow v[5][1] = 1$, $s[6][1] \leftarrow 0$
6	2	5	0	$i > \ell$: $v[6][2] \leftarrow v[5][2] = 5$, $s[6][2] \leftarrow 0$
6	3	8	0	$i > \ell$: $v[6][3] \leftarrow v[5][3] = 8$, $s[6][3] \leftarrow 0$
6	4	10	0	$i > \ell$: $v[6][4] \leftarrow v[5][4] = 10$, $s[6][4] \leftarrow 0$

6	5	13	0	$i > \ell$: $v[6][5] \leftarrow v[5][5] = 13$, $s[6][5] \leftarrow 0$
6	6	16	0	$p[6] + v[6][0] = 14 + 0 \leq v[5][6] = 16$: $v[6][6] \leftarrow v[5][6] = 16$, $s[6][6] \leftarrow 0$

These are the matrices v and s obtained during the execution of the DP algorithm:

$v[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
$i=1$	0	1	2	3	4	5	6
$i=2$	0	1	5	6	10	11	15
$i=3$	0	1	5	8	10	13	16
$i=4$	0	1	5	8	10	13	16
$i=5$	0	1	5	8	10	13	16
$i=6$	0	1	5	8	10	13	16

$s[i][\ell]$	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
$i=0$	0	0	0	0	0	0	0
$i=1$	0	1	1	1	1	1	1
$i=2$	0	0	1	1	1	1	1
$i=3$	0	0	0	1	0	1	1
$i=4$	0	0	0	0	0	0	0
$i=5$	0	0	0	0	0	0	0
$i=6$	0	0	0	0	0	0	0

The optimal value of the RCP instance is $v[6][6] = 16$.

4. Using the matrix s, we can reconstruct an optimal solution as follows:

Iteration	i	ℓ	$s[i][\ell]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$x[6]$	Action
1	6	6	0	0	0	0	0	0	0	$i \leftarrow 5$
2	5	6	0	0	0	0	0	0	0	$i \leftarrow 4$
3	4	6	0	0	0	0	0	0	0	$i \leftarrow 3$
4	3	6	1	0	0	1	0	0	0	$\ell \leftarrow 3$
5	3	3	1	0	0	2	0	0	0	$\ell \leftarrow 0$

Thus, the vector

$$(0, 0, 2, 0, 0, 0)$$

is an optimal solution. The optimal value is

$$v[6][6] = 16.$$

DP for the coin change problem

Exercise 21.1

Consider the following CCP instance, where the amount of change to return is $c = 6$, and the denominations are

$$w = (1, 4).$$

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the array v and the array s.

2. Using the array s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the amount of change k, the cells $v[k]$ and $s[k]$, and the operation performed; $s[k]$ is the index j of the denomination of the coin used. In case of ties the smallest index is kept, because m is updated only when the value is strictly smaller.

k	$v[k]$	$s[k]$	Action
0	0	0	$v[0] \leftarrow 0$, $s[0] \leftarrow 0$

1	1	1	$\min\{1 + v[0]\} = \min\{1\} = 1$
2	2	1	$\min\{1 + v[1]\} = \min\{2\} = 2$
3	3	1	$\min\{1 + v[2]\} = \min\{3\} = 3$
4	1	2	$\min\{1 + v[3], 1 + v[0]\} = \min\{4, 1\} = 1$
5	2	1	$\min\{1 + v[4], 1 + v[1]\} = \min\{2, 2\} = 2$
6	3	1	$\min\{1 + v[5], 1 + v[2]\} = \min\{3, 3\} = 3$

This is the array v after the execution of the DP algorithm:

k	0	1	2	3	4	5	6
$v[k]$	0	1	2	3	1	2	3

This is the array s after the execution of the DP algorithm:

k	0	1	2	3	4	5	6
$s[k]$	0	1	1	1	2	1	1

2. We have the following array s:

k	0	1	2	3	4	5	6
$s[k]$	0	1	1	1	2	1	1

We can reconstruct an optimal solution by starting from $k = 6$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$k \leftarrow k - w_{s[k]}$
–	–	–	0	0	$k = 6$
1	6	1	1	0	$k = 6 - 1 = 5$
2	5	1	2	0	$k = 5 - 1 = 4$
3	4	2	2	1	$k = 4 - 4 = 0$

Thus, the vector

$$(2, 1)$$

is an optimal solution: its total value is $2 \cdot 1 + 1 \cdot 4 = 6 = c$. The optimal value is

$$v[6] = 3.$$

Exercise 21.2

Consider the following CCP instance, where the amount of change to return is $c = 11$, and

the denominations are

$$w = (3, 5).$$

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the array v and the array s.

2. Using the array s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the amount of change k, the cells $v[k]$ and $s[k]$, and the operation performed; $s[k]$ is the index j of the denomination of the coin used. In case of ties the smallest index is kept, because m is updated only when the value is strictly smaller. The value $+\infty$ denotes an amount of change that cannot be returned, for which $s[k] = 0$.

k	$v[k]$	$s[k]$	Action
0	0	0	$v[0] \leftarrow 0$, $s[0] \leftarrow 0$
1	$+\infty$	0	no denomination $w_j \leq 1$: $v[1] \leftarrow +\infty$
2	$+\infty$	0	no denomination $w_j \leq 2$: $v[2] \leftarrow +\infty$
3	1	1	$\min\{1 + v[0]\} = \min\{1\} = 1$
4	$+\infty$	0	$\min\{1 + v[1]\} = \min\{+\infty\} = +\infty$
5	1	2	$\min\{1 + v[2], 1 + v[0]\} = \min\{+\infty, 1\} = 1$
6	2	1	$\min\{1 + v[3], 1 + v[1]\} = \min\{2, +\infty\} = 2$
7	$+\infty$	0	$\min\{1 + v[4], 1 + v[2]\} = \min\{+\infty, +\infty\} = +\infty$
8	2	1	$\min\{1 + v[5], 1 + v[3]\} = \min\{2, 2\} = 2$
9	3	1	$\min\{1 + v[6], 1 + v[4]\} = \min\{3, +\infty\} = 3$
10	2	2	$\min\{1 + v[7], 1 + v[5]\} = \min\{+\infty, 2\} = 2$
11	3	1	$\min\{1 + v[8], 1 + v[6]\} = \min\{3, 3\} = 3$

This is the array v after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9	10	11
$v[k]$	0	$+\infty$	$+\infty$	1	$+\infty$	1	2	$+\infty$	2	3	2	3

This is the array s after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9	10	11
$s[k]$	0	0	0	1	0	2	1	0	1	1	2	1

2. We have the following array s:

k	0	1	2	3	4	5	6	7	8	9	10	11
$s[k]$	0	0	0	1	0	2	1	0	1	1	2	1

We can reconstruct an optimal solution by starting from $k = 11$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$k \leftarrow k - w_{s[k]}$
–	–	–	0	0	$k = 11$
1	11	1	1	0	$k = 11 - 3 = 8$
2	8	1	2	0	$k = 8 - 3 = 5$
3	5	2	2	1	$k = 5 - 5 = 0$

Thus, the vector

$$(2, 1)$$

is an optimal solution: its total value is $2 \cdot 3 + 1 \cdot 5 = 11 = c$. The optimal value is

$$v[11] = 3.$$

Exercise 21.3

Consider the following CCP instance, where the amount of change to return is $c = 7$, and

the denominations are

$$w = (2, 4).$$

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the array v and the array s.

2. Using the array s, reconstruct an optimal solution in integer notation (x_1, x_2), or explain why the instance is infeasible, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the amount of change k, the cells $v[k]$ and $s[k]$, and the operation performed; $s[k]$ is the index j of the denomination of the coin used. In case of ties the smallest index is kept, because m is updated only when the value is strictly smaller. The value $+\infty$ denotes an amount of change that cannot be returned, for which $s[k] = 0$.

k	$v[k]$	$s[k]$	Action
0	0	0	$v[0] \leftarrow 0$, $s[0] \leftarrow 0$
1	$+\infty$	0	no denomination $w_j \leq 1$: $v[1] \leftarrow +\infty$
2	1	1	$\min\{1 + v[0]\} = \min\{1\} = 1$
3	$+\infty$	0	$\min\{1 + v[1]\} = \min\{+\infty\} = +\infty$
4	1	2	$\min\{1 + v[2], 1 + v[0]\} = \min\{2, 1\} = 1$
5	$+\infty$	0	$\min\{1 + v[3], 1 + v[1]\} = \min\{+\infty, +\infty\} = +\infty$
6	2	1	$\min\{1 + v[4], 1 + v[2]\} = \min\{2, 2\} = 2$
7	$+\infty$	0	$\min\{1 + v[5], 1 + v[3]\} = \min\{+\infty, +\infty\} = +\infty$

This is the array v after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7
$v[k]$	0	$+\infty$	1	$+\infty$	1	$+\infty$	2	$+\infty$

This is the array s after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7
$s[k]$	0	0	1	0	2	0	1	0

2. Since $v[7] = +\infty$, the instance is **infeasible**: no combination of coins of denominations $w = (2, 4)$ returns an amount of change exactly equal to $c = 7$. Indeed, every denomination is even, so every amount of change that can be returned is even, while $c = 7$ is odd.

   Since $s[7] = 0$, there is no solution to reconstruct. The optimal value is

   $$v[7] = +\infty.$$

Exercise 21.4

Consider the following CCP instance, where the amount of change to return is $c = 12$, and the denominations are

$$w = (3, 7).$$

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the array v and the array s.

2. Using the array s, reconstruct an optimal solution in integer notation (x_1, x_2), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the amount of change k, the cells $v[k]$ and $s[k]$, and the operation performed; $s[k]$ is the index j of the denomination of the coin used. In case of ties the smallest index is kept, because m is updated only when the value is strictly smaller. The value $+\infty$ denotes an amount of change that cannot be returned, for which $s[k] = 0$.

k	$v[k]$	$s[k]$	Action
0	0	0	$v[0] \leftarrow 0$, $s[0] \leftarrow 0$
1	$+\infty$	0	no denomination $w_j \leq 1$: $v[1] \leftarrow +\infty$
2	$+\infty$	0	no denomination $w_j \leq 2$: $v[2] \leftarrow +\infty$
3	1	1	$\min\{1 + v[0]\} = \min\{1\} = 1$
4	$+\infty$	0	$\min\{1 + v[1]\} = \min\{+\infty\} = +\infty$
5	$+\infty$	0	$\min\{1 + v[2]\} = \min\{+\infty\} = +\infty$
6	2	1	$\min\{1 + v[3]\} = \min\{2\} = 2$
7	1	2	$\min\{1 + v[4], 1 + v[0]\} = \min\{+\infty, 1\} = 1$
8	$+\infty$	0	$\min\{1 + v[5], 1 + v[1]\} = \min\{+\infty, +\infty\} = +\infty$
9	3	1	$\min\{1 + v[6], 1 + v[2]\} = \min\{3, +\infty\} = 3$
10	2	1	$\min\{1 + v[7], 1 + v[3]\} = \min\{2, 2\} = 2$
11	$+\infty$	0	$\min\{1 + v[8], 1 + v[4]\} = \min\{+\infty, +\infty\} = +\infty$
12	4	1	$\min\{1 + v[9], 1 + v[5]\} = \min\{4, +\infty\} = 4$

This is the array v after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9	10	11	12
$v[k]$	0	$+\infty$	$+\infty$	1	$+\infty$	$+\infty$	2	1	$+\infty$	3	2	$+\infty$	4

This is the array s after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9	10	11	12
$s[k]$	0	0	0	1	0	0	1	2	0	1	1	0	1

2. We have the following array s:

k	0	1	2	3	4	5	6	7	8	9	10	11	12
$s[k]$	0	0	0	1	0	0	1	2	0	1	1	0	1

We can reconstruct an optimal solution by starting from $k = 12$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$k \leftarrow k - w_{s[k]}$
–	–	–	0	0	$k = 12$
1	12	1	1	0	$k = 12 - 3 = 9$
2	9	1	2	0	$k = 9 - 3 = 6$
3	6	1	3	0	$k = 6 - 3 = 3$
4	3	1	4	0	$k = 3 - 3 = 0$

Thus, the vector

$$(4, 0)$$

is an optimal solution: its total value is $4 \cdot 3 = 12 = c$. The optimal value is

$$v[12] = 4.$$

Exercise 21.5

Consider the following CCP instance, where the amount of change to return is $c = 9$, and

the denominations are

$$w = (2, 3, 7).$$

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the array v and the array s.

2. Using the array s, reconstruct an optimal solution in integer notation (x_1, x_2, x_3), and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the amount of change k, the cells $v[k]$ and $s[k]$, and the operation performed; $s[k]$ is the index j of the denomination of the coin used. In case of ties the smallest index is kept, because m is updated only when the value is strictly smaller. The value $+\infty$ denotes an amount of change that cannot be returned, for which $s[k] = 0$.

k	$v[k]$	$s[k]$	Action
0	0	0	$v[0] \leftarrow 0$, $s[0] \leftarrow 0$
1	$+\infty$	0	no denomination $w_j \leq 1$: $v[1] \leftarrow +\infty$
2	1	1	$\min\{1 + v[0]\} = \min\{1\} = 1$
3	1	2	$\min\{1 + v[1], 1 + v[0]\} = \min\{+\infty, 1\} = 1$
4	2	1	$\min\{1 + v[2], 1 + v[1]\} = \min\{2, +\infty\} = 2$
5	2	1	$\min\{1 + v[3], 1 + v[2]\} = \min\{2, 2\} = 2$
6	2	2	$\min\{1 + v[4], 1 + v[3]\} = \min\{3, 2\} = 2$
7	1	3	$\min\{1 + v[5], 1 + v[4], 1 + v[0]\} = \min\{3, 3, 1\} = 1$
8	3	1	$\min\{1 + v[6], 1 + v[5], 1 + v[1]\} = \min\{3, 3, +\infty\} = 3$
9	2	1	$\min\{1 + v[7], 1 + v[6], 1 + v[2]\} = \min\{2, 3, 2\} = 2$

This is the array v after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9
$v[k]$	0	$+\infty$	1	1	2	2	2	1	3	2

This is the array s after the execution of the DP algorithm:

k	0	1	2	3	4	5	6	7	8	9
$s[k]$	0	0	1	2	1	1	2	3	1	1

2. We have the following array s:

k	0	1	2	3	4	5	6	7	8	9
$s[k]$	0	0	1	2	1	1	2	3	1	1

We can reconstruct an optimal solution by starting from $k = 9$ as follows:

Iteration	k	$s[k]$	$x[1]$	$x[2]$	$x[3]$	$k \leftarrow k - w_{s[k]}$
–	–	–	0	0	0	$k = 9$
1	9	1	1	0	0	$k = 9 - 2 = 7$
2	7	3	1	0	1	$k = 7 - 7 = 0$

Thus, the vector

$$(1, 0, 1)$$

is an optimal solution: its total value is $1 \cdot 2 + 1 \cdot 7 = 9 = c$. The optimal value is

$$v[9] = 2.$$

Dynamic programming for lot sizing

Exercise 22.1

Consider the following ULS instance with $m = 4$ periods, where the demands, the setup costs, the variable production costs, and the holding costs are:

t	1	2	3	4
d_t	30	10	40	10
sc_t	50	80	70	60
vc_t	2	3	1	2
hc_t	1	2	2	1

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrix c, of the vector v, and of the vector r.

2. Using the vector r, reconstruct an optimal solution $(x_1, x_2, \ldots, x_m)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   First, the matrix c is computed. For each step, the table shows the values of j and t,

the current value of u, the cell $c[j][t]$, and the operation performed; the cell $c[j][t]$ is highlighted in blue when assigned.

j	t	u	$c[j][t]$	Action
1	1	2	110	$u \leftarrow vc[1] = 2$, $g \leftarrow sc[1]+vc[1] \cdot d[1] = 50+2 \cdot 30 = 110$, $c[1][1] \leftarrow 110$
1	2	3	140	$u \leftarrow u+hc[1] = 2+1 = 3$, $g \leftarrow g+u \cdot d[2] = 110+3 \cdot 10 = 140$, $c[1][2] \leftarrow 140$
1	3	5	340	$u \leftarrow u+hc[2] = 3+2 = 5$, $g \leftarrow g+u \cdot d[3] = 140+5 \cdot 40 = 340$, $c[1][3] \leftarrow 340$
1	4	7	410	$u \leftarrow u+hc[3] = 5+2 = 7$, $g \leftarrow g+u \cdot d[4] = 340+7 \cdot 10 = 410$, $c[1][4] \leftarrow 410$
2	2	3	110	$u \leftarrow vc[2] = 3$, $g \leftarrow sc[2]+vc[2] \cdot d[2] = 80+3 \cdot 10 = 110$, $c[2][2] \leftarrow 110$
2	3	5	310	$u \leftarrow u+hc[2] = 3+2 = 5$, $g \leftarrow g+u \cdot d[3] = 110+5 \cdot 40 = 310$, $c[2][3] \leftarrow 310$
2	4	7	380	$u \leftarrow u+hc[3] = 5+2 = 7$, $g \leftarrow g+u \cdot d[4] = 310+7 \cdot 10 = 380$, $c[2][4] \leftarrow 380$
3	3	1	110	$u \leftarrow vc[3] = 1$, $g \leftarrow sc[3]+vc[3] \cdot d[3] = 70+1 \cdot 40 = 110$, $c[3][3] \leftarrow 110$
3	4	3	140	$u \leftarrow u+hc[3] = 1+2 = 3$, $g \leftarrow g+u \cdot d[4] = 110+3 \cdot 10 = 140$, $c[3][4] \leftarrow 140$
4	4	2	80	$u \leftarrow vc[4] = 2$, $g \leftarrow sc[4]+vc[4] \cdot d[4] = 60+2 \cdot 10 = 80$, $c[4][4] \leftarrow 80$

This is the matrix c:

$c[j][t]$	$t=1$	$t=2$	$t=3$	$t=4$
$j=1$	110	140	340	410
$j=2$	–	110	310	380
$j=3$	–	–	110	140
$j=4$	–	–	–	80

For each step of the DP algorithm, the table shows the values of t and j, the cells $v[t]$ and $r[t]$, the current value of b, and the current value of a. The cell of b is highlighted in red when updated; the cell of a is highlighted in blue when updated; the cells of $v[t]$ and $r[t]$ are highlighted in blue when assigned.

t	j	$v[t]$	$r[t]$	b	a	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
1	1	–	–	110	1	$v[0]+c[1][1] = 0+110 = 110 < +\infty$: $b \leftarrow 110$, $a \leftarrow 1$
	–	110	1	110	1	$v[1] \leftarrow b = 110$, $r[1] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
2	1	–	–	140	1	$v[0]+c[1][2] = 0+140 = 140 < +\infty$: $b \leftarrow 140$, $a \leftarrow 1$
	2	–	–	140	1	$v[1]+c[2][2] = 110+110 = 220 > 140$: no update
	–	140	1	140	1	$v[2] \leftarrow b = 140$, $r[2] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
3	1	–	–	340	1	$v[0]+c[1][3] = 0+340 = 340 < +\infty$: $b \leftarrow 340$, $a \leftarrow 1$
	2	–	–	340	1	$v[1]+c[2][3] = 110+310 = 420 > 340$: no update
	3	–	–	250	3	$v[2]+c[3][3] = 140+110 = 250 < 340$: $b \leftarrow 250$, $a \leftarrow 3$
	–	250	3	250	3	$v[3] \leftarrow b = 250$, $r[3] \leftarrow a = 3$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
4	1	–	–	410	1	$v[0]+c[1][4] = 0+410 = 410 < +\infty$: $b \leftarrow 410$, $a \leftarrow 1$
	2	–	–	410	1	$v[1]+c[2][4] = 110+380 = 490 > 410$: no update
	3	–	–	280	3	$v[2]+c[3][4] = 140+140 = 280 < 410$: $b \leftarrow 280$, $a \leftarrow 3$
	4	–	–	280	3	$v[3]+c[4][4] = 250+80 = 330 > 280$: no update
	–	280	3	280	3	$v[4] \leftarrow b = 280$, $r[4] \leftarrow a = 3$

This is the vector v after the execution of the DP algorithm:

	$t=0$	$t=1$	$t=2$	$t=3$	$t=4$
$v[t]$	0	110	140	250	280

This is the vector r after the execution of the DP algorithm:

	$t=1$	$t=2$	$t=3$	$t=4$
$r[t]$	1	1	3	3

2. We have the following vector r:

	$t=1$	$t=2$	$t=3$	$t=4$
$r[t]$	1	1	3	3

We can reconstruct an optimal solution as follows:

Iteration	t	$r[t]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$t \leftarrow r[t]-1$
–	–	–	0	0	0	0	$t=4$
1	4	3	0	0	50	0	$t=3-1=2$
2	2	1	40	0	50	0	$t=1-1=0$

- $t=4$, $r[4]=3$: the last production run of periods $1,\dots,4$ is in period 3 and covers the demands of periods $3,\dots,4$, so $x[3] \leftarrow d_3+d_4 = 40+10 = 50$
- $t=2$, $r[2]=1$: the last production run of periods $1,\dots,2$ is in period 1 and covers the demands of periods $1,\dots,2$, so $x[1] \leftarrow d_1+d_2 = 30+10 = 40$

Thus, the vector

$$(40,0,50,0)$$

is an optimal solution. The setups are in periods 1, 3, the inventories are $(s_1,\dots,s_4) = (10,0,10,0)$, and the total cost is

$$\underbrace{50+70}_{\text{setup}} + \underbrace{2\cdot 40+3\cdot 0+1\cdot 50+2\cdot 0}_{\text{production}} + \underbrace{1\cdot 10+2\cdot 0+2\cdot 10+1\cdot 0}_{\text{holding}} = 120+130+30 = 280.$$

The optimal value is

$$v[4] = 280.$$

Exercise 22.2

Consider the following ULS instance with $m = 4$ periods, where the demands, the setup costs, the variable production costs, and the holding costs are:

t	1	2	3	4
d_t	30	40	10	20
sc_t	50	80	80	50
vc_t	1	3	2	1
hc_t	1	1	1	2

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrix c, of the vector v, and of the vector r.

2. Using the vector r, reconstruct an optimal solution $(x_1, x_2, \ldots, x_m)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   First, the matrix c is computed. For each step, the table shows the values of j and t, the current value of u, the cell $c[j][t]$, and the operation performed; the cell $c[j][t]$ is highlighted in blue when assigned.

j	t	u	$c[j][t]$	Action
1	1	1	80	$u \leftarrow vc[1] = 1$, $g \leftarrow sc[1]+vc[1] \cdot d[1] = 50+1 \cdot 30 = 80$, $c[1][1] \leftarrow 80$
1	2	2	160	$u \leftarrow u+hc[1] = 1+1 = 2$, $g \leftarrow g+u \cdot d[2] = 80+2 \cdot 40 = 160$, $c[1][2] \leftarrow 160$
1	3	3	190	$u \leftarrow u+hc[2] = 2+1 = 3$, $g \leftarrow g+u \cdot d[3] = 160+3 \cdot 10 = 190$, $c[1][3] \leftarrow 190$
1	4	4	270	$u \leftarrow u+hc[3] = 3+1 = 4$, $g \leftarrow g+u \cdot d[4] = 190+4 \cdot 20 = 270$, $c[1][4] \leftarrow 270$

2	2	3	200	$u \leftarrow vc[2] = 3$, $g \leftarrow sc[2]+vc[2] \cdot d[2] = 80+3 \cdot 40 = 200$, $c[2][2] \leftarrow 200$
2	3	4	240	$u \leftarrow u+hc[2] = 3+1 = 4$, $g \leftarrow g+u \cdot d[3] = 200+4 \cdot 10 = 240$, $c[2][3] \leftarrow 240$
2	4	5	340	$u \leftarrow u+hc[3] = 4+1 = 5$, $g \leftarrow g+u \cdot d[4] = 240+5 \cdot 20 = 340$, $c[2][4] \leftarrow 340$
3	3	2	100	$u \leftarrow vc[3] = 2$, $g \leftarrow sc[3]+vc[3] \cdot d[3] = 80+2 \cdot 10 = 100$, $c[3][3] \leftarrow 100$
3	4	3	160	$u \leftarrow u+hc[3] = 2+1 = 3$, $g \leftarrow g+u \cdot d[4] = 100+3 \cdot 20 = 160$, $c[3][4] \leftarrow 160$
4	4	1	70	$u \leftarrow vc[4] = 1$, $g \leftarrow sc[4]+vc[4] \cdot d[4] = 50+1 \cdot 20 = 70$, $c[4][4] \leftarrow 70$

This is the matrix c:

$c[j][t]$	$t=1$	$t=2$	$t=3$	$t=4$
$j=1$	80	160	190	270
$j=2$	–	200	240	340
$j=3$	–	–	100	160
$j=4$	–	–	–	70

For each step of the DP algorithm, the table shows the values of t and j, the cells $v[t]$ and $r[t]$, the current value of b, and the current value of a. The cell of b is highlighted in red when updated; the cell of a is highlighted in blue when updated; the cells of $v[t]$ and $r[t]$ are highlighted in blue when assigned.

t	j	$v[t]$	$r[t]$	b	a	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty,\ a \leftarrow 0$
1	1	–	–	80	1	$v[0]+c[1][1] = 0+80 = 80 < +\infty$: $b \leftarrow 80,\ a \leftarrow 1$
	–	80	1	80	1	$v[1] \leftarrow b = 80,\ r[1] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty,\ a \leftarrow 0$
2	1	–	–	160	1	$v[0]+c[1][2] = 0+160 = 160 < +\infty$: $b \leftarrow 160,\ a \leftarrow 1$
	2	–	–	160	1	$v[1]+c[2][2] = 80+200 = 280 > 160$: no update
	–	160	1	160	1	$v[2] \leftarrow b = 160,\ r[2] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty,\ a \leftarrow 0$
3	1	–	–	190	1	$v[0]+c[1][3] = 0+190 = 190 < +\infty$: $b \leftarrow 190,\ a \leftarrow 1$
	2	–	–	190	1	$v[1]+c[2][3] = 80+240 = 320 > 190$: no update
	3	–	–	190	1	$v[2]+c[3][3] = 160+100 = 260 > 190$: no update
	–	190	1	190	1	$v[3] \leftarrow b = 190,\ r[3] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty,\ a \leftarrow 0$
4	1	–	–	270	1	$v[0]+c[1][4] = 0+270 = 270 < +\infty$: $b \leftarrow 270,\ a \leftarrow 1$
	2	–	–	270	1	$v[1]+c[2][4] = 80+340 = 420 > 270$: no update
	3	–	–	270	1	$v[2]+c[3][4] = 160+160 = 320 > 270$: no update
	4	–	–	260	4	$v[3]+c[4][4] = 190+70 = 260 < 270$: $b \leftarrow 260,\ a \leftarrow 4$
	–	260	4	260	4	$v[4] \leftarrow b = 260,\ r[4] \leftarrow a = 4$

This is the vector v after the execution of the DP algorithm:

	$t=0$	$t=1$	$t=2$	$t=3$	$t=4$
$v[t]$	0	80	160	190	260

This is the vector r after the execution of the DP algorithm:

	$t=1$	$t=2$	$t=3$	$t=4$
$r[t]$	1	1	1	4

2. We have the following vector r:

	$t=1$	$t=2$	$t=3$	$t=4$
$r[t]$	1	1	1	4

We can reconstruct an optimal solution as follows:

Iteration	t	$r[t]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$t \leftarrow r[t]-1$
–	–	–	0	0	0	0	$t=4$
1	4	4	0	0	0	20	$t=4-1=3$
2	3	1	80	0	0	20	$t=1-1=0$

- $t=4$, $r[4]=4$: period 4 produces only its own demand, so $x[4] \leftarrow d_4 = 20$
- $t=3$, $r[3]=1$: the last production run of periods $1,\dots,3$ is in period 1 and covers the demands of periods $1,\dots,3$, so $x[1] \leftarrow d_1+d_2+d_3 = 30+40+10 = 80$

Thus, the vector

$$(80,0,0,20)$$

is an optimal solution. The setups are in periods 1, 4, the inventories are $(s_1,\dots,s_4) = (50,10,0,0)$, and the total cost is

$$\underbrace{50+50}_{\text{setup}} + \underbrace{1\cdot 80+3\cdot 0+2\cdot 0+1\cdot 20}_{\text{production}} + \underbrace{1\cdot 50+1\cdot 10+1\cdot 0+2\cdot 0}_{\text{holding}} = 100+100+60 = 260.$$

The optimal value is

$$v[4] = 260.$$

Exercise 22.3

Consider the following ULS instance with $m = 5$ periods, where the demands, the setup costs, the variable production costs, and the holding costs are:

t	1	2	3	4	5
d_t	10	30	40	30	40
sc_t	70	80	50	80	50
vc_t	1	1	1	1	2
hc_t	1	1	2	1	2

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrix c, of the vector v, and of the vector r.

2. Using the vector r, reconstruct an optimal solution $(x_1, x_2, \ldots, x_m)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   First, the matrix c is computed. For each step, the table shows the values of j and t, the current value of u, the cell $c[j][t]$, and the operation performed; the cell $c[j][t]$ is highlighted in blue when assigned.

j	t	u	$c[j][t]$	Action
1	1	1	80	$u \leftarrow vc[1] = 1$, $g \leftarrow sc[1]+vc[1] \cdot d[1] = 70+1 \cdot 10 = 80$, $c[1][1] \leftarrow 80$
1	2	2	140	$u \leftarrow u+hc[1] = 1+1 = 2$, $g \leftarrow g+u \cdot d[2] = 80+2 \cdot 30 = 140$, $c[1][2] \leftarrow 140$
1	3	3	260	$u \leftarrow u+hc[2] = 2+1 = 3$, $g \leftarrow g+u \cdot d[3] = 140+3 \cdot 40 = 260$, $c[1][3] \leftarrow 260$
1	4	5	410	$u \leftarrow u+hc[3] = 3+2 = 5$, $g \leftarrow g+u \cdot d[4] = 260+5 \cdot 30 = 410$, $c[1][4] \leftarrow 410$

1	5	6	650	$u \leftarrow u+hc[4] = 5+1 = 6$, $g \leftarrow g+u \cdot d[5] = 410+6 \cdot 40 = 650$, $c[1][5] \leftarrow 650$
2	2	1	110	$u \leftarrow vc[2] = 1$, $g \leftarrow sc[2]+vc[2] \cdot d[2] = 80+1 \cdot 30 = 110$, $c[2][2] \leftarrow 110$
2	3	2	190	$u \leftarrow u+hc[2] = 1+1 = 2$, $g \leftarrow g+u \cdot d[3] = 110+2 \cdot 40 = 190$, $c[2][3] \leftarrow 190$
2	4	4	310	$u \leftarrow u+hc[3] = 2+2 = 4$, $g \leftarrow g+u \cdot d[4] = 190+4 \cdot 30 = 310$, $c[2][4] \leftarrow 310$
2	5	5	510	$u \leftarrow u+hc[4] = 4+1 = 5$, $g \leftarrow g+u \cdot d[5] = 310+5 \cdot 40 = 510$, $c[2][5] \leftarrow 510$
3	3	1	90	$u \leftarrow vc[3] = 1$, $g \leftarrow sc[3]+vc[3] \cdot d[3] = 50+1 \cdot 40 = 90$, $c[3][3] \leftarrow 90$
3	4	3	180	$u \leftarrow u+hc[3] = 1+2 = 3$, $g \leftarrow g+u \cdot d[4] = 90+3 \cdot 30 = 180$, $c[3][4] \leftarrow 180$
3	5	4	340	$u \leftarrow u+hc[4] = 3+1 = 4$, $g \leftarrow g+u \cdot d[5] = 180+4 \cdot 40 = 340$, $c[3][5] \leftarrow 340$
4	4	1	110	$u \leftarrow vc[4] = 1$, $g \leftarrow sc[4]+vc[4] \cdot d[4] = 80+1 \cdot 30 = 110$, $c[4][4] \leftarrow 110$
4	5	2	190	$u \leftarrow u+hc[4] = 1+1 = 2$, $g \leftarrow g+u \cdot d[5] = 110+2 \cdot 40 = 190$, $c[4][5] \leftarrow 190$
5	5	2	130	$u \leftarrow vc[5] = 2$, $g \leftarrow sc[5]+vc[5] \cdot d[5] = 50+2 \cdot 40 = 130$, $c[5][5] \leftarrow 130$

This is the matrix c:

$c[j][t]$	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
$j=1$	80	140	260	410	650
$j=2$	–	110	190	310	510
$j=3$	–	–	90	180	340
$j=4$	–	–	–	110	190
$j=5$	–	–	–	–	130

For each step of the DP algorithm, the table shows the values of t and j, the cells $v[t]$ and $r[t]$, the current value of b, and the current value of a. The cell of b is highlighted in red when updated; the cell of a is highlighted in blue when updated; the cells of $v[t]$ and $r[t]$ are highlighted in blue when assigned.

t	j	$v[t]$	$r[t]$	b	a	Action
–	–	0	–	–	–	$v[0] \leftarrow 0$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
1	1	–	–	80	1	$v[0]+c[1][1] = 0+80 = 80 < +\infty$: $b \leftarrow 80$, $a \leftarrow 1$
	–	80	1	80	1	$v[1] \leftarrow b = 80$, $r[1] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
2	1	–	–	140	1	$v[0]+c[1][2] = 0+140 = 140 < +\infty$: $b \leftarrow 140$, $a \leftarrow 1$
	2	–	–	140	1	$v[1]+c[2][2] = 80+110 = 190 > 140$: no update
	–	140	1	140	1	$v[2] \leftarrow b = 140$, $r[2] \leftarrow a = 1$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
3	1	–	–	260	1	$v[0]+c[1][3] = 0+260 = 260 < +\infty$: $b \leftarrow 260$, $a \leftarrow 1$
	2	–	–	260	1	$v[1]+c[2][3] = 80+190 = 270 > 260$: no update
	3	–	–	230	3	$v[2]+c[3][3] = 140+90 = 230 < 260$: $b \leftarrow 230$, $a \leftarrow 3$
	–	230	3	230	3	$v[3] \leftarrow b = 230$, $r[3] \leftarrow a = 3$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
4	1	–	–	410	1	$v[0]+c[1][4] = 0+410 = 410 < +\infty$: $b \leftarrow 410$, $a \leftarrow 1$
	2	–	–	390	2	$v[1]+c[2][4] = 80+310 = 390 < 410$: $b \leftarrow 390$, $a \leftarrow 2$
	3	–	–	320	3	$v[2]+c[3][4] = 140+180 = 320 < 390$: $b \leftarrow 320$, $a \leftarrow 3$
	4	–	–	320	3	$v[3]+c[4][4] = 230+110 = 340 > 320$: no update
	–	320	3	320	3	$v[4] \leftarrow b = 320$, $r[4] \leftarrow a = 3$
	–	–	–	$+\infty$	0	$b \leftarrow +\infty$, $a \leftarrow 0$
5	1	–	–	650	1	$v[0]+c[1][5] = 0+650 = 650 < +\infty$: $b \leftarrow 650$, $a \leftarrow 1$
	2	–	–	590	2	$v[1]+c[2][5] = 80+510 = 590 < 650$: $b \leftarrow 590$, $a \leftarrow 2$
	3	–	–	480	3	$v[2]+c[3][5] = 140+340 = 480 < 590$: $b \leftarrow 480$, $a \leftarrow 3$
	4	–	–	420	4	$v[3]+c[4][5] = 230+190 = 420 < 480$: $b \leftarrow 420$, $a \leftarrow 4$
	5	–	–	420	4	$v[4]+c[5][5] = 320+130 = 450 > 420$: no update
	–	420	4	420	4	$v[5] \leftarrow b = 420$, $r[5] \leftarrow a = 4$

This is the vector v after the execution of the DP algorithm:

	$t=0$	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
$v[t]$	0	80	140	230	320	420

This is the vector r after the execution of the DP algorithm:

	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
$r[t]$	1	1	3	3	4

2. We have the following vector r:

	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
$r[t]$	1	1	3	3	4

We can reconstruct an optimal solution as follows:

Iteration	t	$r[t]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$x[5]$	$t \leftarrow r[t]-1$
–	–	–	0	0	0	0	0	$t=5$
1	5	4	0	0	0	70	0	$t=4-1=3$
2	3	3	0	0	40	70	0	$t=3-1=2$
3	2	1	40	0	40	70	0	$t=1-1=0$

- $t=5$, $r[5]=4$: the last production run of periods $1,\dots,5$ is in period 4 and covers the demands of periods $4,\dots,5$, so $x[4] \leftarrow d_4+d_5=30+40=70$
- $t=3$, $r[3]=3$: period 3 produces only its own demand, so $x[3] \leftarrow d_3=40$
- $t=2$, $r[2]=1$: the last production run of periods $1,\dots,2$ is in period 1 and covers the demands of periods $1,\dots,2$, so $x[1] \leftarrow d_1+d_2=10+30=40$

Thus, the vector

$$(40,0,40,70,0)$$

is an optimal solution. The setups are in periods 1, 3, 4, the inventories are $(s_1,\dots,s_5)=(30,0,0,40,0)$, and the total cost is

$$\underbrace{70+50+80}_{\text{setup}}+\underbrace{1\cdot 40+1\cdot 0+1\cdot 40+1\cdot 70+2\cdot 0}_{\text{production}}$$
$$+\underbrace{1\cdot 30+1\cdot 0+2\cdot 0+1\cdot 40+2\cdot 0}_{\text{holding}}=200+150+70=420.$$

The optimal value is

$$v[5]=420.$$

Exercise 22.4

Consider the following CLS instance with $m = 3$ periods, where the demands, the setup costs, the variable production costs, the holding costs, and the capacities are:

t	1	2	3
d_t	1	2	3
sc_t	10	5	8
vc_t	1	1	2
hc_t	1	1	2
cap_t	2	4	4

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and r.

2. Using the matrix r, reconstruct an optimal solution $(x_1, x_2, \ldots, x_m)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of t and k, the cells $v[t][k]$ and $r[t][k]$, and the operation performed. The cells are computed for increasing values of t and, for each fixed value of t, for increasing values of $k \in \{0, 1, \ldots, D_{t+1,m}\}$. We recall that $f_t(0) = 0$ and $f_t(x) = sc_t + vc_t\, x$ for $x \geq 1$.

t	k	$v[t][k]$	$r[t][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $r[0][0] \leftarrow 0$
0	1	$+\infty$	0	$v[0][1] \leftarrow +\infty$, $r[0][1] \leftarrow 0$
0	2	$+\infty$	0	$v[0][2] \leftarrow +\infty$, $r[0][2] \leftarrow 0$
0	3	$+\infty$	0	$v[0][3] \leftarrow +\infty$, $r[0][3] \leftarrow 0$
0	4	$+\infty$	0	$v[0][4] \leftarrow +\infty$, $r[0][4] \leftarrow 0$
0	5	$+\infty$	0	$v[0][5] \leftarrow +\infty$, $r[0][5] \leftarrow 0$

0	6	$+\infty$	0	$v[0][6] \leftarrow +\infty$, $r[0][6] \leftarrow 0$
1	0	11	1	$\min\{v[0][1]+f_1(0), v[0][0]+f_1(1)\}+hc_1 \cdot 0 = \min\{+\infty, 0+11\}+1 \cdot 0 = 11$; first minimum at $x = 1$
1	1	13	2	$\min\{v[0][2]+f_1(0), v[0][1]+f_1(1), v[0][0]+f_1(2)\}+hc_1 \cdot 1 = \min\{+\infty, +\infty, 0+12\}+1 \cdot 1 = 13$; first minimum at $x = 2$
1	2	$+\infty$	0	the candidates use the cells $v[0][1], \ldots, v[0][3]$, which are all $+\infty$: $v[1][2] \leftarrow +\infty$
1	3	$+\infty$	0	the candidates use the cells $v[0][2], \ldots, v[0][4]$, which are all $+\infty$: $v[1][3] \leftarrow +\infty$
1	4	$+\infty$	0	the candidates use the cells $v[0][3], \ldots, v[0][5]$, which are all $+\infty$: $v[1][4] \leftarrow +\infty$
1	5	$+\infty$	0	the candidates use the cells $v[0][4], \ldots, v[0][6]$, which are all $+\infty$: $v[1][5] \leftarrow +\infty$
2	0	18	2	$\min\{v[1][2]+f_2(0), v[1][1]+f_2(1), v[1][0]+f_2(2)\}+hc_2 \cdot 0 = \min\{+\infty, 13+6, 11+7\}+1 \cdot 0 = 18$; first minimum at $x = 2$
2	1	20	3	$\min\{v[1][3]+f_2(0), v[1][2]+f_2(1), v[1][1]+f_2(2), v[1][0]+f_2(3)\}+hc_2 \cdot 1 = \min\{+\infty, +\infty, 13+7, 11+8\}+1 \cdot 1 = 20$; first minimum at $x = 3$
2	2	22	4	$\min\{v[1][4]+f_2(0), v[1][3]+f_2(1), v[1][2]+f_2(2), v[1][1]+f_2(3), v[1][0]+f_2(4)\}+hc_2 \cdot 2 = \min\{+\infty, +\infty, +\infty, 13+8, 11+9\}+1 \cdot 2 = 22$; first minimum at $x = 4$
2	3	25	4	$\min\{v[1][5]+f_2(0), v[1][4]+f_2(1), v[1][3]+f_2(2), v[1][2]+f_2(3), v[1][1]+f_2(4)\}+hc_2 \cdot 3 = \min\{+\infty, +\infty, +\infty, +\infty, 13+9\}+1 \cdot 3 = 25$; first minimum at $x = 4$
3	0	25	0	$\min\{v[2][3]+f_3(0), v[2][2]+f_3(1), v[2][1]+f_3(2), v[2][0]+f_3(3)\}+hc_3 \cdot 0 = \min\{25+0, 22+10, 20+12, 18+14\}+2 \cdot 0 = 25$; first minimum at $x = 0$

These are the matrices v and r obtained during the execution of the DP algorithm (a dot marks a pair (t,k) with $k > D_{t+1,m}$, which is not a state):

$v[t][k]$	0	1	2	3	4	5	6
$t=0$	0	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$
$t=1$	11	13	$+\infty$	$+\infty$	$+\infty$	$+\infty$	·
$t=2$	18	20	22	25	·	·	·
$t=3$	25	·	·	·	·	·	·

$r[t][k]$	0	1	2	3	4	5	6
$t=0$	0	0	0	0	0	0	0
$t=1$	1	2	0	0	0	0	·
$t=2$	2	3	4	4	·	·	·
$t=3$	0	·	·	·	·	·	·

The optimal value of the CLS instance is $v[3][0] = 25$.

2. Using the matrix r, we can reconstruct an optimal solution as follows:

Iteration	t	k	$r[t][k]$	$x[1]$	$x[2]$	$x[3]$	$k \leftarrow k + d[t] - x[t]$
–	–	–	–	0	0	0	$t=3,\ k=0$
1	3	0	0	0	0	0	$k = 0+3-0 = 3$
2	2	3	4	0	4	0	$k = 3+2-4 = 1$
3	1	1	2	2	4	0	$k = 1+1-2 = 0$

Thus, the vector

$$(2, 4, 0)$$

is an optimal solution. The setups are in periods 1, 2, the inventories are $(s_1, \ldots, s_3) = (1, 3, 0)$, and the total cost is

$$\underbrace{10+5}_{\text{setup}} + \underbrace{1\cdot 2 + 1\cdot 4 + 2\cdot 0}_{\text{production}} + \underbrace{1\cdot 1 + 1\cdot 3 + 2\cdot 0}_{\text{holding}} = 15 + 6 + 4 = 25.$$

The optimal value is

$$v[3][0] = 25.$$

Exercise 22.5

Consider the following CLS instance with $m = 4$ periods, where the demands, the setup costs, the variable production costs, the holding costs, and the capacities are:

t	1	2	3	4
d_t	3	1	1	2
sc_t	5	6	10	10
vc_t	2	1	1	1
hc_t	1	1	2	2
cap_t	4	3	3	2

Questions

1. Show the execution of the bottom-up DP algorithm by reporting, for each iteration, the values of the matrices v and r.

2. Using the matrix r, reconstruct an optimal solution $(x_1, x_2, \ldots, x_m)$, and report the optimal value.

Solution

1. Step-by-step execution of the bottom-up DP algorithm

   For each step of the DP algorithm, the table shows the values of t and k, the cells $v[t][k]$ and $r[t][k]$, and the operation performed. The cells are computed for increasing values of t and, for each fixed value of t, for increasing values of $k \in \{0, 1, \ldots, D_{t+1,m}\}$. We recall that $f_t(0) = 0$ and $f_t(x) = sc_t + vc_t\, x$ for $x \geq 1$.

t	k	$v[t][k]$	$r[t][k]$	Action
0	0	0	0	$v[0][0] \leftarrow 0$, $r[0][0] \leftarrow 0$
0	1	$+\infty$	0	$v[0][1] \leftarrow +\infty$, $r[0][1] \leftarrow 0$
0	2	$+\infty$	0	$v[0][2] \leftarrow +\infty$, $r[0][2] \leftarrow 0$
0	3	$+\infty$	0	$v[0][3] \leftarrow +\infty$, $r[0][3] \leftarrow 0$
0	4	$+\infty$	0	$v[0][4] \leftarrow +\infty$, $r[0][4] \leftarrow 0$
0	5	$+\infty$	0	$v[0][5] \leftarrow +\infty$, $r[0][5] \leftarrow 0$

0	6	$+\infty$	0	$v[0][6] \leftarrow +\infty$, $r[0][6] \leftarrow 0$
0	7	$+\infty$	0	$v[0][7] \leftarrow +\infty$, $r[0][7] \leftarrow 0$
1	0	11	3	$\min\{v[0][3]+f_1(0), v[0][2]+f_1(1), v[0][1]+f_1(2), v[0][0]+f_1(3)\}+hc_1 \cdot 0 = \min\{+\infty, +\infty, +\infty, 0+11\}+1 \cdot 0 = 11$; first minimum at $x = 3$
1	1	14	4	$\min\{v[0][4]+f_1(0), v[0][3]+f_1(1), v[0][2]+f_1(2), v[0][1]+f_1(3), v[0][0]+f_1(4)\}+hc_1 \cdot 1 = \min\{+\infty, +\infty, +\infty, +\infty, 0+13\}+1 \cdot 1 = 14$; first minimum at $x = 4$
1	2	$+\infty$	0	the candidates use the cells $v[0][1], \ldots, v[0][5]$, which are all $+\infty$: $v[1][2] \leftarrow +\infty$
1	3	$+\infty$	0	the candidates use the cells $v[0][2], \ldots, v[0][6]$, which are all $+\infty$: $v[1][3] \leftarrow +\infty$
1	4	$+\infty$	0	the candidates use the cells $v[0][3], \ldots, v[0][7]$, which are all $+\infty$: $v[1][4] \leftarrow +\infty$
2	0	14	0	$\min\{v[1][1]+f_2(0), v[1][0]+f_2(1)\}+hc_2 \cdot 0 = \min\{14+0, 11+7\}+1 \cdot 0 = 14$; first minimum at $x = 0$
2	1	20	2	$\min\{v[1][2]+f_2(0), v[1][1]+f_2(1), v[1][0]+f_2(2)\}+hc_2 \cdot 1 = \min\{+\infty, 14+7, 11+8\}+1 \cdot 1 = 20$; first minimum at $x = 2$
2	2	22	3	$\min\{v[1][3]+f_2(0), v[1][2]+f_2(1), v[1][1]+f_2(2), v[1][0]+f_2(3)\}+hc_2 \cdot 2 = \min\{+\infty, +\infty, 14+8, 11+9\}+1 \cdot 2 = 22$; first minimum at $x = 3$
2	3	26	3	$\min\{v[1][4]+f_2(0), v[1][3]+f_2(1), v[1][2]+f_2(2), v[1][1]+f_2(3)\}+hc_2 \cdot 3 = \min\{+\infty, +\infty, +\infty, 14+9\}+1 \cdot 3 = 26$; first minimum at $x = 3$
3	0	20	0	$\min\{v[2][1]+f_3(0), v[2][0]+f_3(1)\}+hc_3 \cdot 0 = \min\{20+0, 14+11\}+2 \cdot 0 = 20$; first minimum at $x = 0$
3	1	24	0	$\min\{v[2][2]+f_3(0), v[2][1]+f_3(1), v[2][0]+f_3(2)\}+hc_3 \cdot 1 = \min\{22+0, 20+11, 14+12\}+2 \cdot 1 = 24$; first minimum at $x = 0$
3	2	30	0	$\min\{v[2][3]+f_3(0), v[2][2]+f_3(1), v[2][1]+f_3(2), v[2][0]+f_3(3)\}+hc_3 \cdot 2 = \min\{26+0, 22+11, 20+12, 14+13\}+2 \cdot 2 = 30$; first minimum at $x = 0$
4	0	30	0	$\min\{v[3][2]+f_4(0), v[3][1]+f_4(1), v[3][0]+f_4(2)\}+hc_4 \cdot 0 = \min\{30+0, 24+11, 20+12\}+2 \cdot 0 = 30$; first minimum at $x = 0$

These are the matrices v and r obtained during the execution of the DP algorithm (a dot marks a pair (t,k) with $k > D_{t+1,m}$, which is not a state):

$v[t][k]$	0	1	2	3	4	5	6	7
$t=0$	0	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$
$t=1$	11	14	$+\infty$	$+\infty$	$+\infty$	·	·	·
$t=2$	14	20	22	26	·	·	·	·
$t=3$	20	24	30	·	·	·	·	·
$t=4$	30	·	·	·	·	·	·	·

$r[t][k]$	0	1	2	3	4	5	6	7
$t=0$	0	0	0	0	0	0	0	0
$t=1$	3	4	0	0	0	·	·	·
$t=2$	0	2	3	3	·	·	·	·
$t=3$	0	0	0	·	·	·	·	·
$t=4$	0	·	·	·	·	·	·	·

The optimal value of the CLS instance is $v[4][0] = 30$.

2. Using the matrix r, we can reconstruct an optimal solution as follows:

Iteration	t	k	$r[t][k]$	$x[1]$	$x[2]$	$x[3]$	$x[4]$	$k \leftarrow k + d[t] - x[t]$
–	–	–	–	0	0	0	0	$t = 4,\ k = 0$
1	4	0	0	0	0	0	0	$k = 0 + 2 - 0 = 2$
2	3	2	0	0	0	0	0	$k = 2 + 1 - 0 = 3$
3	2	3	3	0	3	0	0	$k = 3 + 1 - 3 = 1$
4	1	1	4	4	3	0	0	$k = 1 + 3 - 4 = 0$

Thus, the vector

$$(4, 3, 0, 0)$$

is an optimal solution. The setups are in periods 1, 2, the inventories are $(s_1, \ldots, s_4) = (1, 3, 2, 0)$, and the total cost is

$$\underbrace{5+6}_{\text{setup}} + \underbrace{2\cdot 4 + 1\cdot 3 + 1\cdot 0 + 1\cdot 0}_{\text{production}} + \underbrace{1\cdot 1 + 1\cdot 3 + 2\cdot 2 + 2\cdot 0}_{\text{holding}} = 11 + 11 + 8 = 30.$$

The optimal value is

$$v[4][0] = 30.$$